VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATION SYSTEMS

FINAL PROJECT REPORT

ENTERPRISE RESOURCE PLANNING

TOPIC:

Kammy chatbox

OVERCOMING THE PARADOX OF CHOICE WITH AN AI-POWERED SALES ASSISTANT FOR CONFIDENT DECISION-MAKING

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We would like to thank Dr. Tran Van Hai Trieu for his in-depth guidance throughout the project. His very interesting presentations have made it clear to us how to understand most of the important knowledge on the ERP system and its application in today's business environment.

The project "Overcoming the paradox of choice and enabling confident decision making with AI-powered sales assistants" gave us an opportunity to apply theoretical knowledge in practice. The stages of project planning and requirements analysis through system implementation and testing rely on the course fundamentals. We have a big thanks to engineer Le Vo Dinh Kha, for his practical insights and constructive feedback, which helped us overcome our challenges and stay focused during the entire development process.

We are proud of what we have achieved, but we also realize that there is always room for growth and continuous improvement. Moreover, his project has not taught us technical skills, but also given us valuable lessons in adaptability, responsibility, and teamwork.

We would like to thank Dr. Tran Van Hai Trieu and Engineer Le Vo Dinh Kha once more for their dedication, encouragement, and generous knowledge sharing. Their support has had a lasting impact on our academic and professional development.

Ho Chi Minh City, June 2025

INSTRUCTOR'S FEEDBACK

ASSIGNMENT AND MEMBER EVALUATION TABLE

Table 1: Assignment and member evaluation table

Name	Student ID	Assignment	Evaluation
Nguyen Thi Thanh Chau	22520150	**Week 1:** Outline Template **Week 2:** Market Research, Case Study Identification, Planning **Week 3:** BMNP Preparation (Shipping process, Good receipt, Stock Issuance Process) **Week 4:** Setting up Database and Coding Core Function O-2-C process **Week 5:** Product Evaluation	**Week 1:** 100% **Week 2:** 100% **Week 3:** 100% **Week 4:** 100% **Week 5:** 100%
Huynh Quang Thinh	22521407	**Week 1:** Design Frontend Interface, ERP Introduction Preparation, Brand Design **Week 2:** Outline Template, Slide, Report **Week 3:** BMNP Preparation (Customer service process, Procurement process) **Week 4:** Proposing New Customer Service Process, Database Design **Week 5:** Optimize Frontend, Slide Preparation	**Week 1:** 100% **Week 2:** 100% **Week 3:** 100% **Week 4:** 100% **Week 5:** 100%
Tran Tinh Dan Thanh	22521368	**Week 1:** Acknowledge Report, Market Research, ERP Research	**Week 1:** 100% **Week 2:** 100%

		Week 2: Dataset Preparation and Progressing (Customers and Mobile Phone Dataset) **Week 3:** BMNP Preparation (Online Sale Process, Offline Sale Process, Supplier Payment Process, Wanrity process) **Week 4:** Proposing New Sales Process **Week 5:** Solution Evaluation, Chapter 5, and Outline Slide	**Week 3:** 100% **Week 4:** 100% **Week 5:** 100%
Nguyen Huy Hoang	22520468	**Week 1:** Setting up AI chatbot, System Architect Preparation **Week 2:** Data Processing and Selecting Technology and Tools. **Week 3:** BMNP Preparation (System Architecture, Inventory Inspection Process, Customer payment process) **Week 4:** Coding core function Product recommendation **Week 5:** Dataset introduction and Implementation Introduction Preparation	**Week 1:** 100% **Week 2:** 100% **Week 3:** 100% **Week 4:** 100% **Week 5:** 100%

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LIST OF ABBREVIATIONS

NO.	ABBREVIATION	CONTENT
1	ERP	Enterprise Resource Planning
2	CRM	Customer Relationship Management
3	O2C	Order to cash
4	RACI	Responsible, Accountable, Consulted, and Informed.
5	AI	Artificial Intelligence
6	CDP	Customer Data Platform
7	WBS	Work Breakdown Structure
8	UI	User Interface
9	BPMN	Process Model and Notation
10	SQL	Structured Query Language
11	RFQ	Request for quotation
12	HTML	HyperText Markup Language
13	ML	Machine Learning
14	UX	Use Experience
15	FAQ	Frequently Asked Questions

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CHAPTER 1: GENERAL INFORMATION

Chapter 1 outlines the brief introduction of our projects, particularly objectives, scopes, stakeholders of this project, and project establishment. Hence, readers are able to know an overview of the product and the key resources involved in the project before delving into the detailed presentation of the solution in the next Chapter.

1.1. ERP Project Overview:

Enterprise Resource Planning systems illustrates integrated software frameworks designed to centralize and streamline an organization's core business functions[1]. Common ERP software modules include product planning, purchasing, inventory, distribution, order tracking, finance, accounting, and other similar things [1]. In our project, we focus on model Customer Relationship Management in ERP via developing Chatbot "Kammy", which can support companies to respond to customers' requirements in real-time and provide the wholesale process for customers, ranging from order to cancellation order. Our group is implementing the "Kammy" chatbot in Vietnam's smartphone industry, especially for Cellphone S company, aiming to address their problems, including Paradox of Choices (Case study 1); Inconsistent Purchase Journey (Case study 2); the expectation of fast, personalized support among Gen Z customers (Case study 3); Manual Consultation Model (Case Study 4). Furthermore, by deploying this integrated approach, the system achieves consistent, high-accuracy data capture and significantly reduces processing time.

1.2. Project Scope:

We determine the scope of Kammy Chatbot via 3 factors, including geography, language, features. In terms of geography, this chatbot is used for Cellphone S Company - a Mobile Phone Retail System in Vietnam. And this Kammy is used in the Vietnamese smartphone market so the main language of our project is VietNamese. With regardless of features, Chatbot allows companies to improve their traditional process in CRM module of ERP (sale and customer services), particularly:

- Product Recommendation: Based on customers' requests and their characteristics, Kammy will analyse and find the best choices for them.
- Product Order: Customers are able to order via chat with Kammy.
- Product Cancel: Customers can cancel the order by sending a "cancel" message.
- Product Payment: Customers are capable of making payment quickly via chatbot by prompting "Payment".

1.3. Project Objectives:

This project aims to enhance the CRM module in the ERP system by integrating Artificial Intelligence (AI) to replace manual consultant staff. The goal is to keep the project's error rate below 20% while enabling the chatbot to answer customer inquiries through phone or other channels, support the Order-to-Cash (O2C) process, and automatically update information in the database.

1.4. Stakeholders

To identify stakeholders as well as their responsibilities, our team uses the RACI matrix to manage human resources effectively. In this project, there are seven core roles, covering all the tasks in the Work Breakdown Table.

In this Matrix, we divides the responsible level into four categories, including R meaning Responsible for completing the task, A standing for Accountable for ensuring the Responsible completes the task, C being Responsible for advising and giving feedback to help the Responsible complete the task, and I for Monitor progress and receive reports after the task is completed.

Table 1.4.1.1: RACI Matrix

Activity	Product Manager	Back-end Developer	Data Analyst	Business Analyst	Front-end Developer	Tester	Financialist	CEO
1.1. Research								
1.1.1. Market Research	A			R				I
1.1.2. Customer Research	A							I
1.2. Stakeholder Engagement				R				I
1.2.1. Stakeholder Identification	R			A				I
1.2.2. Stakeholder Communication Setup	A			R				I
1.2.3. Stakeholder Goal Identification	A			R				I

1.2.4. Stakeholder Scope Identification	**A**			**R**				**I**
1.3. Resource Identification							**R**	**I**
1.3.1. Total Budget Identification	**A**			**C**				**I**
1.3.2. Human Resource Identification	**R**						**R**	**I**
1.3.3. Development Timeline Identification	**A/R**							**I**
2. Project Planning Phase								
2.1. Scope Planning	**A/R**							**I**
2.2. Time Planning	**A/R**							**I**
2.3. Cost Planning	**A/R**							**I**

2.4. Risk Planning	**A/R**							**I**
3. Requirement Analysis Phase				**R**				
3.1. Gather Customers' Requirement	**A**			**R**				**I**
3.2. User Stories	**A**			**R**				**I**
3.3. System Analysis & Design	**A**							**I**
4. Development Phase								
4.1. Product Backlogs	**A/R**							**I**
4.2. Human Resource Preparation	**A/R**							**I**
4.3. Sprint 1								**I**

4.3.1. Sprint 1 Planning								**I**
4.3.2. Sprint 1 Implementation								**I**
4.3.2.1. Sprint 1 Scrum Daily Meetings	**A/R**							**I**
4.3.2.2. Prototype Creation	**A**			**C**	**R**			**I**
4.3.2.3. Database Initialization and Setup	**A**	**C**	**R**	**C**				**I**
4.3.2.4. Sprint 1 Document Updates	**A**			**R**				**I**
4.3.3. Sprint 1 Review	**A**			**R**				**I**
4.3.3.1. Completed Tasks Sprint 1 Review	**A/R**	**I**	**I**	**I**	**I**	**I**	**I**	**I**

4.3.3.2. Collection of Feedback from Stakeholders on Sprint 1	**A/R**	**I**	**I**	**I**	**I**	**I**	**I**	**I**
4.3.3.3. Discussion of the Lessons Learned from Sprint 1	**A/R**	**I**	**I**	**I**	**I**	**I**	**I**	**I**
4.3.4. Sprint 1 Retrospect								**I**
4.3.4.1. Sprint 1 Product Backlog Refinement	**A/R**	**I**	**I**	**I**	**I**	**I**	**I**	**I**
4.3.4.2. Sprint 1 Team Performance Evaluation	**A/R**	**I**	**I**	**I**	**I**	**I**	**I**	**I**
4.4. Sprint 2								**I**
4.4.1. Sprint 2 Planning								**I**

4.4.1.1. Sprint 2 Goal and Deliverables Definition	**A/R**	**I**	**I**	**I**	**I**	**I**	**I**	**I**
4.4.1.2. Sprint 2 Sprint Backlog Creation	**A/R**	**I**	**I**	**I**	**I**	**I**	**I**	**I**
4.4.2. Sprint 2 Implementation								**I**
4.4.2.1. Sprint 2 Scrum Daily Meetings	**A/R**	**I**	**I**	**I**	**I**	**I**	**I**	**I**
4.4.2.2. Sprint 2 Front End	**A**	**C**	**C**	**C**	**R**			**I**
4.4.2.3. Sprint 2 Back End	**A**	**R**	**C**	**C**	**C**			**I**
4.4.3. Sprint 2 Review								**I**
4.4.3.1. Completed Tasks Sprint 2 Review	**R**							**I**

4.4.3.2. Collection of Feedback from Stakeholders on Sprint 2	**R**							**I**
4.4.3.3. Discussion of the Lessons Learned from Sprint 2	**R**							**I**
4.4.4. Sprint 2 Retrospect								**I**
4.4.4.1. Sprint 2 Product Backlog Refinement	**R**							**I**
4.4.4.2. Sprint 2 Team Performance Evaluation	**R**							**I**
4.5. Sprint 3								**I**
4.5.1. Sprint 3 Planning								**I**

4.5.1.1. Sprint 3 Goal and Deliverables Definition	**A/R**							**I**
4.5.1.2. Sprint 3 Sprint Backlog Creation	**A.R**							**I**
4.5.2. Sprint 3 Implementation								**I**
4.5.2.1. Sprint 3 Scrum Daily Meetings	**A/R**	**I**	**I**	**I**	**I**	**I**	**I**	**I**
4.5.2.2. Sprint 3 Front End	**A**	**C**	**C**	**C**	**R**			**I**
4.5.2.3. Sprint 2 Back End	**A**	**R**	**C**	**C**	**C**			**I**
4.5.3. Sprint 3 Review								**I**
4.5.3.1. Completed Tasks Sprint 3 Review	**A/R**							**I**

4.5.3.2. Collection of Feedback from Stakeholders on Sprint 3	**A/R**							**I**
4.5.3.3. Discussion of the Lessons Learned from Sprint 3	**A/R**							**I**
4.5.4. Sprint 3 Retrospect								**I**
4.5.4.1. Sprint 3 Product Backlog Refinement	**A/R**							**I**
4.5.4.2. Sprint 3 Team Performance Evaluation	**A/R**							**I**
4.6. Sprint 4								**I**
4.6.1. Sprint 4 Planning								**I**

4.6.1.1. Sprint 4 Goal and Deliverables Definition	**A/R**							**I**
4.6.1.2. Sprint 4 Sprint Backlog Creation	**A/R**							**I**
4.6.2. Sprint 4 Implementation								**I**
4.6.2.1. Sprint 4 Scrum Daily Meetings	**A/R**	**I**	**I**	**I**	**I**	**I**	**I**	**I**
4.6.2.2. Sprint 4 Front End	**A**	**C**	**C**	**C**	**R**			**I**
4.6.2.3. Sprint 4 Back End	**A**	**R**	**C**	**C**	**C**			**I**
4.6.3. Sprint 4 Review								**I**
4.6.3.1. Completed Tasks Sprint 4 Review	**A/R**							**I**

4.6.3.2. Collection of Feedback from Stakeholders on Sprint 4	**A/R**							**I**
4.6.3.3. Discussion of the Lessons Learned from Sprint 4	**A/R**							**I**
4.6.4. Sprint 4 Retrospect								**I**
4.6.4.1. Sprint 4 Product Backlog Refinement	**A/R**							**I**
4.6.4.2. Sprint 4 Team Performance Evaluation	**A/R**							**I**
4.7. Sprint 5								**I**
4.7.1. Sprint 5 Planning								**I**

4.7.1.1. Sprint 5 Goal and Deliverables Definition	**A/R**							**I**
4.7.1.2. Sprint 5 Sprint Backlog Creation	**A/R**							**I**
4.7.2. Sprint 5 Implementation								**I**
4.7.2.1. Unit Test	**A**	**C**	**C**	**R**	**C**			**I**
4.7.2.2. Integration Test	**A**	**C**	**C**	**R**	**C**			**I**
4.7.2.3. System Test	**A**	**C**	**C**	**R**	**C**			**I**
4.7.2.4. Training Materials Creation	**A**	**C**	**C**	**R**	**C**			**I**
4.7.3. Sprint 5 Review								**I**

4.7.3.1. Completed Tasks Sprint 5 Review	**A/R**							**I**
4.7.3.2. Collection of Feedback from Stakeholders on Sprint 5	**A/R**							**I**
4.7.3.3. Discussion of the Lessons Learned from Sprint 5	**A/R**							**I**
4.7.4. Sprint 5 Retrospect								**I**
4.7.4.1. Sprint 5 Product Backlog Refinement	**A/R**							**I**
4.7.4.2. Sprint 5 Team Performance Evaluation	**A/R**							**I**
5. Final Phase								

5.1. Final Meeting	**A**	**I**	**I**	**I**	**R**	**I**	**I**	**I**
5.2. Review Final Project	**A**	**I**	**I**	**I**	**R**	**I**	**I**	**I**
5.3. Delivery Project for Customers	**A**	**I**	**I**	**I**	**R**	**I**	**I**	**I**

After stakeholders determination, we set up these roles for each member in our groups, and the table below will show about our members' responsibilities as well as their contact information:

Table 1.4.1.2: Task Responsibilities Table

SNo	Name	Roles	Contacts
1	Nguyễn Thị Thanh Châu	CEO, Tester, Business Analyst	Email: 22520150@gm.uit.edu.vn Phone: 0945814112
2	Nguyễn Huy Hoàng	Back-end Developer, Tester	Email: 22520468@gm.uit.edu.vn Phone:
3	Huỳnh Quang Thịnh	Front-end Developers, Tester	Email: 22521407@gm.uit.edu.vn Phone: 0847271104
4	Trần Tịnh Đan Thanh	Product Manager, Data Analyst, Financilist	Email: 22521368@gm.uit.edu.vn Phone:

1.5. Project Layouts

In order to ensure that the ERP project is executed and monitored effectively, it is vital to build a consistent report's composition. The main purpose of this action is to develop a clear, organized, and consistent report which helps stakeholders easily capture the content. This report is divided into five main chapters, each of which presents the major phase of the project from introduction to evaluation, including: General information, Theoretical overview, Business process system, Proposed algorithms and solutions, Conclusion and future directions.

Chapter 1: General information

The purpose of this chapter is to clarify the scope, objectives, and stakeholders in the ERP's project development. The RACI methodological taken in this project is used to define roles and responsibilities which enables structured task assignment and accountability. Additionally, a combination of structured WBS and time estimation techniques was applied to project analysis and implementation. Thus, all methods provide the foundation for managing the project systematically.

Chapter 2: Theoretical overview

This chapter mentioned rapid growth of ERP systems, from MRP/MRP II to ERP's cloud-based and integrated with AI and automation. Besides, this part also refers to technologies and tools utilised to support system development and deployment, for instance: OpenAI, Node.js, Oracle DB, GitHub, and VSCode. Moreover, this section sets out the context in which this research was conducted. These are challenges that the mobile phone retail sector has to face, particularly in managing the paradox of choice. The section focused on four case studies to determine that AI can transform sales and customer service processes.

Chapter 3: Business process system

This chapter focused on the development of the BPMN processes for modeling workflows in customer service, sales, payment, stock insurance, shipping, and warranty. In addition, the workflow comprises key functions of suppliers such as procurement, inventory checks, goods receipt, and payment are detailed. To automate manual steps within sales and service processes, this section suggested a suitable model - Kammy chatbot. The specifications mentioned in this chapter were used to design an ERP system aligned with the refined process models.

Chapter 4: Proposed algorithms and solutions

This chapter presents algorithms solution including the proposed system integrates Kammy chatbot which is designed to improve the sale process in ERP system. Furthermore, the system architecture included an AI model powered by OpenAI, a backend server, and a structured database schema which is clearly explained in this section. Data for this study were collected from Kaggle such as mobile phone dataset, and customer's dataset. This project implements an end-end system for Order-to-Cash (O2C) process, and personalized recommendations.

Chapter 5: Conclusion and future directions

This final chapter summarized the achievements of the project, assessing how well the set objectives have been met. Moreover, this section proposed the future direction which would enhance and expand the project in the near future.

1.6. Project Plan - Work Assignment Table:

In this section, our group provides two perspectives of the work assignment table, including WBS tree and WBS Table. In the project, we utilize three methods, such as Three-Point Estimating, Expert Judgment and Analogous Estimating, so as to set up duration, and determinate time.

1.6.1. Task Duration Estimation Approach:

a. Description of Task Estimation Methodology:

To categorize methods that use to calculate duration for each task, we define task characteristics as the following table:

Table 1.6.1.1: Methods for Calculating Duration

Type of Task	Estimation Method	Applicable When	Reason	Formula / Implementation
Completely new task	Three-Point Estimating, Expert Judgment	The team has never done it before, e.g., designing a new UI, developing backend for a new module	Requires risk assessment and uncertainty handling using three scenarios: optimistic, most likely, pessimistic	(tO + tM + tP) / 3 Note: Our team determines these estimates as follows: the pessimistic time (tP) is based on the working time of the slowest-performing team member, the optimistic time (tO) is based on the fastest-performing team member, and the most likely time (tM) can be derived from expert judgment.
Similar task previously done, but with significant changes in	Expert Judgment	E.g., re-implementing a known feature using a new framework, changing the method, or in a different domain	Previous experience is not fully applicable → expert input needed to estimate effort in the new	Based on expert opinions and evaluations

tools, technology, approach, or domain			context	
Simple, familiar task, can reuse existing templates, source code, or previous structures	Analogous Estimating	E.g., minor edits, reusing a known function, updating from existing source code	Easy to estimate based on past similar tasks, fast and accurate within a similar context	Estimate based on the duration of previously similar tasks

b. Applying Method for Each Task

This is the task that we define to each method for calculating duration

Table 1.6.2: Applying Method to Calculate Duration Of each Phase

Type of Task	Method
Phase 1	Analogous Estimating
Phase 2 and Phase 3	Expert Judgment
Phase 4: Sprint Planning, Sprint Review, Sprint Retrospect	Expert Judgment
Phase 4: Sprint Implementation	Three-Point Estimating, Expert Judgment
Phase 4: Human Resource Preparation, Product Backlog	Analogous Estimating
Phase 5	Analogous Estimating

1.6.2. Work Assignment Tree:

The first perspective provides the reader the whole view of all planning tasks, and in this tree, we eliminate some detail description and just represent the important parts.

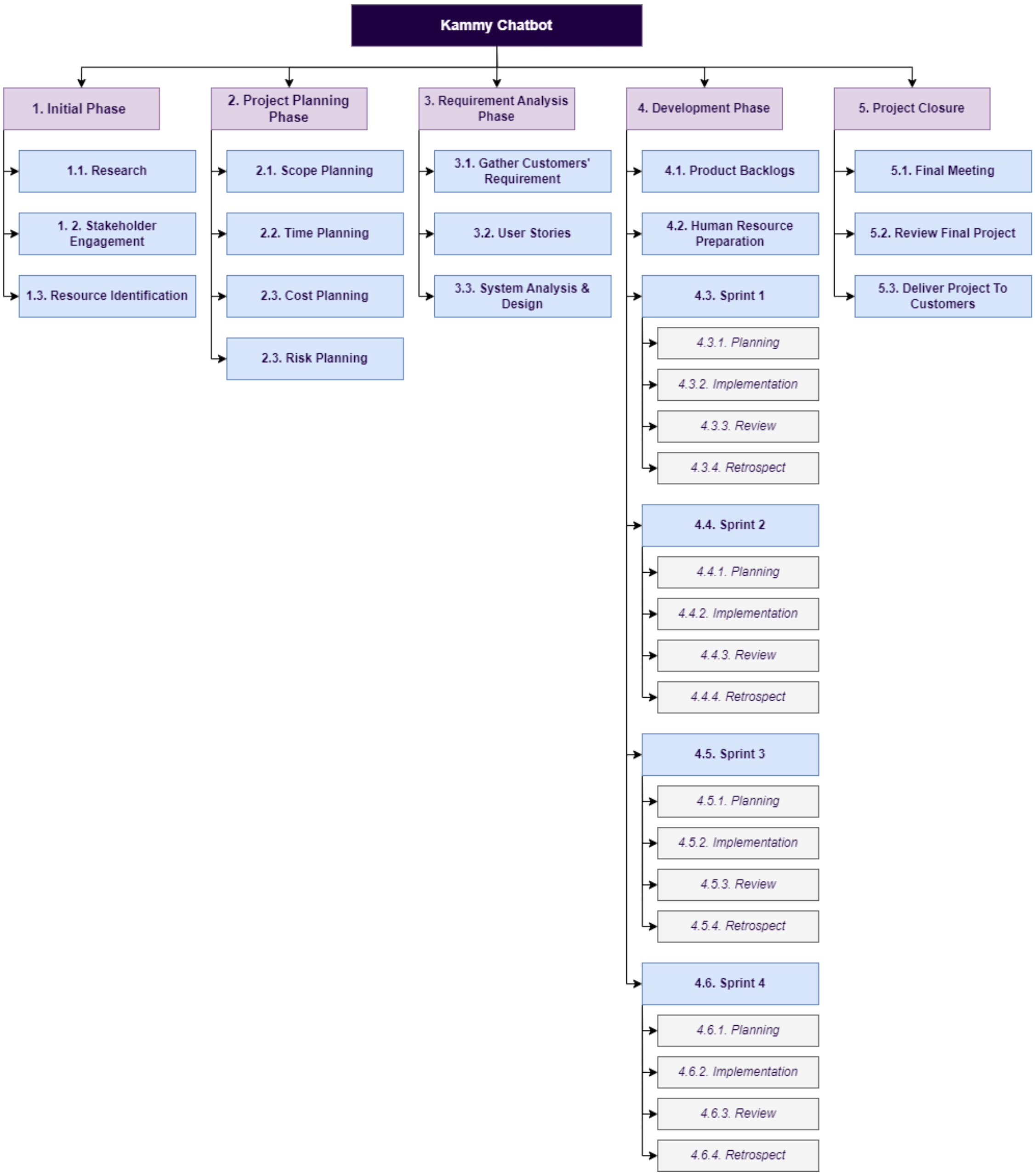

Figure 1.6.2: Work Assignment Tree

1.6.3. Work Assignment Table:

The second perspective illustrates a detailed view of task planning in this project, including start date, end date and duration.

Table 1.6.3: Work Assignment

Activity	Start Date	Finish Date	Duration (Unit: Day)
1. Initial Phase	2025-04-12	2025-04-20	8
1.1. Research	2025-04-12	2025-04-14	2
1.1.1. Market Research	2025-04-12	2025-04-13	1
1.1.2. Customer Research	2025-04-12	2025-04-14	2
1.2. Stakeholder Engagement	2025-04-14	2025-04-21	7
1.2.1. Stakeholder Identification	2025-04-14	2025-04-15	1
1.2.2. Stakeholder Communication Setup	2025-04-15	2025-04-16	1
1.2.3. Stakeholder Goal Identification	2025-04-16	2025-04-17	1
1.2.4. Stakeholder Scope Identification	2025-04-16	2025-04-19	3
1.3. Resource Identification	2025-04-19	2025-04-20	1
1.3.1. Total Budget Identification	2025-04-19	2025-04-20	1
1.3.2. Human Resource Identification	2025-04-19	2025-04-20	1
1.3.3. Development Timeline Identification	2025-04-19	2025-04-20	1
2. Project Planning Phase	2025-04-12	2025-04-23	11
2.1. Scope Planning	2025-04-12	2025-04-15	3

2.1.1. Project Charter	2025-04-12	2025-04-14	2
2.1.2. Scope Statement	2025-04-12	2025-04-13	1
2.1.3. Work Down Structure	2025-04-12	2025-04-15	3
2.2. Time Planning	2025-04-15	2025-04-21	6
2.2.1. Milestone Table	2025-04-15	2025-04-17	2
2.2.2. Milestone Schedule	2025-04-17	2025-04-19	2
2.2.3. AON Diagram	2025-04-19	2025-04-21	2
2.3. Cost Planning	2025-04-19	2025-04-22	3
2.3.1. Cost Table	2025-04-19	2025-04-20	1
2.3.2. Cost Baseline	2025-04-19	2025-04-22	3
2.4. Risk Planning	2025-04-15	2025-04-23	8
2.4.1. Risk Breakdown Structure	2025-04-15	2025-04-17	2
2.4.2. Risk Analysis	2025-04-17	2025-04-20	3
2.4.3. Risk Reduction Plan	2025-04-20	2025-04-23	3
3. Requirement Analysis Phase	2025-04-23	2025-05-03	**10**
3.1. Gather Customers' Requirement	2025-04-23	2025-04-25	2
3.2. User Stories	2025-04-23	2025-04-25	2
3.2.1. User Stories List	2025-04-23	2025-04-24	1
3.2.2. Stakeholder - Involved User Story Finalization	2025-04-24	2025-04-26	2

3.3. System Analysis & Design	2025-04-26	2025-05-03	7
3.3.1. UML Diagrams	2025-04-26	2025-05-03	7
3.3.2. System Modeling Diagrams	2025-04-26	2025-05-02	6
4. Development Phase	2025-05-03	2025-06-29	57
4.1. Product Backlogs	2025-05-03	2025-05-09	6
4.1.1. User Story Decomposition into Tasks	2025-05-03	2025-05-06	3
4.1.2. Selection of Tasks for Backlog Creation	2025-05-06	2025-05-09	3
4.2. Human Resource Preparation	2025-05-09	2025-05-11	2
4.2.1. Role Assignment	2025-05-09	2025-05-11	2
4.2.2. Task Assignment	2025-05-09	2025-05-10	1
4.3. Sprint 1	2025-05-11	2025-05-25	14
4.3.1. Sprint 1 Planning	**2025-05-11**	**2025-05-12**	**1**
4.3.1.1. Sprint 1 Goal and Deliverables Definition	2025-05-11	2025-05-12	1
4.3.1.2. Sprint 1 Sprint Backlog Creation	2025-05-11	2025-05-12	1
4.3.2. Sprint 1 Implementation	2025-05-12	2025-05-22	10
4.3.2.1. Sprint 1 Scrum Daily Meetings	2025-05-12	2025-05-13	1
4.3.2.2. Prototype Creation	2025-05-12	2025-05-22	10
4.3.2.3. Database Initialization and Setup	2025-05-12	2025-05-16	4

4.3.2.4. Sprint 1 Document Updates	2025-05-12	2025-05-14	2
4.3.3. Sprint 1 Review	2025-05-22	2025-05-24	2
4.3.3.1. Completed Tasks Sprint 1 Review	2025-05-22	2025-05-23	1
4.3.3.2. Collection of Feedback from Stakeholders on Sprint 1	2025-05-22	2025-05-24	2
4.3.3.3. Discussion of the Lessons Learned from Sprint 1	2025-05-22	2025-05-23	1
4.3.4. Sprint 1 Retrospect	2025-05-24	2025-05-25	1
4.3.4.1. Sprint 1 Product Backlog Refinement	2025-05-24	2025-05-25	1
4.3.4.2. Sprint 1 Team Performance Evaluation	2025-05-24	2025-05-25	1
4.4. Sprint 2	2025-05-25	2025-06-02	8
4.4.1. Sprint 2 Planning	**2025-05-25**	**2025-05-26**	**1**
4.4.1.1. Sprint 2 Goal and Deliverables Definition	2025-05-25	2025-05-26	1
4.4.1.2. Sprint 2 Sprint Backlog Creation	2025-05-25	2025-05-26	1
4.4.2. Sprint 2 Implementation	**2025-05-26**	**2025-05-30**	**4**
4.4.2.1. Sprint 2 Scrum Daily Meetings	2025-05-26	2025-05-27	1
4.4.2.2. Sprint 2 Front End	2025-05-26	2025-05-28	2

4.4.2.2.1. Home Page Integration with Chatbot Widget	2025-05-26	2025-05-28	2
4.4.2.2.2. Chatbot Overlay UI	2025-05-26	2025-05-28	2
4.4.2.2.3. Chatbot Floating Button	2025-05-26	2025-05-28	2
4.4.2.2.4. Chat History View (Web Embedded)	2025-05-26	2025-05-28	2
4.4.2.2.5. Notification UI (Voucher, Promo)	2025-05-26	2025-05-28	2
4.4.2.2.6. Admin Panel UI (Product & Promotion Management)	2025-05-26	2025-05-28	2
4.4.2.2.7. Chatbot Quick Questions Template	2025-05-26	2025-05-28	2
4.4.2.2.8. Chatbot Personalized Recommendation Card UI	2025-05-26	2025-05-28	2
4.4.2.2.9. Product List View	2025-05-26	2025-05-28	2
4.4.2.3. Sprint 2 Back End	2025-05-26	2025-05-30	4
4.4.2.3.1. Auth API	2025-05-26	2025-05-30	4
4.4.2.3.2. User Role Management	2025-05-26	2025-05-29	3
4.4.2.3.3. Chat Session Logger	2025-05-26	2025-05-29	3
4.4.2.3.4. Product Management API	2025-05-26	2025-05-29	3
4.4.2.3.5. Promotion Engine	2025-05-26	2025-05-30	4
4.4.2.3.6. Notification Handle	2025-05-26	2025-05-29	3

4.4.2.3.7. DB Schema: Users, Products, Promotions, Conversations	2025-05-26	2025-05-30	4
4.4.3. Sprint 2 Review	**2025-05-30**	**2025-06-01**	**2**
4.4.3.1. Completed Tasks Sprint 2 Review	2025-05-30	2025-05-31	1
4.4.3.2. Collection of Feedback from Stakeholders on Sprint 2	2025-05-30	2025-06-01	2
4.4.3.3. Discussion of the Lessons Learned from Sprint 2	2025-05-30	2025-05-31	1
4.4.4. Sprint 2 Retrospect	**2025-06-01**	**2025-06-02**	**1**
4.4.4.1. Sprint 2 Product Backlog Refinement	2025-06-01	2025-06-02	1
4.4.4.2. Sprint 2 Team Performance Evaluation	2025-06-01	2025-06-02	1
4.5. Sprint 3	2025-06-02	2025-06-10	8
4.5.1. Sprint 3 Planning	**2025-06-02**	**2025-06-03**	**1**
4.5.1.1. Sprint 3 Goal and Deliverables Definition	2025-06-02	2025-06-03	1
4.5.1.2. Sprint 3 Sprint Backlog Creation	2025-06-02	2025-06-03	1
4.5.2. Sprint 3 Implementation	**2025-06-03**	**2025-06-07**	**4**
4.5.2.1. Sprint 3 Scrum Daily Meetings	2025-06-03	2025-06-04	1
4.5.2.2. Sprint 3 Front End	2025-06-03	2025-06-07	4

4.5.2.2.1. Conversation Launcher Button	2025-06-03	2025-06-06	3
4.5.2.2.2. Dynamic Response Block UI	2025-06-03	2025-06-07	4
4.5.2.2.3. User Info Collection Form	2025-06-03	2025-06-04	1
4.5.2.2.4. Loading & Typing Animation	2025-06-03	2025-06-06	3
4.5.2.2.5. Mobile Responsiveness	2025-06-03	2025-06-07	4
4.5.2.3. Sprint 2 Back End	2025-06-03	2025-06-07	4
4.5.2.3.1. Chat Flow Builder (Sale)	2025-06-03	2025-06-07	4
4.5.2.3.2. Intent Recognition & Routing	2025-06-03	2025-06-07	4
4.5.2.3.3. Lead Capture Logic	2025-06-03	2025-06-05	2
4.5.2.3.4. Product Recommendation AP	2025-06-03	2025-06-07	4
4.5.2.3.5. CRM Integration Stub	2025-06-03	2025-06-07	4
4.5.3. Sprint 3 Review	**2025-06-07**	**2025-06-09**	**2**
4.5.3.1. Completed Tasks Sprint 3 Review	2025-06-07	2025-06-08	1
4.5.3.2. Collection of Feedback from Stakeholders on Sprint 3	2025-06-07	2025-06-09	2
4.5.3.3. Discussion of the Lessons Learned from Sprint 3	2025-06-07	2025-06-08	1
4.5.4. Sprint 3 Retrospect	**2025-06-09**	**2025-06-10**	**1**
4.5.4.1. Sprint 3 Product Backlog Refinement	2025-06-09	2025-06-10	1

4.5.4.2. Sprint 3 Team Performance Evaluation	2025-06-09	2025-06-10	1
4.6. Sprint 4	2025-06-10	2025-06-18	8
4.6.1. Sprint 4 Planning	**2025-06-10**	**2025-06-11**	**1**
4.6.1.1. Sprint 4 Goal and Deliverables Definition	2025-06-10	2025-06-11	1
4.6.1.2. Sprint 4 Sprint Backlog Creation	2025-06-10	2025-06-11	1
4.6.2. Sprint 4 Implementation	**2025-06-11**	**2025-06-15**	**4**
4.6.2.1. Sprint 4 Scrum Daily Meetings	2025-06-11	2025-06-12	1
4.6.2.2. Sprint 4 Front End	2025-06-11	2025-06-13	2
4.6.2.2.1. Promotion Notification UI	2025-06-11	2025-06-13	2
4.6.2.2.2. Smart Product Carousel UI	2025-06-11	2025-06-13	2
4.6.2.2.3. User Feedback Form in Chat	2025-06-11	2025-06-13	2
4.6.2.2.4. Campaign Popup Triggered by Kammy	2025-06-11	2025-06-13	2
4.6.2.2.5. UI Tùy biến theo Persona	2025-06-11	2025-06-13	2
4.6.2.3. Sprint 4 Back End	2025-06-11	2025-06-15	4
4.6.2.3.1. Product Recommendation Engine	2025-06-11	2025-06-13	2
4.6.2.3.2. User Interaction Data Storage	2025-06-11	2025-06-14	3

4.6.2.3.3. Third-party Integration for Marketing	2025-06-11	2025-06-15	4
4.6.2.3.4. DB: Users, Promotions, Product Recommendations	2025-06-11	2025-06-15	4
4.6.3. Sprint 4 Review	**2025-06-15**	**2025-06-17**	**2**
4.6.3.1. Completed Tasks Sprint 4 Review	2025-06-15	2025-06-16	1
4.6.3.2. Collection of Feedback from Stakeholders on Sprint 4	2025-06-15	2025-06-17	2
4.6.3.3. Discussion of the Lessons Learned from Sprint 4	2025-06-15	2025-06-16	1
4.6.4. Sprint 4 Retrospect	**2025-06-17**	**2025-06-18**	**1**
4.6.4.1. Sprint 4 Product Backlog Refinement	2025-06-17	2025-06-18	1
4.6.4.2. Sprint 4 Team Performance Evaluation	2025-06-17	2025-06-18	1
4.7. Sprint 5	2025-06-18	2025-06-29	11
4.7.1. Sprint 5 Planning	**2025-06-18**	**2025-06-19**	**1**
4.7.1.1. Sprint 5 Goal and Deliverables Definition	2025-06-18	2025-06-19	1
4.7.1.2. Sprint 5 Sprint Backlog Creation	2025-06-18	2025-06-19	1
4.7.2. Sprint 5 Implementation	**2025-06-19**	**2025-06-26**	**7**
4.7.2.1. Unit Test	2025-06-19	2025-06-24	5

4.7.2.2. Integration Test	2025-06-19	2025-06-24	5
4.7.2.3. System Test	2025-06-19	2025-06-26	7
4.7.2.4. Training Materials Creation	2025-06-19	2025-06-24	5
4.7.3. Sprint 5 Review	**2025-06-26**	**2025-06-28**	**2**
4.7.3.1. Completed Tasks Sprint 5 Review	2025-06-26	2025-06-27	1
4.7.3.2. Collection of Feedback from Stakeholders on Sprint 5	2025-06-26	2025-06-28	2
4.7.3.3. Discussion of the Lessons Learned from Sprint 5	2025-06-26	2025-06-27	1
4.7.4. Sprint 5 Retrospect	**2025-06-28**	**2025-06-29**	**1**
4.7.4.1. Sprint 5 Product Backlog Refinement	2025-06-28	2025-06-29	1
4.7.4.2. Sprint 5 Team Performance Evaluation	2025-06-28	2025-06-29	1
5. Final Phase	2025-06-29	2025-07-03	4
5.1. Final Meeting	2025-06-29	2025-06-30	1
5.2. Review Final Project	2025-06-29	2025-07-02	3
5.3. Close Project	2025-07-02	2025-07-03	1

Once the necessary resources are fully prepared, our group moves to the next step to build our solution AI Chatbot "Kammy" and the ways we develop it will be mentioned as a detail in the next chapter.

CHAPTER 2: THEORETICAL OVERVIEW

Chapter 2 covers the fundamental elements of ERP systems throughout the implementation process, beginning with an introduction to its theoretical groundwork. It then explores the technologies applied in the development of the system and the historical progression of these technologies. Finally, the hands-on applications and limitations of ERP integrations are illustrated through real-world problems.

2.1. ERP introduction

2.1.1. Overview of ERP:

Enterprise Resource Planning (ERP) is a unified software ecosystem utilized by organizations to streamline and automate core processes in business environments. ERP's goal is to consolidate data, enhance operational visibility, and enable effortless data exchange across all departments within an organization. ERP systems are software platforms designed to automate core business processes, including finance, human resources, manufacturing, supply chain, services, procurement, and others, such as:

- Manufacturing: Some of the functions include engineering, capacity, workflow management, quality control, bills of material, manufacturing process, etc. [1]
- Financials: Accounts payable, accounts receivable, fixed assets, general ledger and cash management, etc. [1]
- Human Resources: Benefits, training, payroll, time and attendance, etc. [1]
- Supply Chain Management: Inventory, supply chain planning, supplier scheduling, order entry, purchasing, etc. [1]
- Projects: Costing, billing, activity management, time, expense, etc. [1]
- Customer Relationship Management: sales and marketing, service, commissions, customer contact, call center support, etc. [1]
- Data Warehouse: A module can be accessed by an organization's customers, suppliers, employees, etc. [1]

- Other similar modules [1]

Figure 2.1.1: Features and Functions of ERP software

(Source: OCD, 2023 – https://ocd.vn/chuc-nang-cua-phan-mem-erp/)

Nowadays, due to the needs of improving efficiency, reducing costs, and adapting to dynamic market conditions, many cloud-based ERP solutions including real-time analytics, automation, and AI-driven insights were offered.

2.1.2. The drive of ERP in Sale Process and Customer Service Process:

The Customer Data Platform (CDP) supports critical functions for many companies' sales and customer service operations through an Enterprise Resource Planning (ERP) system. Order management and customer information synchronization as well as how the sale is accounted for and post-sale support are details revealed in this ERP. The capability to store transaction histories and purchasing behaviors is supported with a view of building long-term loyalty.

In practice, the true value of an ERP system has never been unleashed. Much of customer care, internal coordination, and omnichannel management remain broken and manual tasks. If ERP systems are not integrated, operational inefficiencies and inconsistent service experiences are likely to occur, as the lack of integration can lead to data redundancy, miscommunication, and delays in decision-making, which affect overall business performance (Wieder, Booth, & Matolcsy, 2006). Where a CDP exists it is acting as a silo which is not deeply embedded within a holistic ERP landscape which severely limits the ability to automate workflows and deliver consistent customer engagement. Therefore, ERP is the base building block, but further integration and automation are required to realize its maximum value in sales and customer service.

2.1.3. The effect of Chatbot on ERP process (CRM module: Sale Process and Customer Service Process)

The development of AI-based chatbots and their implantation in ERP systems have improved the modules of sales as well as the customer service module, leading to increased efficiency and satisfaction. Chatbots being smart interfaces proactively engage users in real-time, making interactions easy and decisions well informed and on time, hence quicker. Mostly in sales domains, this integration impacts a lot by reducing workloads on the side of employees; it also reduces the rate at which potential clients drop off and offers individualized product recommendations to customers. In that sense, chatbots guide users through O2C processes whereby they remove the existing frictions during online shopping; hence it reduces abandonment by using carts. It does not just improve customer satisfaction but also leads to better conversion rates within the sales funnel (Melnyk et al., 2024).

Chatbots not only drive sales but also improve the efficiency of customer service. Inquiries that can be easily automated and answered immediately reduce waiting times on the part of customers, so that human agents can concentrate more on critical issues. This makes for a much better and smoother experience for customers as well as improving the quality of service. In addition, chatbots run conversations 24/7 with attendant support at any time hence raising customer engagement. Integrating chatbots in ERP systems for customer service will also reduce human resource pressure and ensure that customers get quick, accurate, and personalized responses (Brightwood et al., 2023).

The role of chatbots in customer service goes beyond handling routine issues; they can also engage proactively, and thus help businesses align with the dynamic requirements of customers. For instance, since chatbots know the inventory situation and the status of orders, they can respond to inquiries about product availability and order status as well as troubleshoot problems, all of which leads to quicker resolution and better satisfaction from customers. Such integrations enable companies to close the gap between what customers expect and what is delivered by way of services, making customer service not only more efficient but also more effective (Nelson et al., 2022).

The implementation of chatbots into both sales and customer service processes in ERP systems is a necessity for today's enterprises. It boosts efficiency by automating mundane activities and makes interactions with customers more speedy and personal. Such automation will help keep the organization ahead in this rapidly moving digital world where sales and customer service processes must drive better customer experiences, brand loyalty, and bottom-line results (Badi, 2023). Hence, these two processes in ERP systems are of utmost importance—they form the base of achieving long-term operational excellence as well as customer satisfaction.

2.2. Technologies used in system building

To build Kammy Chatbot, our team make a decision to utilize useful and effective tools as well as technologies, such as:

- **Git / GitHub:** For version control and collaborative development, ensuring that all code changes are tracked and managed efficiently.
- **Visual Studio Code:** As the primary code editor due to its versatility and support for multiple programming languages and tools.
- **HTML, CSS:** Employed as the front-end framework to build a dynamic and responsive user interface.
- **Oracle Database:** Used as the backend database system to handle large volumes of data with high performance and reliability.

- **Node.js / Express:** For the server-side logic, providing a lightweight and efficient platform for handling web requests.
- **Open AI:** For the server-side logic, providing a lightweight and efficient platform for handling web requests.
- **Flask Framework:** With backend we utilize flask for making functions behind the frontend

The following diagram is our solution architecture, which supports for improving sale process and customer services process in ERP systems:

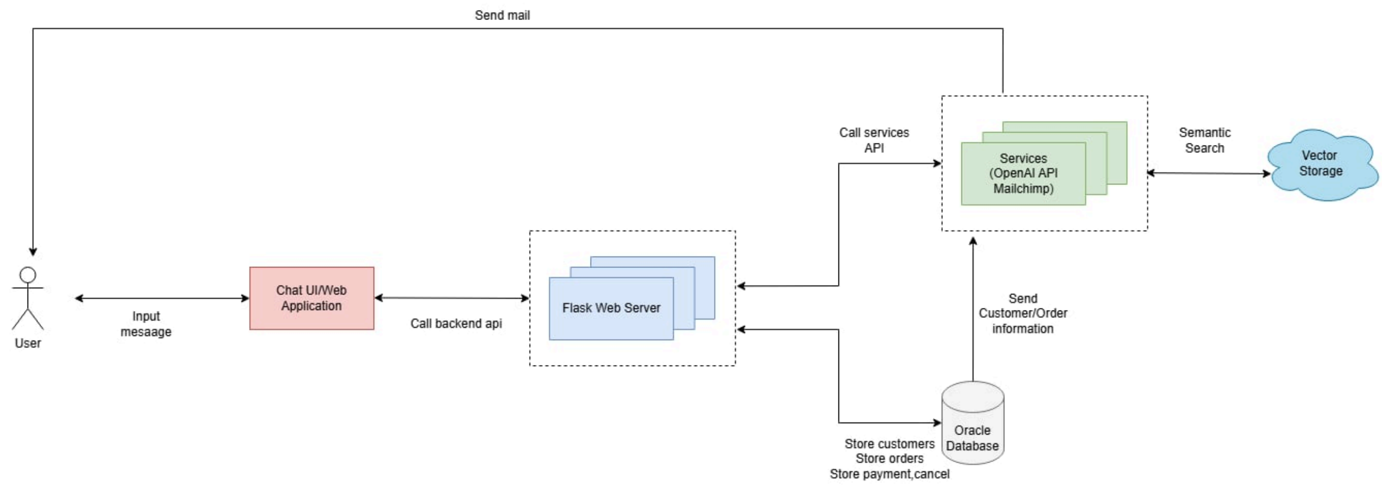

Figure 2.2.1: The architect of "Kammy Chatbot"

2.3 Description of the problem applying ERP to the project topic

2.3.1. Context:

a. E-Commerce and AI chatbot:

Although E-Commerce is changing all aspects in our lives, as well as, make a significant contribution to promoting sale of customers for company through online payment, online order, cancel order, and so on, it witness some drawbacks and challenging, particularly the requirement that the system need to deliver an effective and personalized customer experience, tailored to individual needs and preferences throughout the customer journey. With the aim to address these challenges, the growing number of enterprises utilize

Artificial Intelligence to optimize marketing efforts and enhance the overall customer experience [14]. The department which can gain most value from AI within a business is the marketing department. Marketing includes three key functions: understanding customers' insights, matching those needs with products and services, fostering customers to make orders. And AI plays a potential role to enhance the effectiveness of these functions [15].

With the advent of technology, AI can enable companies, and customers to make decisions quickly and effectively since it has the ability to process data faster and reduce human mistakes [15]. Technology is capable of learning, representing, storing, and continuously refining their knowledge based on past experiences and current information, allowing them to real-time response and accuracy answers [16]. Some methods, such as text sentiments and AI/Machine Learning are used to categorise customers' feeling and need, supporting business companies to understand their customers clearly.

b. Mobile industry:

- In the world:

The smartphone market has been experiencing significant growth, driven by continuous technological advancements and rising consumer demand. In fact, according to Statista, global smartphone shipments reached over 1.2 billion units in 2023, reflecting the market's robust expansion.

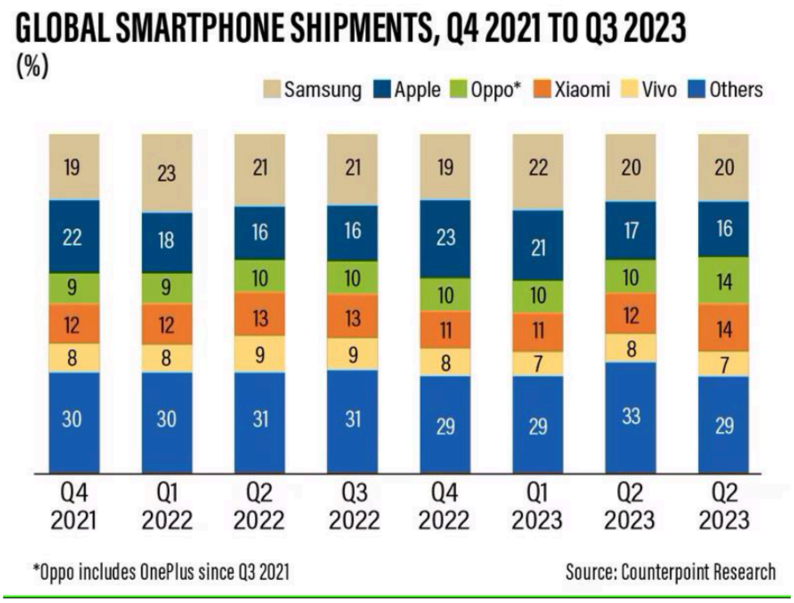

Figure 2.3.1.1: Global Smartphone Shipments

(Source: Counterpoint Research. Global Smartphone Shipments, Q4 2021 to Q3)

New smartphone models are now launched not only annually but almost every month, as brands race to capture market share with improved cameras, faster processors, and 5G capabilities. At the same time, online sales channels are expanding rapidly, reshaping consumer purchasing behavior. Research by eMarketer highlights that e-commerce sales of electronics, including smartphones, grew by 17% in 2023 alone. To capitalize on this trend, smartphone companies are increasingly distributing their products through leading e-commerce platforms. With this flourishing market, the smartphone industry has been developed across the globe.

- In Vietnam:

In Vietnam, the smartphone market has witnessed significant and active growth in recent years. “People spend more money on smartphones”- Sharing with Lao Dong, a representative of the CellPhoneS system said that by the end of November 2024, the phone market had grown in quantity by 20% [7]. Moreover, according to market research firm Statista, in 2025 revenue in the smartphone market in Vietnam will reach 4.1 billion USD [7]. In addition, Vietnam is among the countries with the highest smartphone penetration rates in the world. According to a report by Statista (2024), over 70% of the Vietnamese population uses smartphones, placing Vietnam in the top ranks globally for smartphone adoption [7]. Besides, competition in this market is steadily growing with the presence of familiar brands, including Samsung, Apple, Xiaomi, OPPO and Vivo, which have continuously released their new smartphone version and promoted their marketing strategy with the aim to attract their customers. These factors illustrate the great interest of Vietnamese consumers on modern technology as well as, Vietnamese smartphone industry is not only growing in scale but also open up many opportunities for businesses in the industry. With the remarle growth of this section in recent years, various online and offline distribution, specialising in mobile electronics have emerged in Vietnam, inducing FPT Shop, The Gioi Dong, Dien May Xanh, Hoang Ha Mobile, etc. In the context of Vietnam's rapidly growing smartphone market, it's not only major corporations but also a place where many innovative and creative retail models have emerged, with Cellphone S standing out as a notable example. Unlike major systems such as Thế Giới Di Động or FPT Retail, which adopted integrated ERP and CRM systems from an early stage, Schannel, which entered the market at a later stage, adopts a hybrid model that combines content creation with commerce, driving purchase conversions through e-commerce affiliate links. However, owing to its relatively young stage in business operations - particularly in customer management, logistics, and internal process optimization - CellphoneS still heavily relies on fragmented and manual tools. Therefore, solutions that improve the ERP process in Cellphone S can allow them to be more competitive with their competitors.

These are the reasons why our team has selected the Smartphone industry and CellPhone S in the Vietnam Market as a primary platform to analyze and leverage for smartphone distribution strategies.

2.3.2. Case study

In Cellphone S operation Process, company use Customer Relationship Management platform (CRM) named Customer Data Platform with various functions:

- Customer Information Management: store and synchronous data, such as personal information, transaction history, purchasing behavior.
- Sale process Management: keep track of access and potential customer management
- Customer Care: Monitor transaction histories and post-sale interactions helps enhance customer satisfaction and build long-term loyalty from their consumers.
- Behavioral Data Analysis: By analyzing customer behavior, businesses can make informed decisions and tailor marketing and sales strategies to each specific segment.

However, in the real operation, all processes in Cellphone S still depend on manual and disconnected tools . This comes from the immaturity in operational management, especially, customer care, content management, internal work coordination, as well as omnichannel sales coordination. In spite of accessing the CDP system, this company still is not integrated strictly with comprehensive ERP architecture. The daily sales and customer-related operations within their ERP system have not yet reached their full potential. Many processes are still handled manually, relying heavily on personnel support rather than being fully automated or optimized. Besides, their main customers are gen Z - a generation always changing their needs, their behavior in sales and customers so if there is no power tool for support companies to improve Sale and Customers Relationships.Therefore, Implementing Chatbot AI in ERP system is a strategy steps for this company since it can brings their benefits in all aspects:

- Automatic Customers Interactions (FAQ, Product, Product consulting, Technique Support, etc)
- Reducing workload, and human resources of Customer Service Department
- Keep track of orders and recommend suitable content or products via CRM data.
- Multichannel data synchronization to improve internal management and decision-making.

Owing to its advantage in being a young organization with a flexible, creative team that is ready to embrace change, Schanel is an ideal base to test and implement modern technology like AI Agent - Chatbot in ERP system. Using AI, especially AI-powered chatbots, brings significant benefits to businesses. For instance, 90% of businesses report faster complaint resolution thanks to AI chatbots, which help handle customer issues promptly and efficiently. Additionally, 61% of companies believe that chatbots significantly boost productivity by automating routine tasks such as follow-ups, allowing employees to focus on more complex work [9]. Hence, using AI enables it to get comprehensive digital transformation as well as support them in four following keys case study:

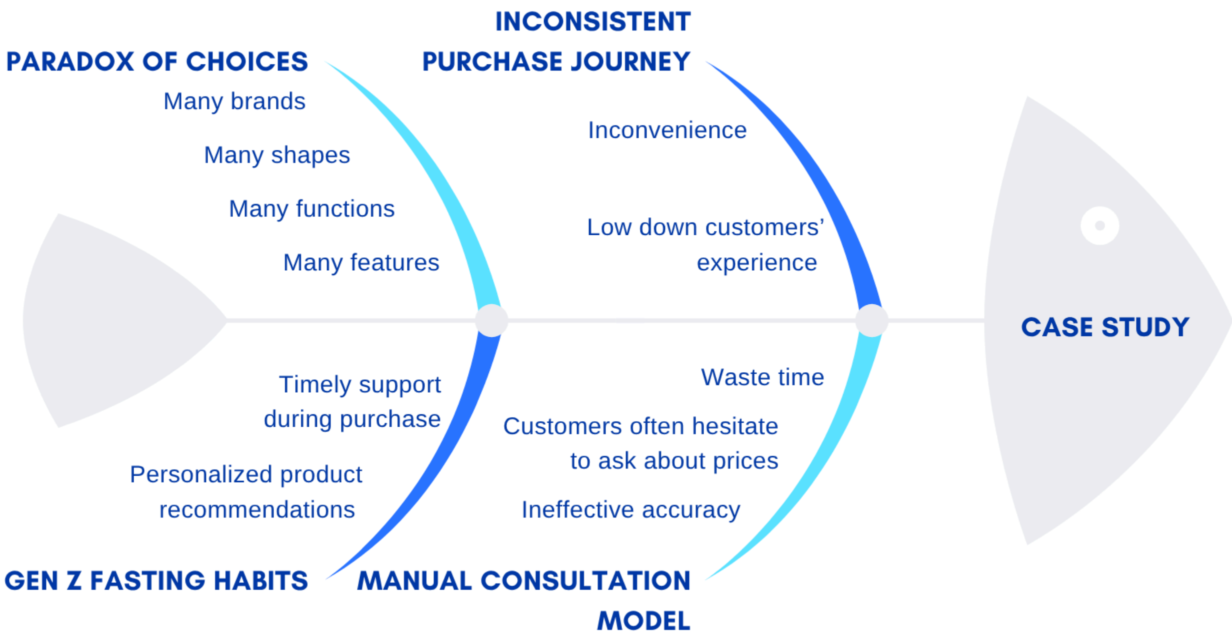

Figure 2.3.2.1: Overview of Case Study via Fishbone Framework

a. Case study 1:

The paradox of choice is a concept introduced by psychologist Barry Schwartz which suggests that the more options we have, the less satisfied we feel with our decision. This phenomenon occurs because having too many choices requires more cognitive effort, leading to decision fatigue and increased regret over our choices [4]. According to an experiment in the Paradox of Choice: Why More Is Less. HarperCollins papers, there were

2 scenarios used to test the impacts of choices on consumer behaviors. In the first case study, although it attracted a lot of visits from customers, the number of purchases was lower. However, in the other case, with just only 6 jams on the display, the purchase rates significantly increased overwhelming the first case. Conclusion, more choices customers have, more stress and dissatisfaction they gain, affecting sales performances of products. Nowadays, this phenomenon is straightforward to catch in the trading market across all business sectors with many types of merchandise, ranging from electronic devices to food products. Especially, smartphone industry is one of them, which releases hundreds of different products, offering a variety of brands, designs, prices, shapes and features ranges. With the high competition in phone industry, it requires leading companies, like Apple, Samsung, and so on, have growing the brands by upgrading the products with many newly features, such as AI, FaceID, etc as well as introduces new versions, allowing them to maintain their markets relevance, attract more consumers, and stay ahead of competitor. However, making a decision for buying one of these digital devices among their brands and version devices, which being suitable with their characteristics may be quite challenging with mobile devices users. Therefore, it is highly advisable for consumers to seek guidance from knowledgeable experts in the smartphone industry to make informed and confident decisions. In short, as product choices continue to expand, personalized advice from industry experts becomes essential in helping consumers navigate complexity and make the most suitable decisions.

In traditional ERP methods, particularly online sale and customer services processes, human consultants or sellers, who have domain knowledge about smartphone industries, are often responsible for helping customers via online channels. They will search and provide useful product information, which is suitable for the customer. However, this approach still has some limitations, including lower response times, inconsistent advice, inaccurate or irrelevant responses. Firstly, staff are unable to respond to their customers twenty-four hours a day. Secondly, since there is various information about smartphones, rangings versions from technologies, it is challenging when the sellers try to choose the most suitable one to introduce to the customers. Thirdly, they may have the necessary knowledge but are often unable to process and access all relevant information effectively, leading to incomplete or insufficient advice; therefore this limits their ability to provide tailored recommendations that match the specific needs of each customer. So, although companies are able to solve the paradox of choices in buying mobile phones of customers by the help of experts in the sales process and customer services process, it is still not effective and helpful ways to solve these problems. Overall, classic sales processes and customer processes in ERP may reduce productivity and efficiency in sales.

b. Case study 2:

Although chatbots can address customers' problems effectively, allowing them to avoid the trap of "choices", they need to face the other challenges during shopping experience, particularly that they have to open a new tab or switch to the new pages so as to select the products, make payment or cancel orders. These actions not only waste the customers' time but also makes the experience feel fragmented, lacking coherence, and can easily cause cónusmers to lose interest in continuing their purchase. According to statistics from Baymard Institute (2023), 69% of consumers leave their shopping carts owing to a complicated or inconsistent purchasing process. Besides, from the previous Global Consumer Insight Pulse Survey in 2023, it highlights the need to streamline the customer purchasing process by eliminating barriers before the initial shopping occurs. This needs the support of empowered AI chatbots that can not only recommend the product but also handle the order in the O2C process, which makes a significant contribution in customer relationship management improvement. By focussing on this point, where chatbots can guide consumers, simplify transactions, and deliver seamless support — businesses can get closer to their customers. Understanding this trend, a variety of chatbot agents, offering a full range of features to support the order to cash process, including AhaChat, Fchat, TPos, and so on, have been increasingly released by developers or technology companies. Besides, the growing trend of automatic pipelines, namely n8n, zapier, code, etc has solved the problems, such as the fragmentation of tasks and processes, leading to poor customer experience. For example, instead of wasting time to search for products and then switching to another tab to make orders or payments, an automatic pipeline will enable customers to do these actions in one tab, such as AI agent or AI chatbot. Reconizating this situation and potential of these features our group equips chatbot with End-to-End Order Management Workflow to empower the sale process and customer process in CRM modules for our solution.

c. Case study 3:

Customer Segments of Cellphone S are dynamic young people, who prioritize personal expression, and demonstrate strong adaptability to emerging technologies. After conducting research, we propose customers' personas of Cellphone S customers, such as:

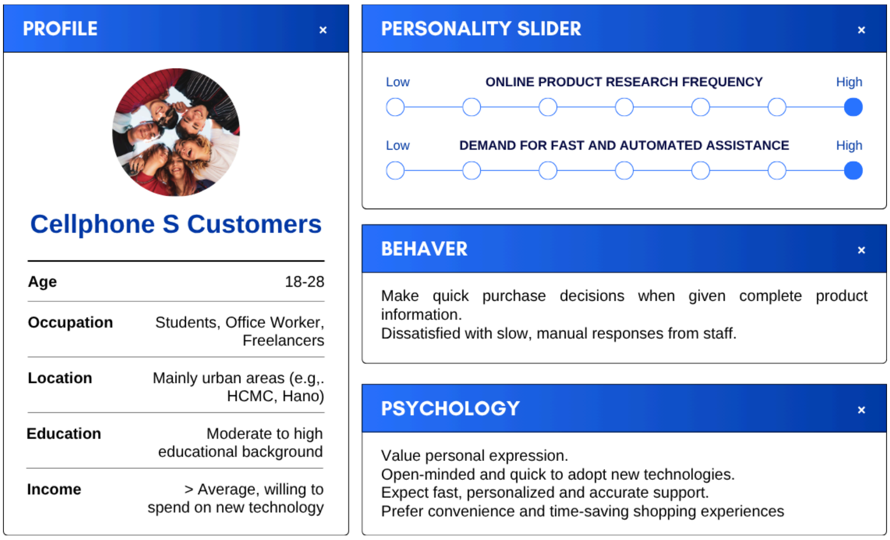

Figure 2.3.2.2: Customer Persona of Cellphone S Company

Via analyzing their demographic, psychology, and behavior, we find that supporting these customers' segments to search for suitable products by quick response and personal answers, can enhance their purchasing experiences and brand satisfaction. Particularly, these customers tend to make quick decisions and need timely and prompt support regarding product information, including pricing, functionality, and features of mobile phones, but in traditional sales and customer service processes, product recommendations for customers based on staff' experiences who have to collect information from a database or google for providing an answer, which waste time and be low accuracy, leading to decrease customer satisfaction. Hence, Virtual Assistant AI is the best option to support customers as it can provide information faster and more accurately, along with personal answers, which improves customer purchasing experiences on the e-commerce platform.

d. Case study 4:

Cellphone S provides a colorful website for their customers to have the best experience when they visit for purchasing products on this platform. As we can see, the main functions in these websites, including electronic devices searching, orders tracking, shopping, payment and staff connection to get more details of products.

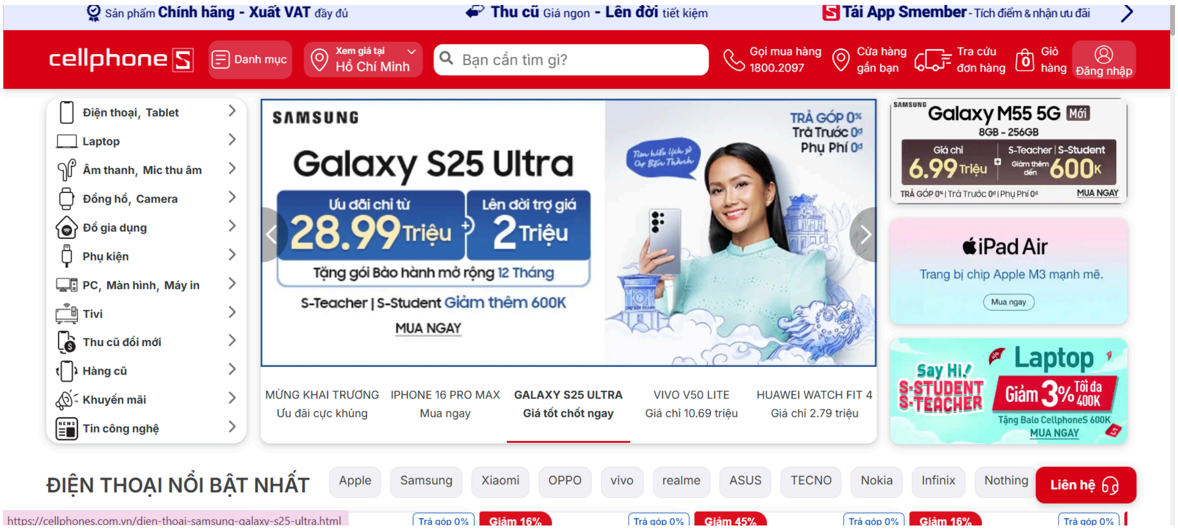

Figure 2.3.2.3: Cellphone S Website Interface 1

However, their company provides various products for mobile phones from many brands like Apple, Xiomi, Samsung, Vivo, etc. Each product has their own features, shape and functions, leading to difficulties for customers who are unsure which option best suits their needs, with numerous after-sales services, particularly screen protector application, phone repair, etc.

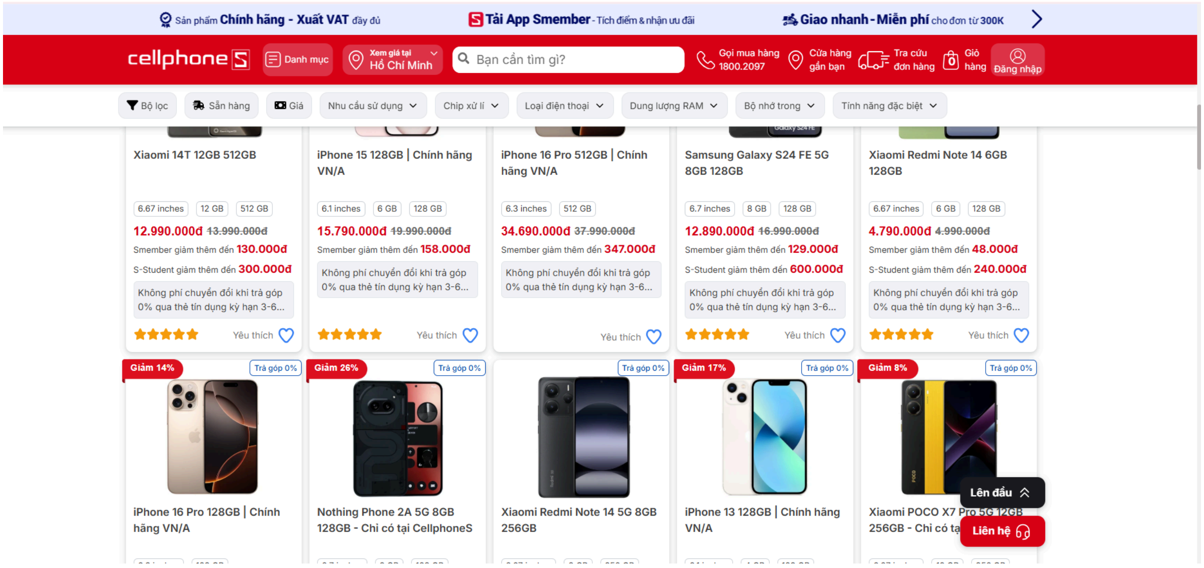

Figure 2.3.2.4: Cellphone S Website Interface 2

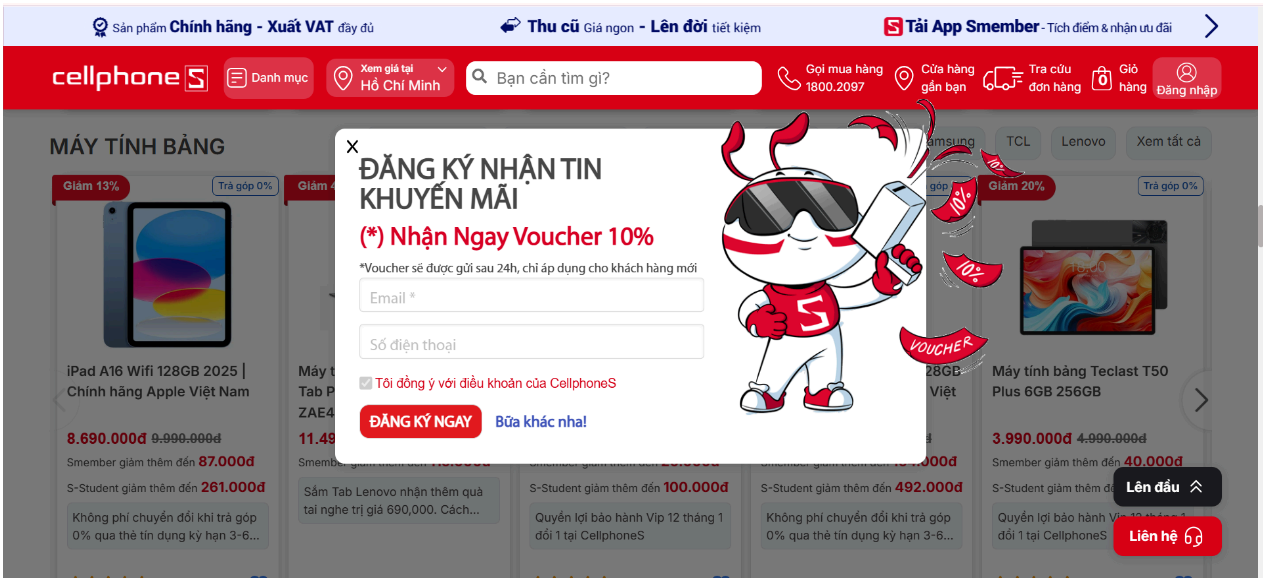

Figure 2.3.2.5: Cellphone S Website Interface 3

Therefore, to help their customers can choose the best ones, they provide a function that chat with online sale staff as the following images:

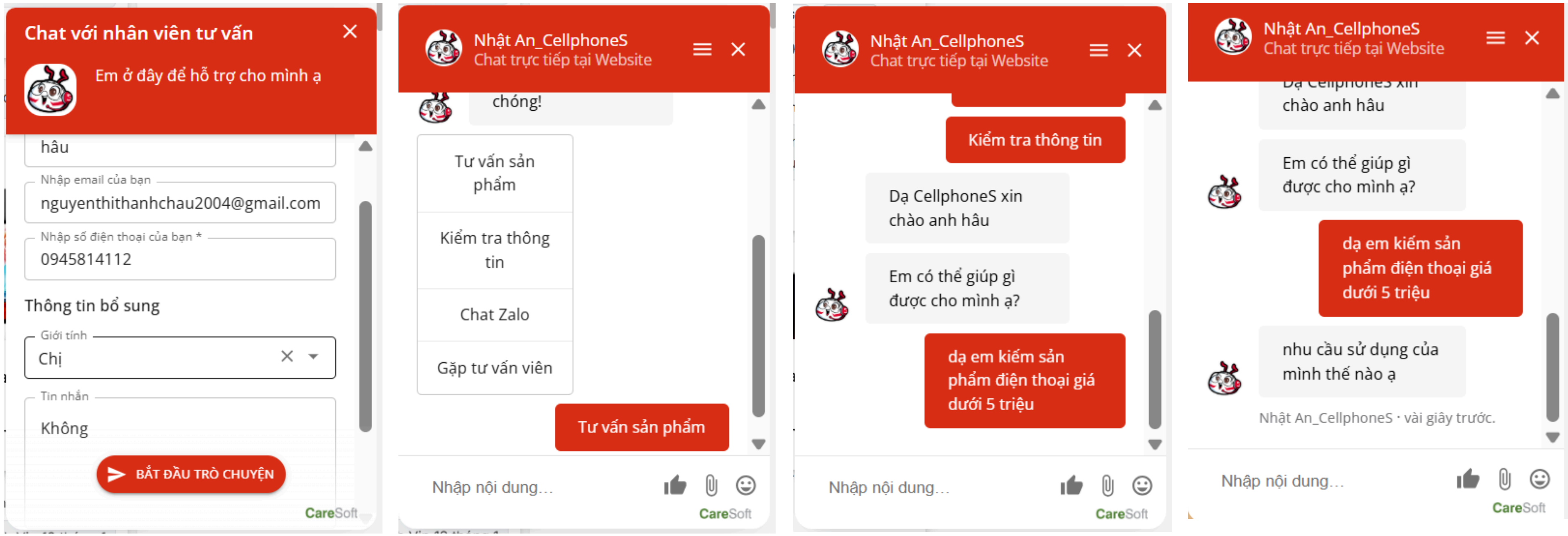

Figure 2.3.2.6: Cellphone S Website Interface For Product Consultant

To chat with sales employees, customers need to fill in their information in the chat box, and then click on "Start conversation" so as to connect with the employee. After that, customers have to wait a long time to receive from staff to start the conversation.

After trying to chat with staff to get consulting, we found that users have to wait a long time to receive responses. Besides, there is no Chat Bot AI in order to assist customers when staff are unavailable. Especially, with the question about prices with human staff can make users feel more shy and embarrassed. Hence, with these factors, it can slow customers' experiences.

And those are the four case studies we have analyzed in this chapter. In the next chapter 3, our group will define deeper insight into each business process of a company operating in consumer electronics retail like, such as Cellphone S by using the BNMP framework. The model we use is adapted from best practices of leading Vietnamese enterprises like Viettel and Thế Giới Di Động (Mobile World), which have successfully implemented end-to-end business process management and digital transformation in their operations.

CHAPTER 3: BUSINESS PROCESS SYSTEM

Chapter 3 illustrates the whole process of ERP system in smartphone sale via BMNP diagrams, including customers services process, sale process, procurement process, stock issuance process, good receipt, inventory inspection process, shipping process, warranty process, supplier payment process and customer payment process. In this part, we will present how BPMN diagrams serve as a standard of business process modeling, so readers are able to gain the visualizable insights of our solution straightforwardly in the following chapters.

3.1. System overview:

In this part, we will show the overview of business processes in an Amazon Company via Business Process Model and Notation (BPMN). This section outlines the main modules of this company, covering from customer services process to warranty process. It reflects 2 perspectives in this business cycle, including customer journey and supplier journey.

From the customer's perspective, customers will be supported by staff, particularly received vouchers, new release of products, etc. Then they are able to purchase products through offline or online channels and make a payment for their shopping accordingly. When customers confirm their orders, systems in ERP will update the quality of products in stocks as well as send them to customers, along with warranty policy.

In terms of supplier perspective, when stock levels fall below the required threshold, company staff will select the best supplier to fulfil their warehouse via the Procurement Process. After finishing the previous step, warehouse staff will take responsibility for receiving goods in order to restock items and inspect goods quality. Once the goods are of sufficient quantity and meet the required quality standards, they will be approved for payment.

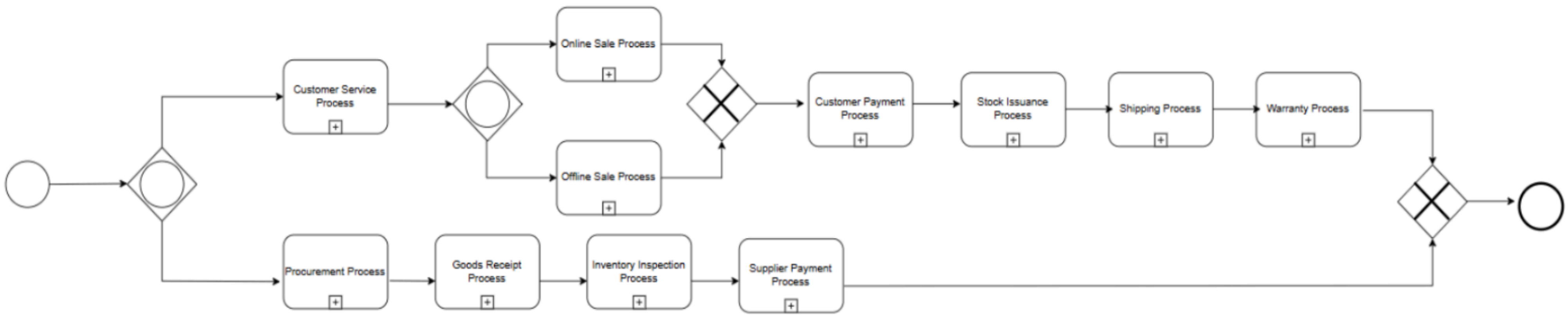

Figure 3.1.1. Overview of Whole ERP Process

In our project, we just improve the CRM module, particularly Customer Service Process, and Online by using our Chatbot named "Kammy".

3.2. Customer Perspectives:

3.2.1. Customer Service Process

a. BPMN Diagram:

This is the traditional Customer Service Process referenced in the Viettel Internship Report:

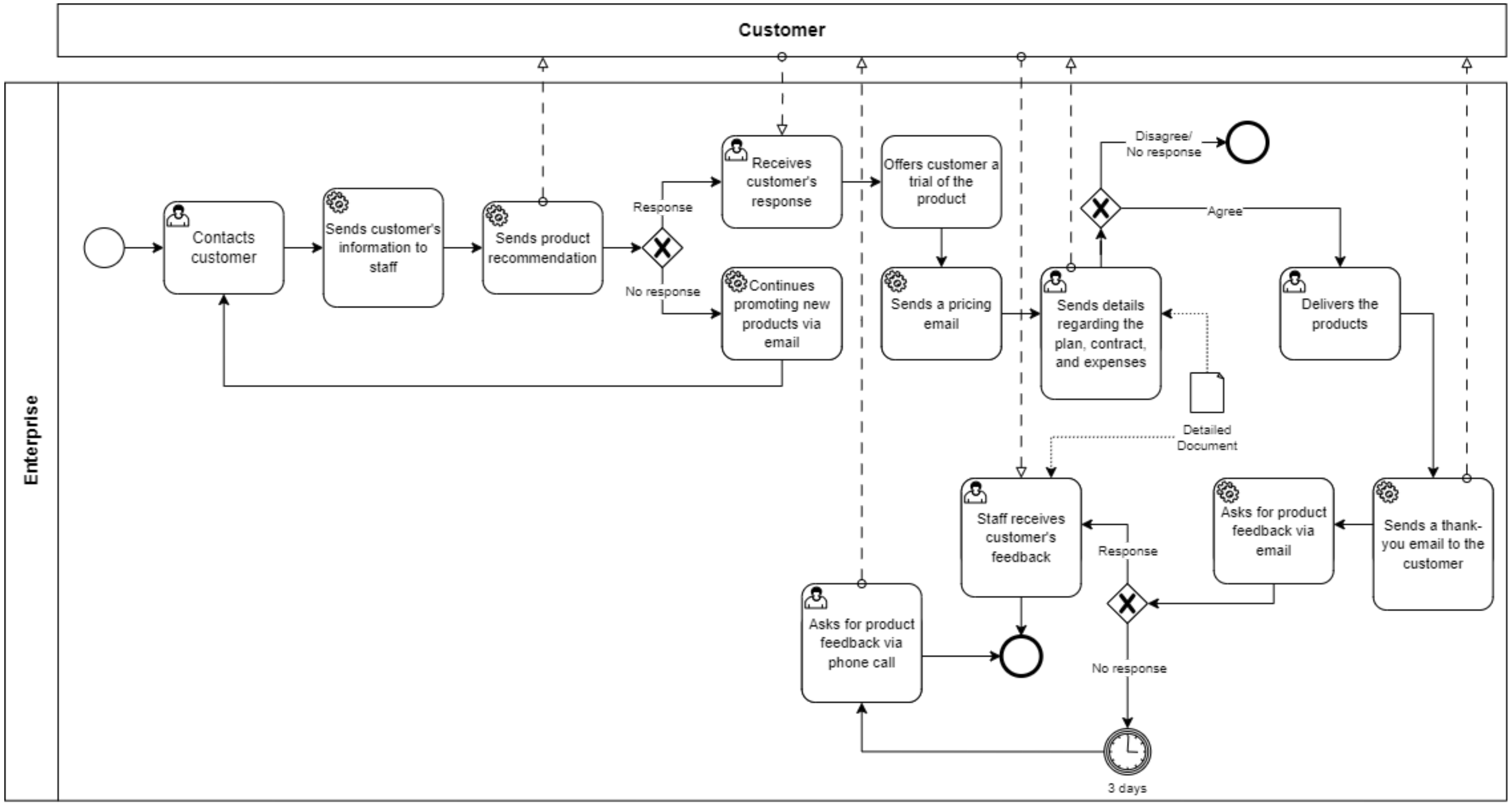

Figure 3.2.1.1: Customer service process

b. Specification:

Table 3.2.1.1: Customer service process BPMN specification

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Contacts customer	–	Customer Service Department	Contacts the registered customer
Sends customer's information to staff	–	Customer Service Department	Sends the customer's information to the staff so they can find relevant products for the customer
Sends product recommendation	–	Customer Service Department	Sends product recommendation to the customer
Customer responds	Exclusive gateway	Customer	The customer receives the recommendation and replies
Customer does not respond	Exclusive gateway	Customer	The customer may receives the recommendation and does not reply
Receives	–	Customer Service	Receives the customer's response to the product recommendation

customer's response		Department	
Continues promoting new products via email	–	Customer Service Department	Recommends new product information via another channel (email) to ensure the customer receives the promotion for new products
Offers customer a trial of the product	–	Customer Service Department	Boosts customer's confidence in decision-making by offering a trial of the product
Sends a pricing email	–	Customer Service Department	Once the trial is offered, a pricing email is sent to the customer
Sends details regarding the plan, contract and expenses	–	Customer Service Department	Details, including product details, pricing breakdown, subscription options, payment methods, contract and expenses, are sent to the customer
Customer agrees	Exclusive gateway	Customer	The customer receives the details about the trial offer and confirm with

with the contract			the purchase
Customer disagrees with the contract or do not respond	Exclusive gateway	Customer	The customer may receives the details about the trial offer and does not process any actions
Delivers the products	–	Logistics Department	Once the order has been confirmed by the customer and verified by the customer service department, it is logged into the ERP system and automatically transferred to the logistics department, where inventory availability is checked, shipping labels are generated, and the order is prepared for dispatch.
Sends a thank-you email to the customer	–	Customer Service Department	Once the product has been delivered, a thank-you email is sent to the customer in order to maintain a strong customer relationships
Asks for product feedback via email	–	Customer Service Department	Encourage customer sharing product using experience via email in order to collect valuable insights for further development of products or

			customer service quality
Customer feeds back	Exclusive gateway	Customer	Customer agrees to take the product feedback
Customer does not feed back	Exclusive gateway	Customer	Customer has not responded the product feedback request email yet
Staff receives customer’s feedback	–	Customer Service Department	Customer service department receives customer’s feedback via email
Asks for product feedback via phone call	–	Customer Service Department	If the customer does not respond within three days after the product feedback request email is sent, a follow-up phone call will be made to request their feedback

c. Innovative BMNP Diagram:

To improve the traditional process, we implement our solution "Kammy" to automate the process, allowing companies to solve the customers' requirements quickly and effectively without time, and human consumption.

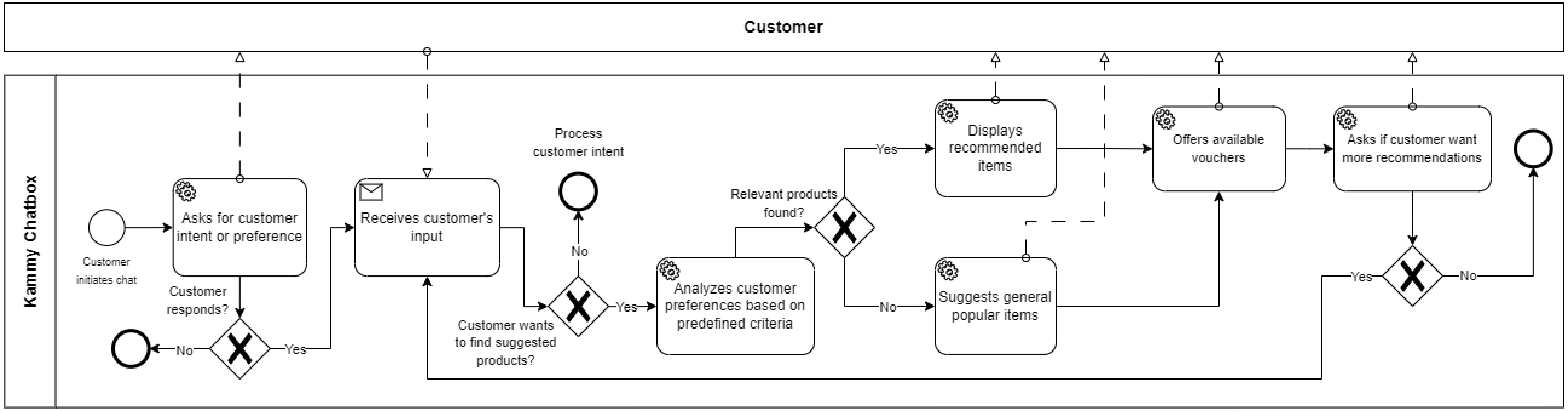

Figure 3.2.1.2: New Customer Service Process

d. Innovative BMNP Specification:

Table 3.2.1.2. New Customer Service Process BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Asks for customer intent or preference	—	Kammy Chatbot	The chatbot asks customer for their intent or preference
Customer responds	Exclusive gateway	Customer	The customer responds
Customer does not responds	Exclusive gateway	Customer	Customer does not responds
Receives customer’s input	—	Kammy Chatbot	The customer responds with their preference
Customer wants to find suggested products	Exclusive gateway	Customer	Customer wants to find suggested products

Customer wants to perform other functions	Exclusive gateway	Customer	Customer wants to perform other functions
Analyzes customer preferences based on predefined criteria	—	Kammy Chatbot	Once the chatbot has received the customer’s product preferences, it analyzes their preferences based on predefined criteria to find relevant product suggestions
Relevant products are found	Exclusive gateway	Kammy Chatbot	There are products that align with the customer’s need
Relevant products are not found	Exclusive gateway	Kammy Chatbot	There are no products that is matched with the customer’s need
Displays recommended items	—	Kammy Chatbot	Recommend found items that are relevant to the customer’s preference

Suggests general popular items	—	Kammy Chatbot	If no recommended products are found, the chatbot suggests general popular items
Offers available vouchers	—	Kammy Chatbot	Once the chatbot has recommended a list of items, it offers currently available voucher in order to enhance customer engagement and encourage purchases
Asks if customer want more recommendations	–	Kammy Chatbot	The chatbot keeps the customer engaged in discovering more products by prompting them to continue the conversation
Customer wants more recommendations	Exclusive gateway	Kammy Chatbot	The customer responds to the chatbot
Customer does not want more recommendations	Exclusive gateway	Kammy Chatbot	The customer does not respond to the chatbot or asks the chatbot to stop the conversation

3.2.2. Online Sale Process:

a. BPMN Diagram:

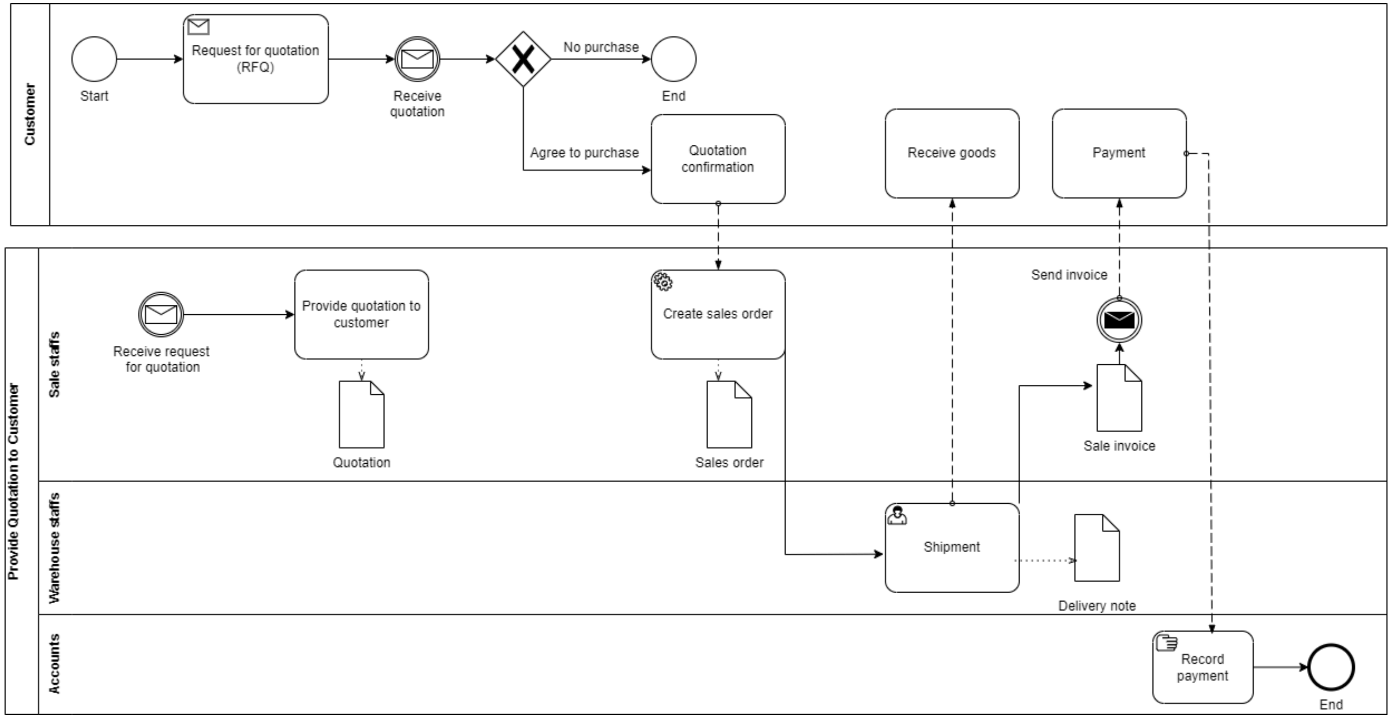

Figure 3.2.2.1: Online Sale Process

b. Specification:

Table 3.2.2.1 Online Sale Process BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Request for quotation (RFQ)	–	Customer	The customer submits a request for quotation.
Receive request for quotation	–	Sales staffs	Sales staff receives the RFQ from the customer.
Provide quotation	–	Sales staffs	Sales staff provides the quotation to the customer.
Receive quotation	–	Customer	The customer receives the quotation.
Agree to purchase	Exclusive gateway	Customer	The customer decides whether to proceed with the purchase.
No purchase	Exclusive gateway	Customer	The customer chooses not to proceed, then the process ends.
Quotation confirmation	–	Customer	The customer confirms acceptance of the quotation.

Create sale order	–	Sales staff	A sales order is created based on the confirmed quotation.
Shipment	–	Warehouse staff	The warehouse staff processes and ships the goods.
Receive goods	–	Customer	The customer receives the goods.
Create sales invoice	–	Sales staff	A sales invoice is generated by sales staff.
Send invoice	–	Sales staff	The invoice is sent to the customer.
Payment	–	Customer	The customer makes a payment according to the invoice.
Record payment	–	Accounts	The accounting department records the payment.

c. Innovative BMNP Diagram:

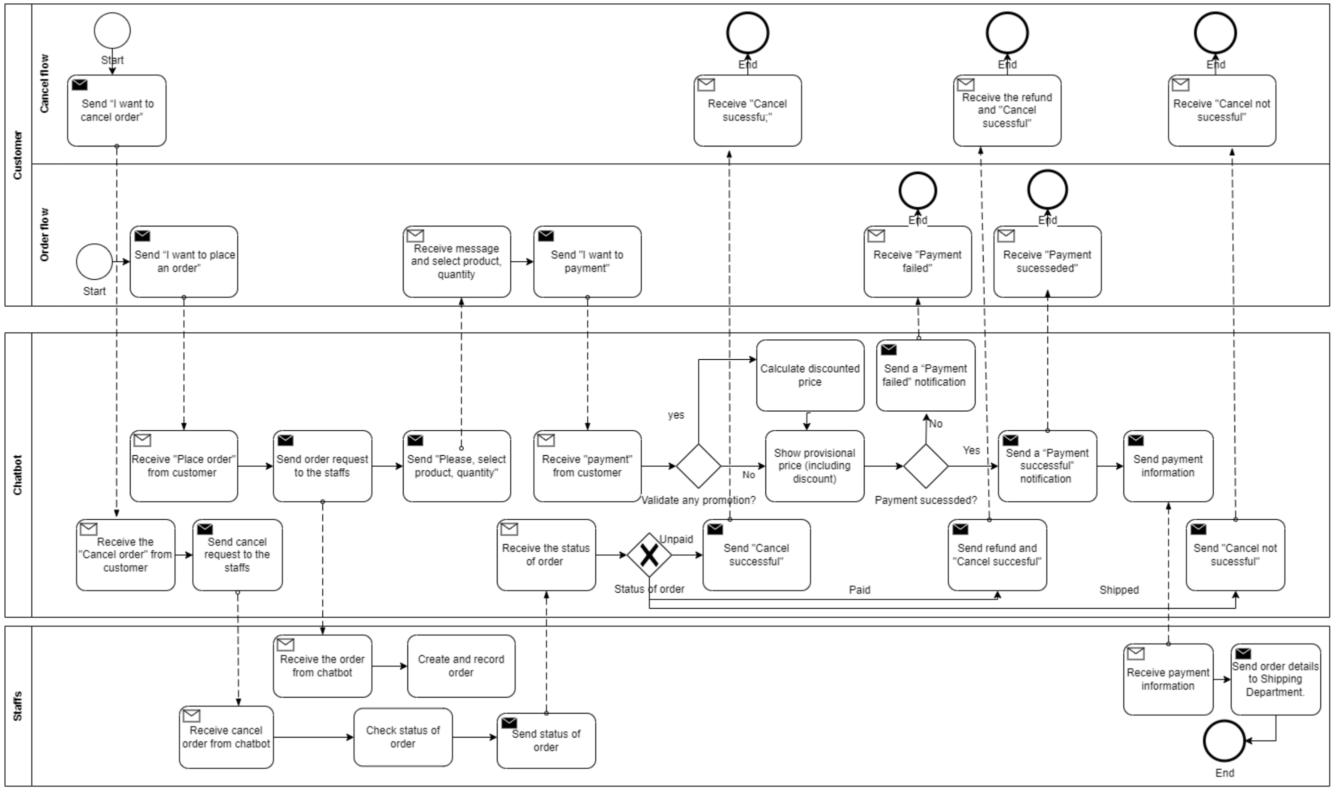

Figure 3.2.2.2: New Online Sale Process

d. Innovative BMNP Specification:

Table 3.2.2.2. New Online Sale Process BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Send "I want to place an order"	–	Customer	Customer initiates an order request.
Receive "Place order"	–	Chatbot	Chatbot receives order request.
Send order request to staff	–	Chatbot	Order forwarded to staff.
Send "Please select product, quantity	–	Chatbot	Prompt customer to select product and quantity.
Receive message and selection	–	Customer	Customer receive and select the product and quantity.

Send "I want to payment"	–	Customer	Customer proceeds to payment.
Receive payment from customer	–	Customer	Payment info received.
Validate promotion?	Exclusive gateway	Chatbot	Checks if any promotion applies.
Calculate discounted price	–	Chatbot	Applies discount if applicable.
Show provisional price	–	Chatbot	Sends final price to customer.
Payment succeeded?	Exclusive gateway	Chatbot	Checks whether the payment was successful.
Send "Payment	–	Chatbot	Notifies customer of failed payment.

failed" notification			
Send "Payment successful" notification	–	Chatbot	Notifies customer of successful payment.
Send payment information	–	Chatbot	Payment data sent to staff.
Receive payment info	–	Staff	Staff receives payment confirmation.
Send order to shipping	–	Staff	Shipping department is notified.
Send "I want to cancel order"	–	Customer	Customer requests to cancel the order.
Receive cancel	–	Chatbot	Chatbot receives a cancel request.

request			
Send cancel request to staff	–	Chatbot	Chatbot sends cancel request to staff.
Receive cancel request	–	Staff	Staff receives a cancel request.
Check order status	–	Staff	Verifies payment or shipping status.
Status of order	Exclusive gateway	Chatbot	Checks if order was paid/unpaid/shipped.
Send "Cancel successful"	–	Chatbot	Send cancel sucessful notification to customer.
Send refund and "Cancel successful"	–	Chatbot	Send cancel sucessful notification and refund to customer.
Send "Cancel not successful"	–	Chatbot	Send cancel unsucessful notification to customer.

Receive "Cancel successful"	–	Customer	Customer receives cancel sucessful notification.
Receive refund and "Cancel successful"	–	Customer	Customer receives cancel sucessful and refund notification.
Receive "Cancel not successful"	–	Customer	Customer receives cancel unsucessful notification.

3.2.3. Offline Sale Process:

a. BPMN Diagram:

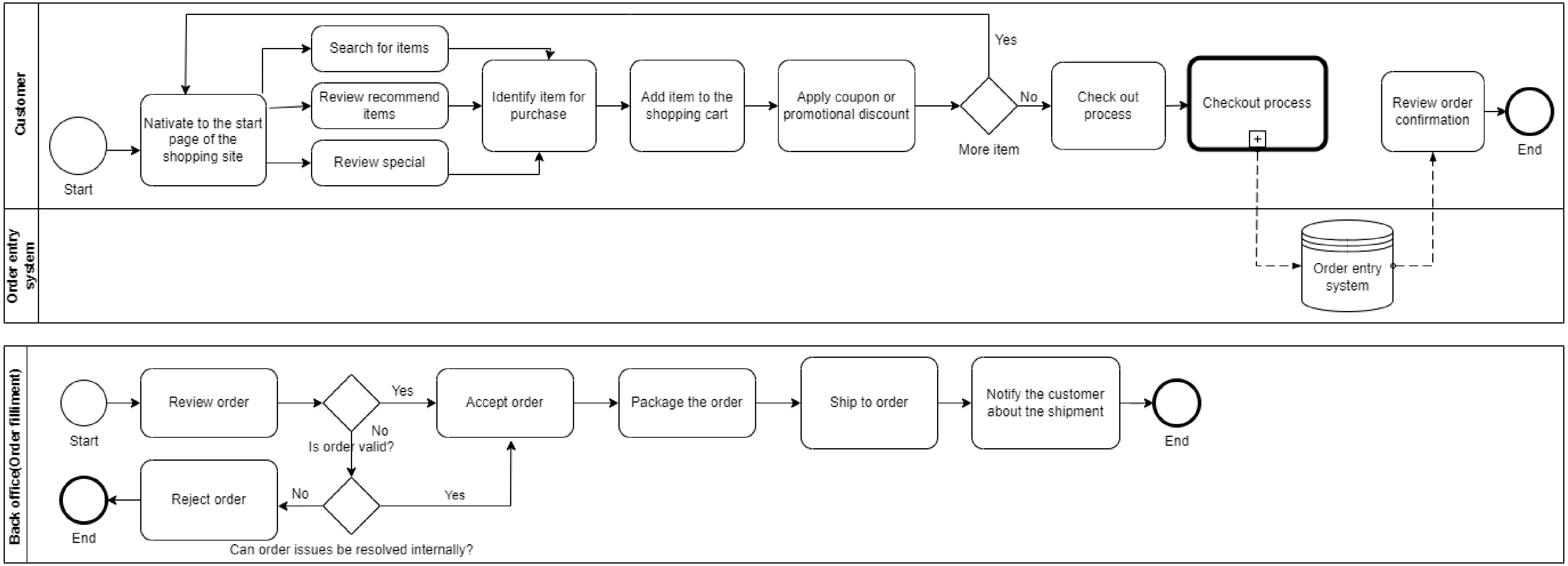

Figure 3.2.3.1: Offline Sale Process

b. Specification:

Table 3.2.3.1: Offline Sale Process BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Navigate to the start page of the shopping site	–	Customer	Customer navigates to the home page of the shopping website.
Search for items	–	Customer	Customer uses the search feature to find items.
Review recommended items	–	Customer	Customer reviews items recommended by the system.
Review special items	–	Customer	Customer reviews special offers or discounted items.
Identify item for purchase	–	Customer	Customer selects an item they want to buy.

Add item to the shopping cart	–	Customer	Customer adds selected item to their shopping cart.
Apply coupon or promotional discount	Exclusive gateway	Customer	Customer applies a coupon or discount code if available.
More items?	Exclusive gateway	Customer	Customer decides whether to add more items or proceed to checkout.
Checkout process	–	Customer	Customer begins the checkout process.
Review order confirmation	–	Customer	Customer reviews the order and confirms the items to be purchased.
Order entry system	–	Order Entry System	The system records and processes the order information.
Review order	–	Back Office	Back Office staff reviews the order for validity and correctness.
Is order valid?	Exclusive gateway	Back Office	Back office checks whether the order is valid.

Accept order	–	Back Office	If the order is valid, it is accepted for further processing.
Reject order	–	Back Office	If the order is invalid, it is rejected.
Can order issues be resolved internally?	–	Back Office	Checks if internal issues can resolve the order.
Package the order	–	Back Office	The accepted order is packaged for shipment.
Ship to order	–	Back Office	The packaged order is shipped to the customer.
Notify the customer about the shipment	–	Back Office	Back office notifies the customer that the order has been shipped.

3.2.4. Customer Payment Process:

a. BPMN Diagram:

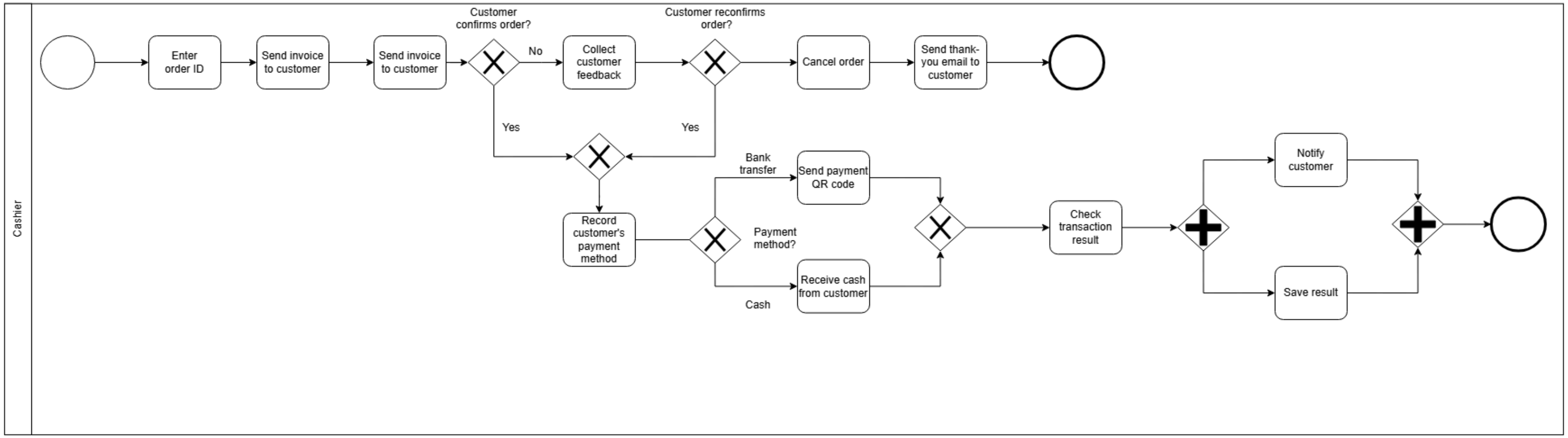

Figure 3.2.4.1: Customer Payment Process

b. Specification:

Table 3.2.4: Customer Payments Process BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Start payment	–	Customer	Customer initiates the payment process
Enter order ID	–	Cashier	Cashier enters the order ID
Send invoice	–	Bot	Bot sends invoice with payment details to customer via PDF file attachment
Confirm payment	–	Customer	Customer confirms whether payment is ready
Verify payment intent	–	Cashier	Cashier checks if the customer intends to proceed with the payment
Send confirmation	–	Bot	Bot sends a confirmation email to the customer upon completing the process
Provide payment	–	Cashier	Cashier offers payment methods for customer to select

options			
Verify transaction result	–	Cashier	Cashier checks the transaction result
Notify customer	–	Bot	Bot notifies the customer about the transaction result and logs it in the system
Complete payment		Cashier	Cashier finalizes the payment process

3.2.5. Stock Insurance Process:

a. BPMN Diagram:

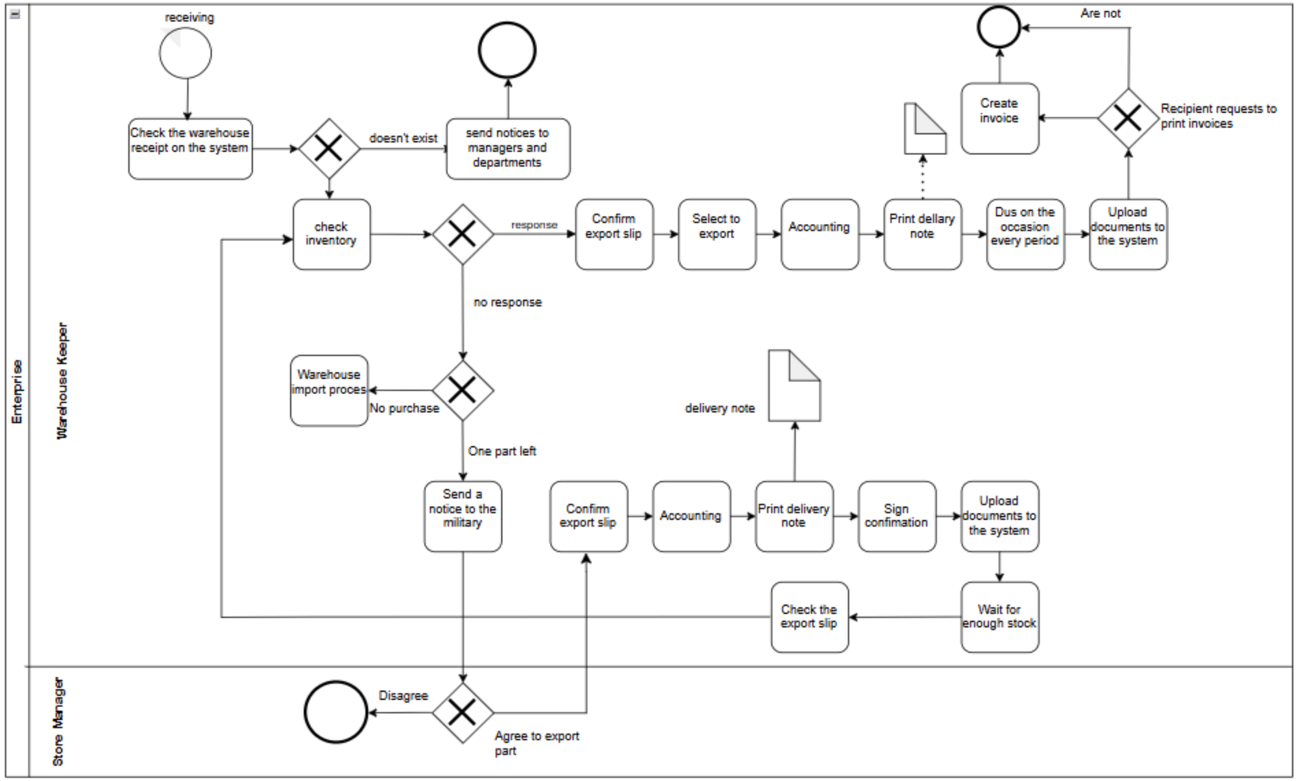

Figure 3.2.5: Stock Insurance Process

b. Specification:

Table 3.2.5: Stock Insurance Process BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Check warehouse receipt	—	Warehouse Keeper	Check if the goods have been received and recorded in the system.
Check inventory	Does the receipt exist?	Warehouse Keeper	If no receipt, check current stock levels for the item.
Send notice to managers	If the receipt doesn't exist	Warehouse Keeper	Notify departments and managers about missing goods.
Warehouse import process	If item not in inventory	Warehouse Keeper	Start the import process if no items are available in stock.
Send notice to the military	If one part left	Warehouse Keeper	Notify the military about limited stock availability.

Agree to export part	Store manager decision	Store Manager	Approve or deny partial item export.
Confirm export slip	If approved	Warehouse Keeper	Confirm and prepare the export slip.
Select to export	—	Warehouse Keeper	Choose the correct items to export.
Accounting	—	Accounting Department	Register the export details in the accounting system.
Print delivery note	—	Warehouse Keeper	Print the delivery note for the selected items.
Sign confirmation	—	Warehouse Keeper	Approve and sign the delivery note.
Upload documents	—	Warehouse	Upload the signed documents to the internal system.

to the system		Keeper	
Check the export slip	—	Warehouse Keeper	Double-check the slip to ensure everything is correct.
Wait for enough stock	If stock is insufficient	Warehouse Keeper	Pause the process until required stock becomes available.
Create invoice	—	System / Staff	Generate an invoice for the exported items.
Recipient requests to print invoice	Are invoices requested or periodic?	Recipient / System	Print invoices if the recipient requests or based on scheduled timing.
Upload invoice documents	—	Staff / System	Upload final invoice documents to the internal system.

3.2.6. Shipping Process:

a. BPMN Diagram:

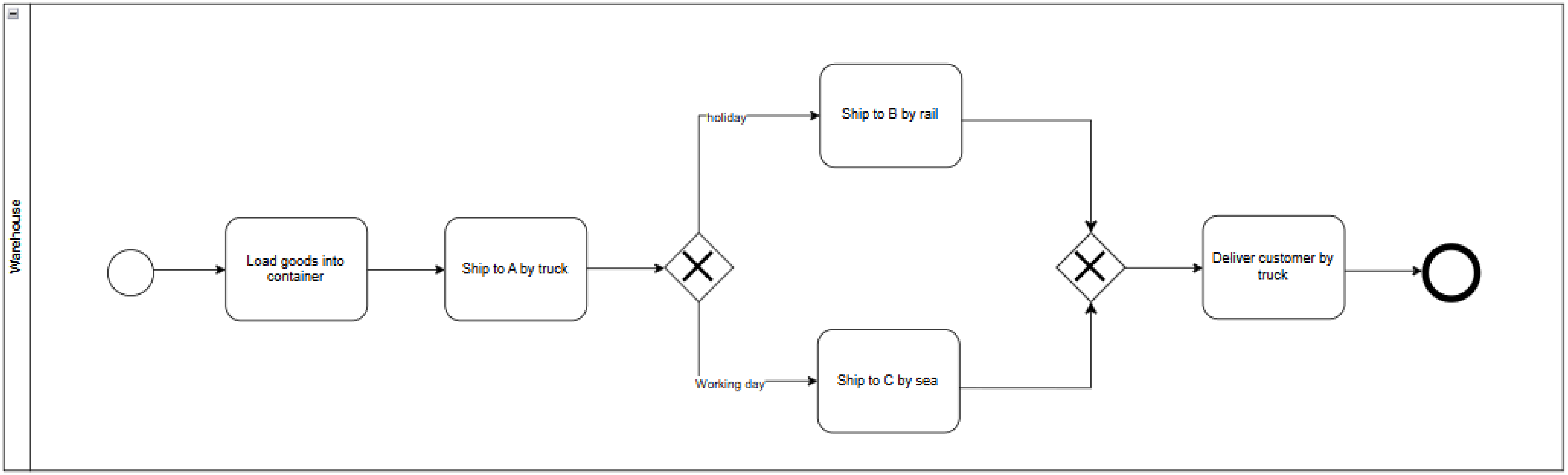

Figure 3.2.6.1: Shipping Process

b. Specification:

Table 3.2.6.1: Shipping Process BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Load goods into container	—	Warehouse	The warehouse staff loads goods into a container in preparation for shipping.
Ship to A by truck	—	Warehouse	Goods are shipped to Location A via truck.
Determine shipment method	Holiday / Working day decision	Warehouse	Staff determines the shipping method based on the calendar: holiday or working day.
Ship to B by rail	—	Warehouse	If it's a holiday, goods are shipped to Location B via rail.
Ship to C by sea	—	Warehouse	If it's a working day, goods are shipped to Location C by sea.
Merge shipments	—	Warehouse	All shipments (B or C) are merged for final delivery.
Deliver customer by truck	—	Warehouse	Goods are delivered to the customer by truck.

End the process	—	Warehouse	The delivery process concludes.

3.2.7. Warranty Process:

a. BPMN Diagram:

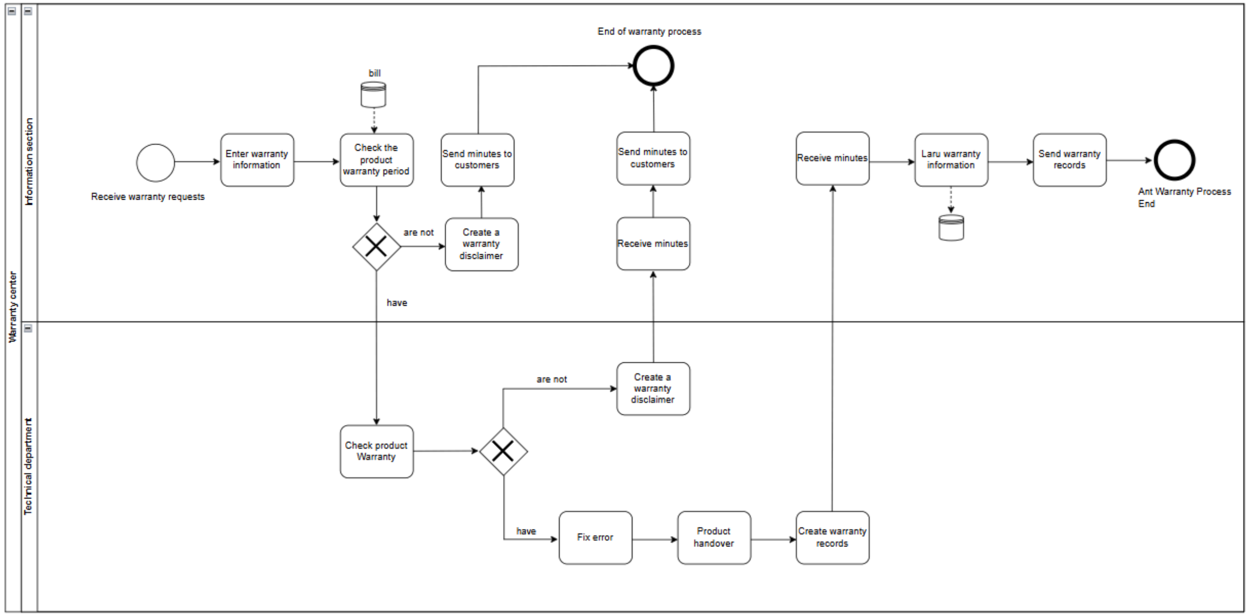

Figure 3.2.7: Warranty Process BPMN

b. Specification:

Table 3.2.7: Warranty Process Specification

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Receive warranty requests	—	Information Section	Receive warranty requests from customers.
Enter warranty information	—	Information Section	Enter warranty information into the system.
Check the product warranty period	—	Information Section	Check the product’s warranty period.
Decision: are not/have	Exclusive Gateway	Warranty Center	Check the result: valid (have) or invalid (are not).
Create a warranty disclaimer	—	Warranty Center	Create a warranty disclaimer if the warranty is invalid.

Check product warranty	—	Technical Department	Inspect the product's warranty status.
Decision: are not/have	Exclusive Gateway	Technical Department	Check the result: repairable (have) or not repairable (are not).
Fix error	—	Technical Department	Perform repairs on the product.
Product handover	—	Technical Department	Hand over the repaired product.
Create warranty records	—	Technical Department	Create warranty records.
Send minutes to customers	—	Information Section	Send repair minutes to customers.
Receive minutes	—	Warranty Center	Receive the repair minutes.

Save warranty information	—	Warranty Center	Save the warranty information.
Send warranty records	—	Warranty Center	Send the warranty records.
End of warranty process	End Event	—	End of the warranty process.

3.3. Suppliers Perspectives:

3.3.1. Procurement Process

a. BPMN Diagram:

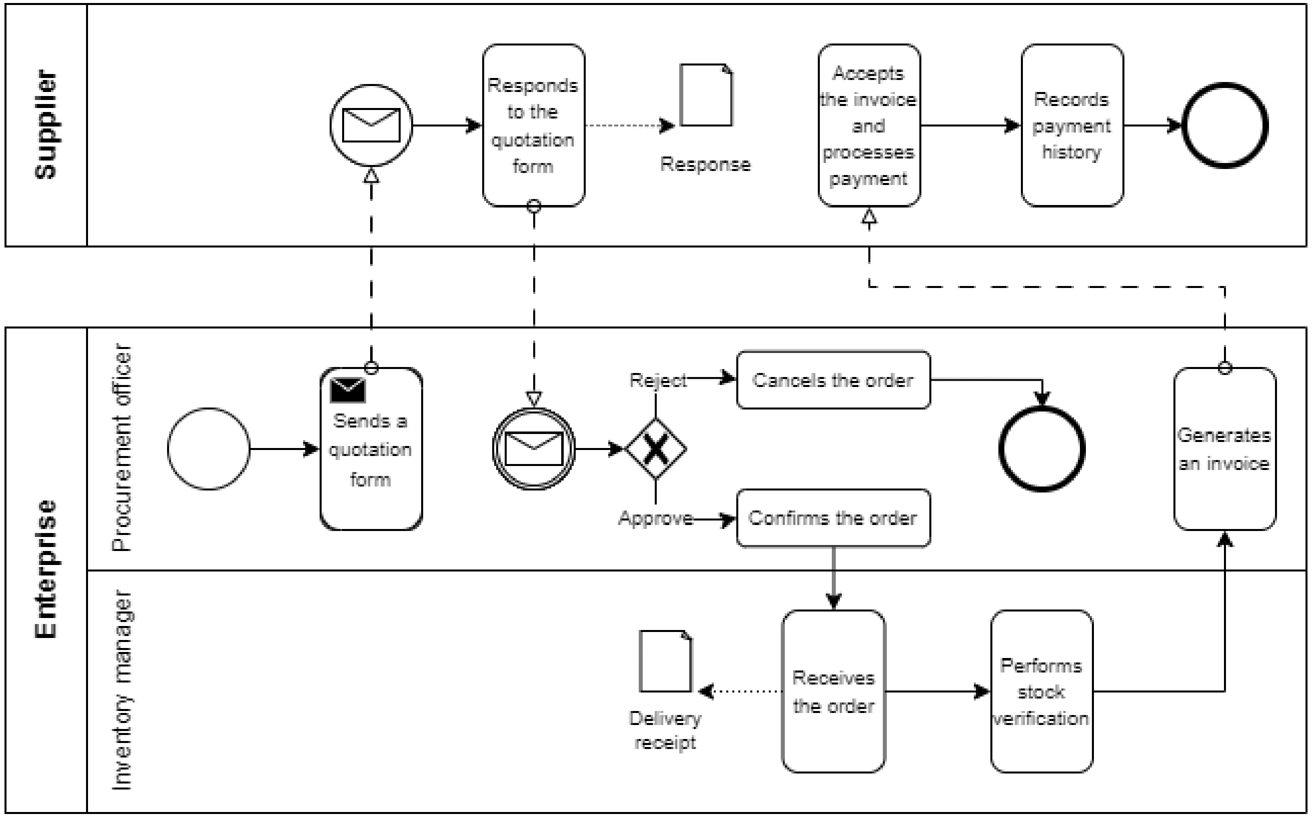

Figure 3.3.1.1: Procurement process

b. Specification:

Table 3.3.1.1: Procurement process specification BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Sends a quotation form	–	Procurement officer	A quotation form is sent to the supplier
Responds to the quotation form	–	Supplier	The supplier responds to the quotation form with a proposal that includes information on pricing, products, payment schedules, delivery timelines and conditions, etc.
The supplier's proposal is rejected	Exclusive gateway	Procurement officer	The supplier's proposal is rejected due to various reasons, such as pricing issues, quality concerns, or better alternatives found.
The supplier's proposal is approved	Exclusive gateway	Procurement officer	The supplier's proposal is accepted by the procurement officer
Cancels the order	–	Procurement officer	After rejecting the supplier's proposal, the procurement officer cancels the order

Confirms the order	–	Procurement officer	After approving the supplier's proposal, the procurement officer confirms the order
Receives the order	–	Inventory manager	The inventory manager receives a purchase order from the procurement department
Performs stock verification	–	Inventory manager	A stock verification process is executed to ensure that the actual stock matches recorded inventory levels
Generates an invoice	–	Procurement officer	The procurement officer creates a formal billing document that requests payment for goods or services provided
Accepts the invoice and processes payment	–	Supplier	The supplier receives the invoice, verifies the details and processes the payment
Records payment history	–	Supplier	The supplier records the payment history for financial tracking, cash flow management, credit assessment, etc.

3.3.2. Good Receipt Process

a. BPMN Diagram:

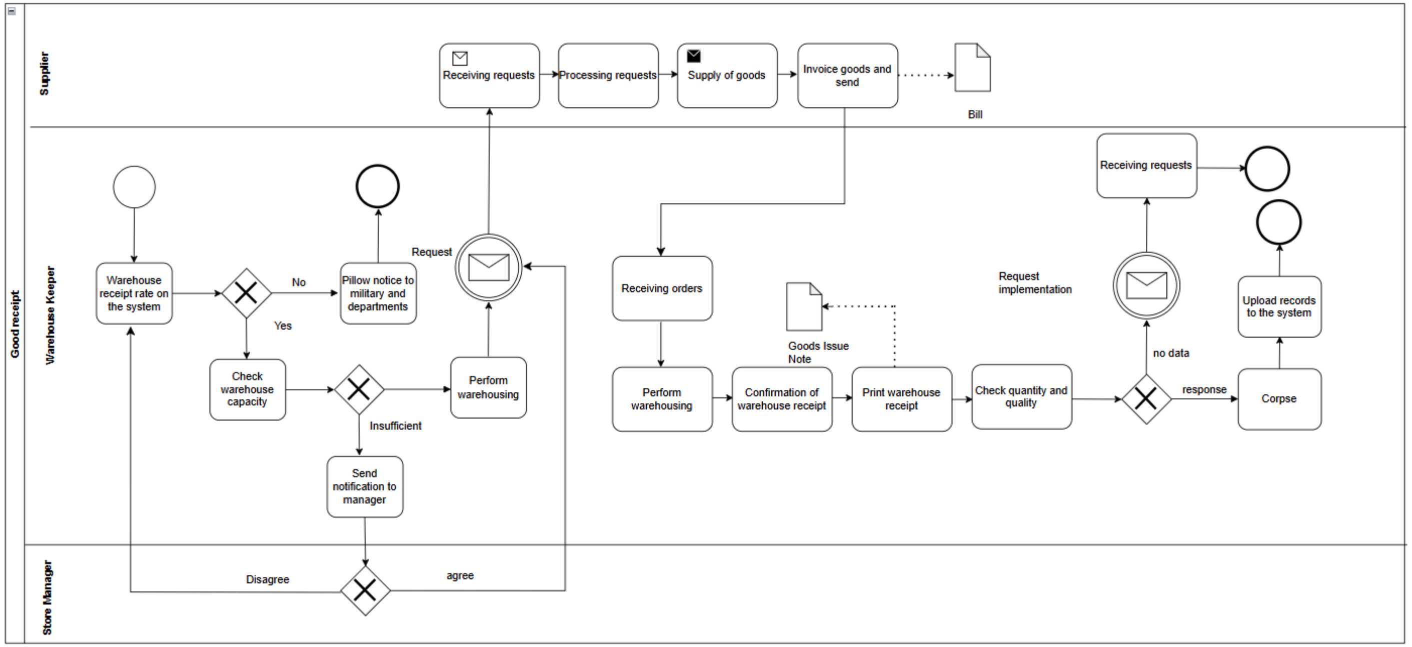

Figure 3.3.2.1: Good Receipt

b. Specification:

Table 3.3.2.1: Good Receipt Process BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Check warehouse receipt status	–	Warehouse Keeper	Verify if the receipt of goods has been recorded in the system.
Check warehouse capacity	Yes / No	Warehouse Keeper	Assess if there is enough space available in the warehouse.
Notify manager	If capacity is insufficient	Warehouse Keeper	Send a request to the store manager for further instructions.
Manager decision	Approve / Reject	Store Manager	Decide whether to approve or deny the warehousing request.
Send internal notice	–	Warehouse Keeper	Notify internal departments about the incoming goods.
Perform	–	Warehouse	Physically move goods into the warehouse.

warehousing		Keeper	
Receive warehousing request	–	Supplier	Accept order request from the retailer.
Process the request	–	Supplier	Review and prepare the requested goods.
Supply goods	–	Supplier	Deliver the goods to the warehouse.
Send invoice	–	Supplier	Issue and send the invoice along with the goods.
Receive goods	–	Warehouse Keeper	Acknowledge receipt of goods from the supplier.
Confirm receipt	–	Warehouse Keeper	Verify that the correct goods were received.
Print warehouse receipt	–	Warehouse Keeper	Generate and print the warehouse receipt (Goods Issue Note).

Check quality and quantity	–	Warehouse Keeper	Inspect the goods for quantity and quality compliance.
Request clarification	No data / Awaiting reply	Warehouse Keeper	Send a follow-up request if information is missing or unclear.
Upload records	–	System Admin / Staff	Upload data or documentation to the system for recordkeeping.
Finalize process (Corpse)	–	System / Admin (assumed)	Close the process or store the data permanently in the system.

3.3.3. Inventory Inspection Process

a. BPMN Diagram:

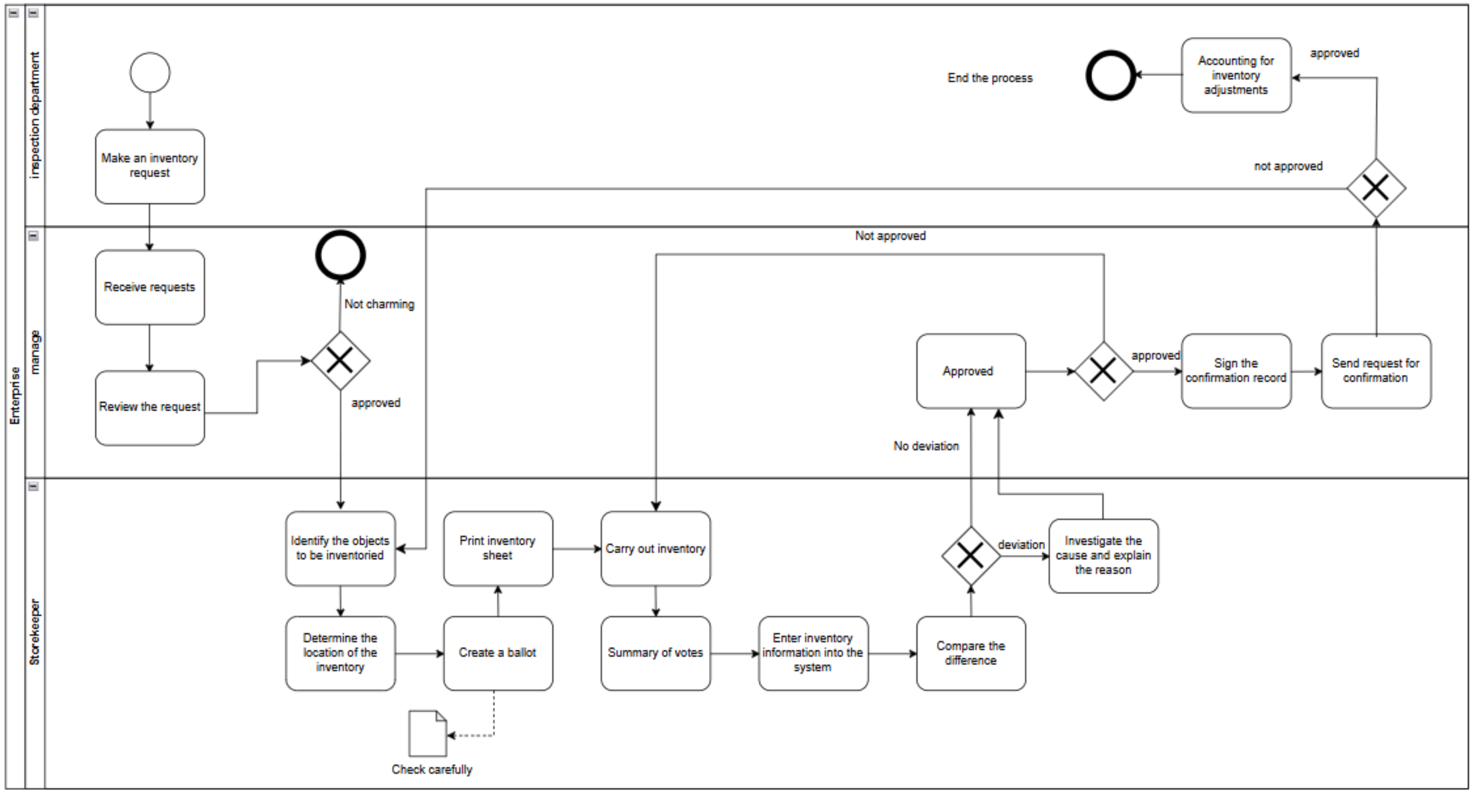

Figure 3.2.3: Inventory Inspection Process

b. Specification:

Table 3.2.3: Inventory Inspection Process BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Make an inventory request	–	Inspection Department	The Inspection Department initiates the process by creating an inventory request.
Receive requests	–	Management	Management receives the inventory request for further review.
Review the request	Approved / Not charming	Management	Management reviews the request and decides whether it is approved or not charming (not satisfying requirements).
Identify the objects to be inventoried	–	Storekeeper	The storekeeper identifies all objects that need to be inventoried.
Determine the location of the inventory	–	Storekeeper	The storekeeper determines the physical location of the items to be inventoried.
Print inventory	–	Storekeeper	The storekeeper prints the inventory sheet that lists all items to be

sheet			inventoried.
Create a ballot	–	Storekeeper	The storekeeper creates a ballot to be used during inventory count.
Carry out inventory	–	Storekeeper	The storekeeper performs the physical inventory count.
Summary of votes	–	Storekeeper	The storekeeper summarizes the votes from the ballot.
Enter inventory information into the system	–	Storekeeper	The storekeeper inputs inventory information into the system.
Compare the difference	Deviation / No deviation	Storekeeper	The storekeeper compares the counted inventory with the expected results and identifies deviations.
Investigate the cause and explain the reason	–	Storekeeper	If there is a deviation, the storekeeper investigates and explains the reason.
Approved	–	Management	If the deviation is resolved, Management approves the results.

Sign the confirmation record	–	Management	Management signs the confirmation record.
Send request for confirmation	–	Management	Management sends a request for confirmation to finalize the process.
Accounting for inventory adjustments	Approved / Not approved	Management	Management makes final accounting adjustments based on inventory discrepancies.
End the process	–	All performers	The process concludes, finalizing all steps.

3.3.4. Supplier Payment Process

a. BPMN Diagram:

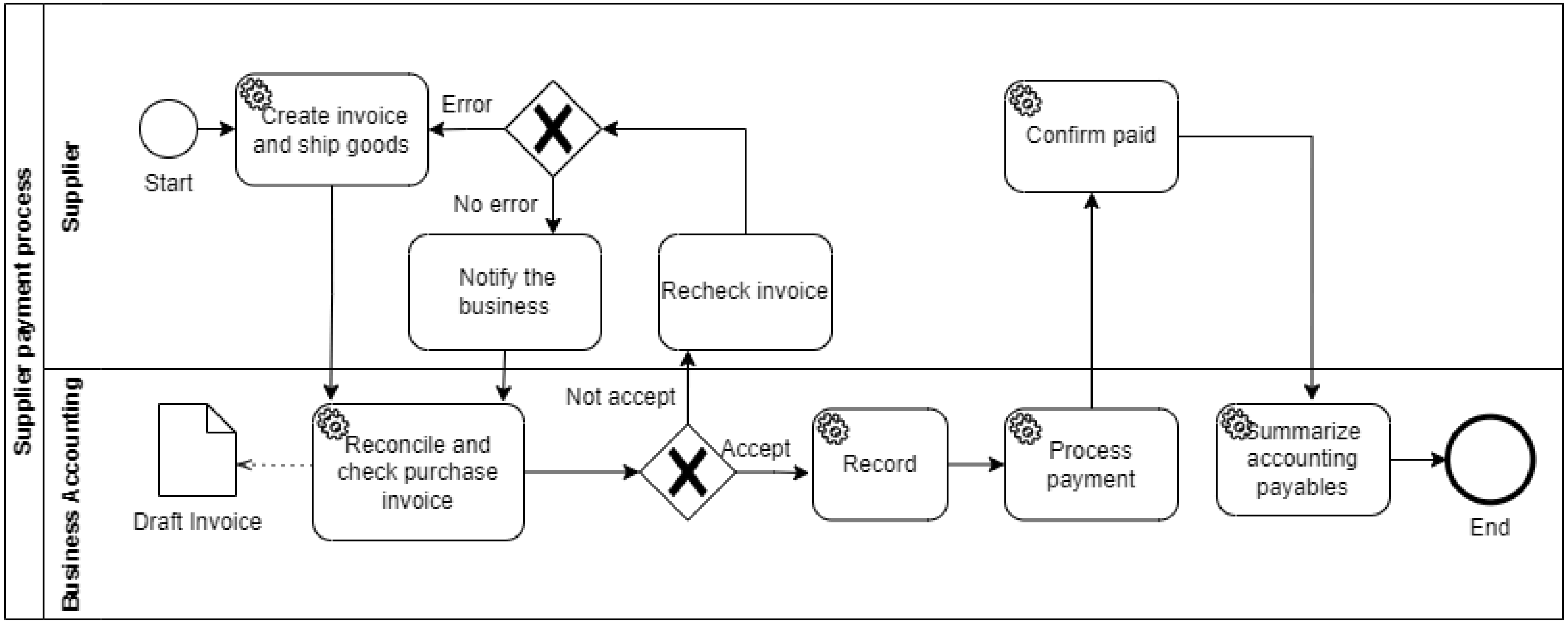

Figure 3.2.4: Supplier Payment Process

b. Specification:

Table 3.2.4: Supplier Payment Process BPMN

ACTIVITY	GATEWAY	PERFORMER	DESCRIPTION
Create invoice and ship goods	–	Supplier	Supplier creates invoice and dispatches goods to the business.
Reconcile and check purchase invoice	–	Business accounting	Accounting staff compares invoice with purchase records.
Error in invoice?	Exclusive gateway	Business accounting	Determines whether invoice contains any errors.
Notify the business	–	Business accounting	Notifies supplier if invoice has errors.
Recheck invoice	–	Supplier	Supplier reviews and corrects the invoice.

Accept invoice?	Exclusive gateway	Business accounting	Is invoice acceptable after reconciliation?
Record	–	Business accounting	Records the approved invoice in the system.
Process payment	–	Business accounting	Initiates payment to the supplier.
Confirm paid	–	Suppliers	Supplier confirms payment has been received.
Summarize accounting payables	–	Business accounting	Business summarizes payment records.

In summary, the conclusion of Chapter 3 is the business process of an electronic company, and in this section, we have mentioned the traditional Sale and CRM process used in that system. To upgrade the old process to be more convenient and effective, we improve it by our solution and visualise detail via diagrams in part 3.2.1 and part 3.2.2. And how chatbot can be applied in this system, will be explained clearly for readers in the next chapter.

CHAPTER 4: PROPOSED ALGORITHMS AND SOLUTIONS

In Chapter 4,we implement the AI model for product recommendations, and the O2C process within the ERP system, focusing on its integration into the sales process and customer service process. This advanced LLM, developed by Open AI, processes text, significantly improving search, the smartphone adapting with customers' requirements as well as, providing the automatic end-to-sale process, detailing the integration steps into ERP systems, and discussing the required architectural adjustments. Our solution aims to improve their sales and customer services process for Cellphone S company by developing AI Chatbot using Open AI API. By the end of this chapter, readers are able to understand why this solution is valuable for Cellphone S Company.

4.1. Technology

4.1.1. Implementation:

Overview:

The implementation of the system is achieved by integrating Flask as the backend, OpenAI for intent detection, Pandas and NumPy for data processing, and an Oracle Database for storing customer and order information.

Development Process:

- **Data Loading and Preprocessing:** The system loads customer and product data from 2 CSV files. For products, a detailed description column is generated in order to be embedded for further semantic search. To explain it, embeddings for each product description are created using OpenAI's text embedding model to enable effective search and recommendation.
- **Intent Detection:** The system uses OpenAI's APi to detect the user's intent. It classifies messages into categories such as "has product intent", "brand", "price range", "battery", and "camera" preferences.

- **Product Filtering and Recommendation:** Based on user's intent, the system embedded user's messages, and then tracks the already product embedded dataset by using semantic search. The output then returns back to response user's intent as top three recommendations.

Order to Cash Management:

- The system allows customers to place orders by capturing their details (name, phone, email, etc.). Orders are assigned unique IDs, and the data is stored in the Oracle Database. It also supports order cancellation and payment processing via APIs integrated with QR code as a payment gateway.

Testing and Validation:

- **Unit Testing**: The individual parts such as data loading, embedding creation, and product filtering were separately tested to verify that each one functioned as anticipated.
- **Integration Testing:** The whole process from customer input, intent recognition, product filtering, and order creation was tested to ensure smooth interaction of the backend, the OpenAI API, and Oracle database.

Deployment:

The solution was implemented as a web application on Flask, with the chatbot interface being integrated into a frontend (HTML, CSS, JavaScript). The product recommendation and order management capabilities are live and operational, with smooth connections to the database and external APIs for payment and product details.

4.1.2 Dataset

We use two datasets from kaggle for this project, including Customers_Lookup.csv and Mobiles_Dataset.csv, aiming to develop Order-to-Cash and Returns and Customer Support features.

In Customers_Lookup.csv, it contains a comprehensive list of mobile phones with detailed information such as product name, brand, specifications (RAM, internal storage, camera, battery), pricing, and customer ratings. With Customer Lookup Dataset (Kaggle), this dataset includes customer details such as full name, date of birth, gender, and email address. For importing and preprocessing data, we follow 4 steps as the below:

- Loading customer and product data
- Generating product ‘description’
- Generating embedding for ‘description’
- Converting embedding to NumPy arrays

a. Load customer and product data

Firstly, the dataset of both mobiles and customers is imported to VScode, such as:

```
# Load dữ liệu
customers_df = pd.read_csv('./data/Customers_Lookup_Processed.csv')
products_df = pd.read_csv('./data/Mobiles_Dataset_Processed.csv')
```

Figure 4.1.2.1: Data Loading

Secondly, customers and products data is store into customers_df and products_df via using pandas library, and the below is our results:

- Customers:

Table 4.1.2.1: Sample Customer Dataset

Customer Key	First Name	Last Name	Birthdate	Gender	Email Address	Annual Income	Education Level	Occupation
11000	JON	YANG	4/8/1966	M	jon24@adventure-works.com	9000000	Bachelors	Professional

- Mobiles:

Table 4.1.2.2: Sample Mobiles Dataset

Company Name	Mobile Name	Mobile Weight	RAM	Front Camera	Back Camera	Processor	Battery Capacity	Price (VND)
Samsung	Galaxy Z Flip 5 512GB	187g	8GB	12MP	12MP + 12MP	Snapdragon	3700mAh	28470000

b. Generate product 'Description'

After importing the dataset, we create an input column for the embedding section.

```
# Tạo mô tả nếu chưa có
if 'Description' not in products_df.columns:
    def generate_description(row):
        return (f"{row['Model Name']} với {row['RAM']} RAM, camera trước {row['Front Camera']}, "
                f"camera sau {row['Back Camera']}, chip {row['Processor']}, pin {row['Battery Capacity']}, "
                f"màn hình {row['Screen Size']}")
    products_df['Description'] = products_df.apply(generate_description, axis=1)
```

Figure 4.1.2.2: Description Product Code

A new column is purposely generated which is a collection of all fields belonging to each phone named 'Description'. This column will be an input for the next part, embedding.

c. Generate embedding for 'Description'

By using Open-AI embedding models , text-embedding-3-small, we generate the Description column into vectors.

```
# Tạo embedding nếu chưa có
if 'embedding' not in products_df.columns or products_df['embedding'].isnull().any():
    def create_embedding(text):
        response = openai.embeddings.create(
            input=text,
            model='text-embedding-3-small'
        )
        return response.data[0].embedding

    print(" 🔹 Đang tạo embedding cho sản phẩm...")
    products_df['embedding'] = products_df['Description'].apply(create_embedding)
    products_df.to_csv('./data/Mobiles_Dataset_Processed.csv', index=False)
    print("✅ Đã lưu xong embedding.")
```

Figure 4.1.2.3: Embedding Description Product Code

The model generates data from the Description column from the previous step into numeric vectors. These vectors allow semantic comparison similarity between the user's question and product description. After that it is stored in a new csv file.

d. Convert embedding to NumPy arrays

When embedding in csv , the results are stored in string format, so we convert them into numeric values in order for further semantic search.

```
# Chuyển sang numpy array
def to_array(e):
    try:
        return np.array(eval(e)) if isinstance(e, str) else np.array(e)
    except:
        return np.zeros(1536)  # fallback nếu lỗi
products_df['embedding'] = products_df['embedding'].apply(to_array)
```

Figure 4.1.2.4: Converting embedding to NumPy arrays Code

By using the NumPy library with eval() function, we convert the string in the description text from the previous part into a vector array.

4.1.3 Database:

This is our ERD before being developed to a Relational Diagram:

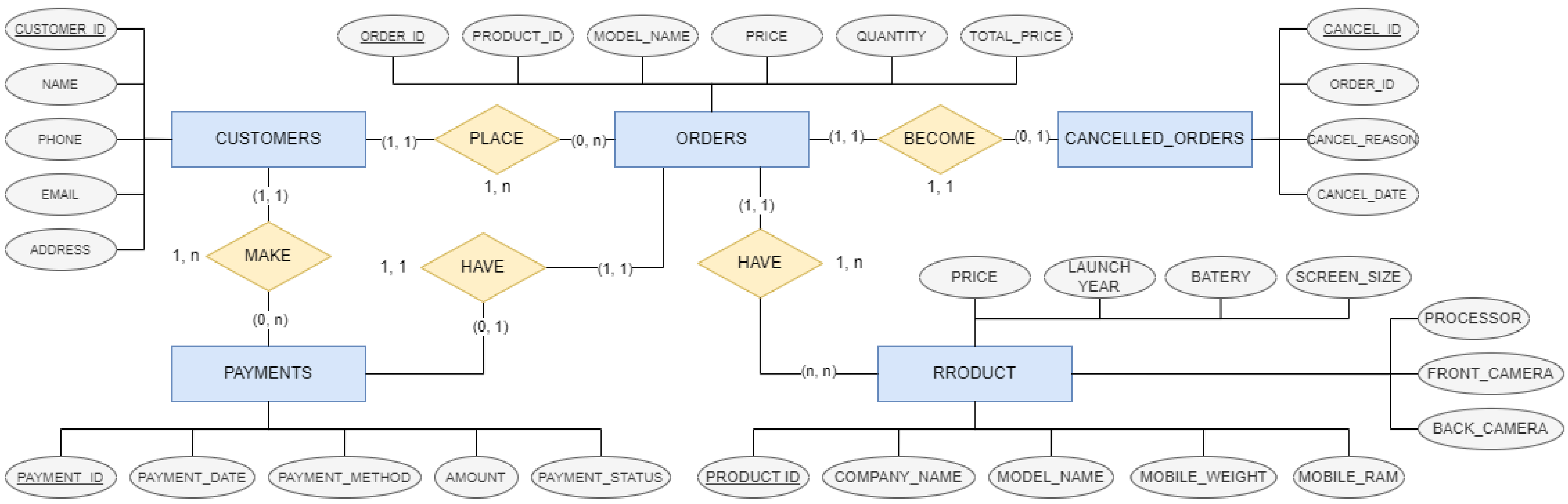

Figure 4.1.3.1: Entity relationship diagram

Beside, processing dataset for training chatbot, we also need a database to process the order, payment and cancelation of customers with the table, such as:

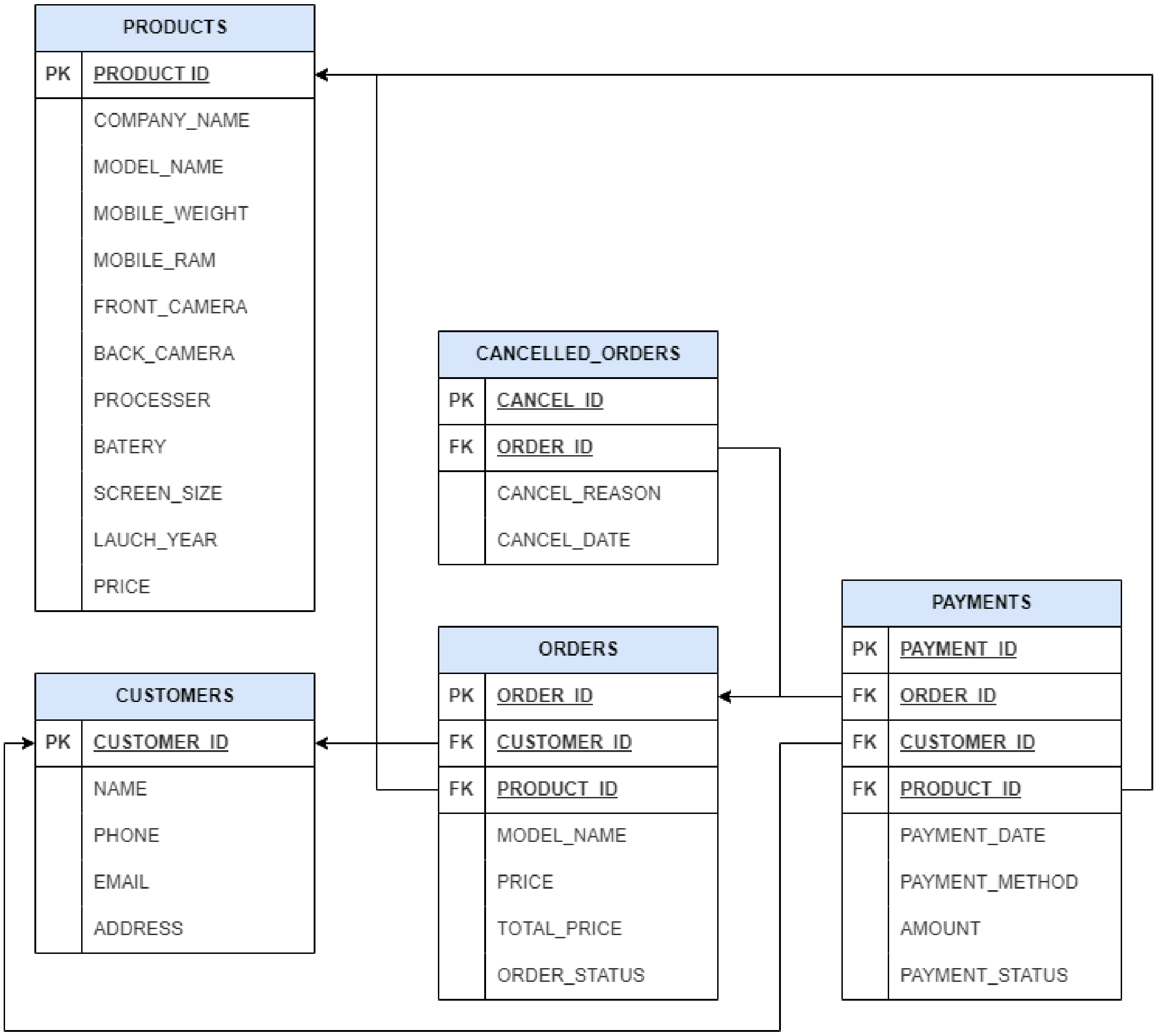

Figure 4.1.3.2: Relational model

Data Description:

Table 4.1.3 Data Description :

Table Name	Column Name	Data Type	Description
CUSTOMERS	CUSTOMER_ID	NUMBER (PK)	Unique identifier for the customer.
	NAME	VARCHAR2(100)	Name of the customer.
	PHONE	VARCHAR2(20)	Customer’s phone number.
	EMAIL	VARCHAR2(100)	Customer’s email address.
	ADDRESS	VARCHAR2(255)	Customer’s address.
ORDERS	ORDER_ID	NUMBER (PK)	Unique identifier for the order.
	CUSTOMER_ID	NUMBER (FK)	Customer ID (references CUSTOMERS).
	PRODUCT_ID	VARCHAR2(20)	Product code.

	MODEL_NAME	VARCHAR2(100)	Name of the product model.
	PRICE	NUMBER	Price of the product.
	QUANTITY	NUMBER	Quantity ordered.
	TOTAL_PRICE	NUMBER	Total price (PRICE * QUANTITY).
	ORDER_STATUS	VARCHAR2(30)	Status of the order: ‘pending’, ‘cancel’, ‘delivered’.
CANCELLED_ORDERS	CANCEL_ID	NUMBER (PK)	Cancellation record ID (auto increment).
	ORDER_ID	NUMBER (FK)	ID of the canceled order (references ORDERS).
	CANCEL_REASON	VARCHAR2(500)	Reason for order cancellation.

	CANCEL_DATE	DATE	Date of cancellation.
PAYMENTS	PAYMENT_ID	NUMBER (PK)	Payment record ID (auto increment).
	ORDER_ID	NUMBER (FK)	Order ID (references ORDERS).
	CUSTOMER_ID	NUMBER (FK)	Customer ID (references CUSTOMERS).
	PAYMENT_DATE	DATE	Payment date, defaults to the current date.
	PAYMENT_METHOD	VARCHAR2(50)	Payment method (e.g., cash, card, bank transfer, e-wallet, etc.).
	AMOUNT	NUMBER	Payment amount.
	PAYMENT_STATUS	VARCHAR2(20)	Status of the payment.

ORDER_ID	ORDER_ID	NUMBER (PK)	Unique identifier for the order.
	CUSTOMER_ID	NUMBER (FK)	Customer ID (references CUSTOMERS).
	PRODUCT_ID	VARCHAR2(20)	Product code.
	MODEL_NAME	VARCHAR2(100)	Name of the product model.
	PRICE	NUMBER	Price of a single unit of the product.
	QUANTITY	NUMBER	Number of units ordered.
	TOTAL_PRICE	NUMBER	Total price (calculated as PRICE * QUANTITY).
	ORDER_STATUS	VARCHAR2(30)	Status of the order: 'pending', 'cancel', 'delivered'. Default: 'pending'.

4.1.4. Result:

a. Product Recommendation:

Our result when we prompt for searching any smartphone is as:

ID001. Customers ask about style of smartphone:

Figure 4.1.4.1: Response of Chabot about style question

ID002. Customers ask about price:

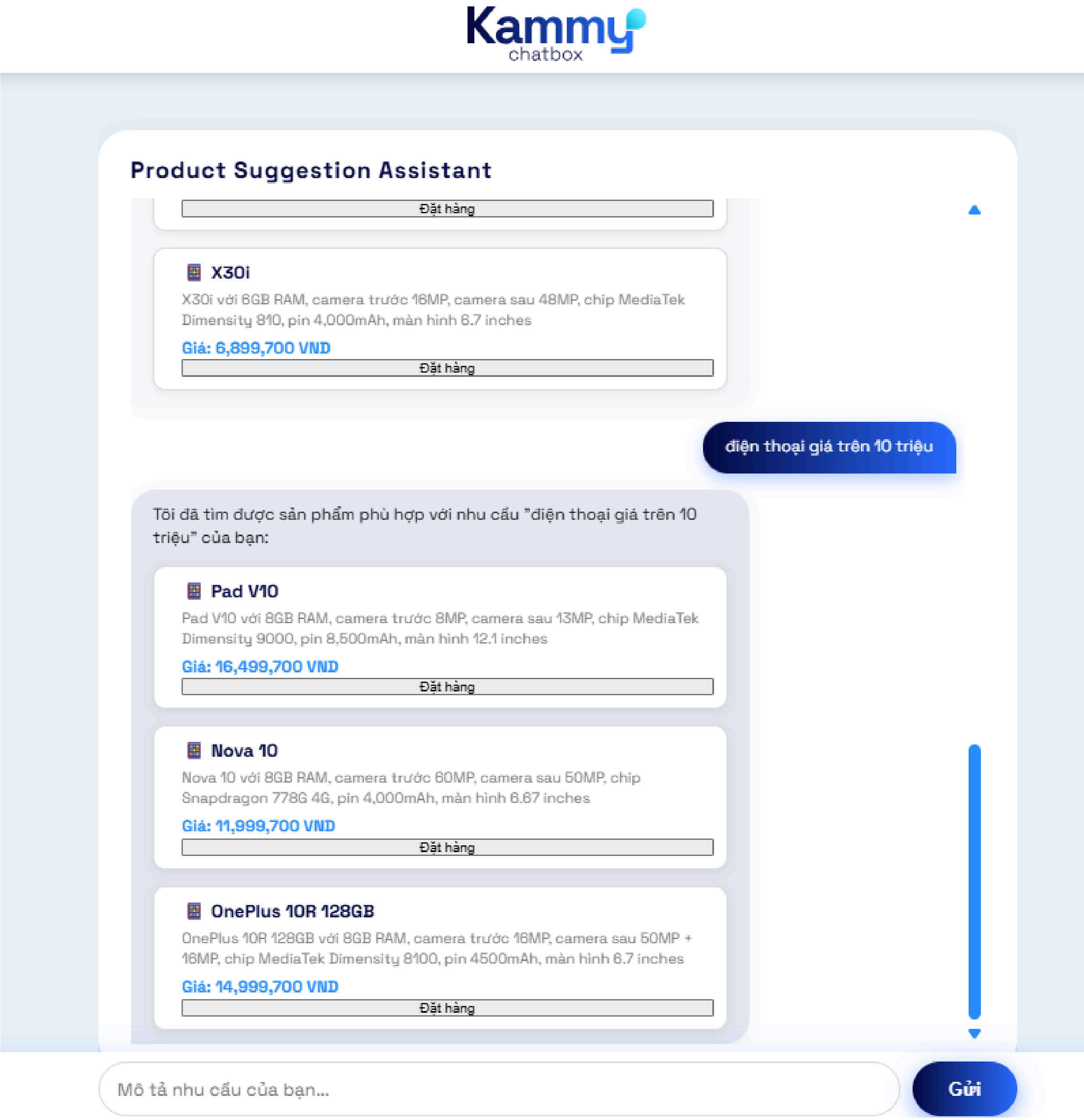

Figure 4.1.4.2: Response of Chabot about price question 1

Figure 4.1.4.3: Response of Chabot about style question 2

ID003. Customers ask about shape:

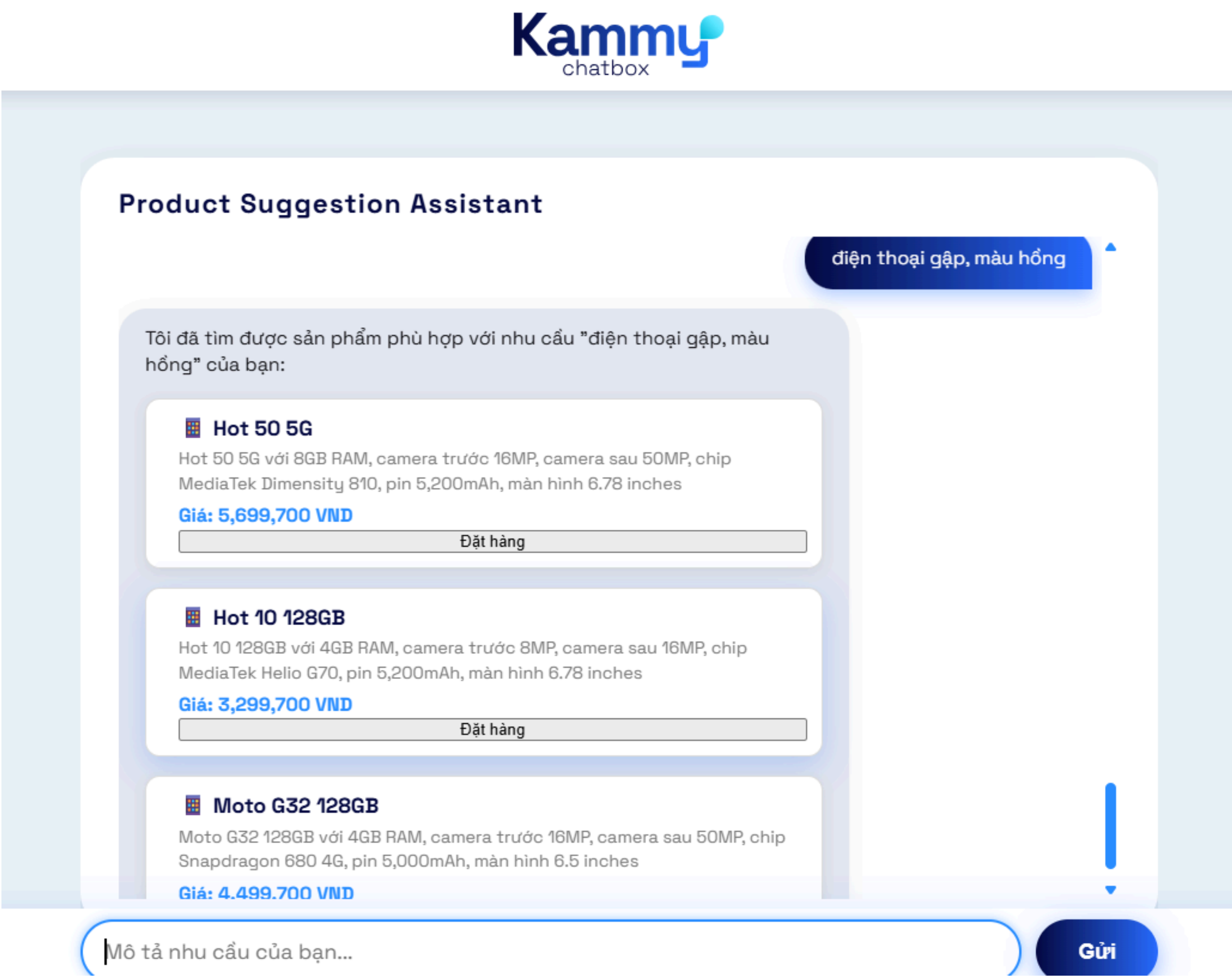

Figure 4.1.4.4: Response of Chabot about shape and design question

ID004. Customers ask about functions:

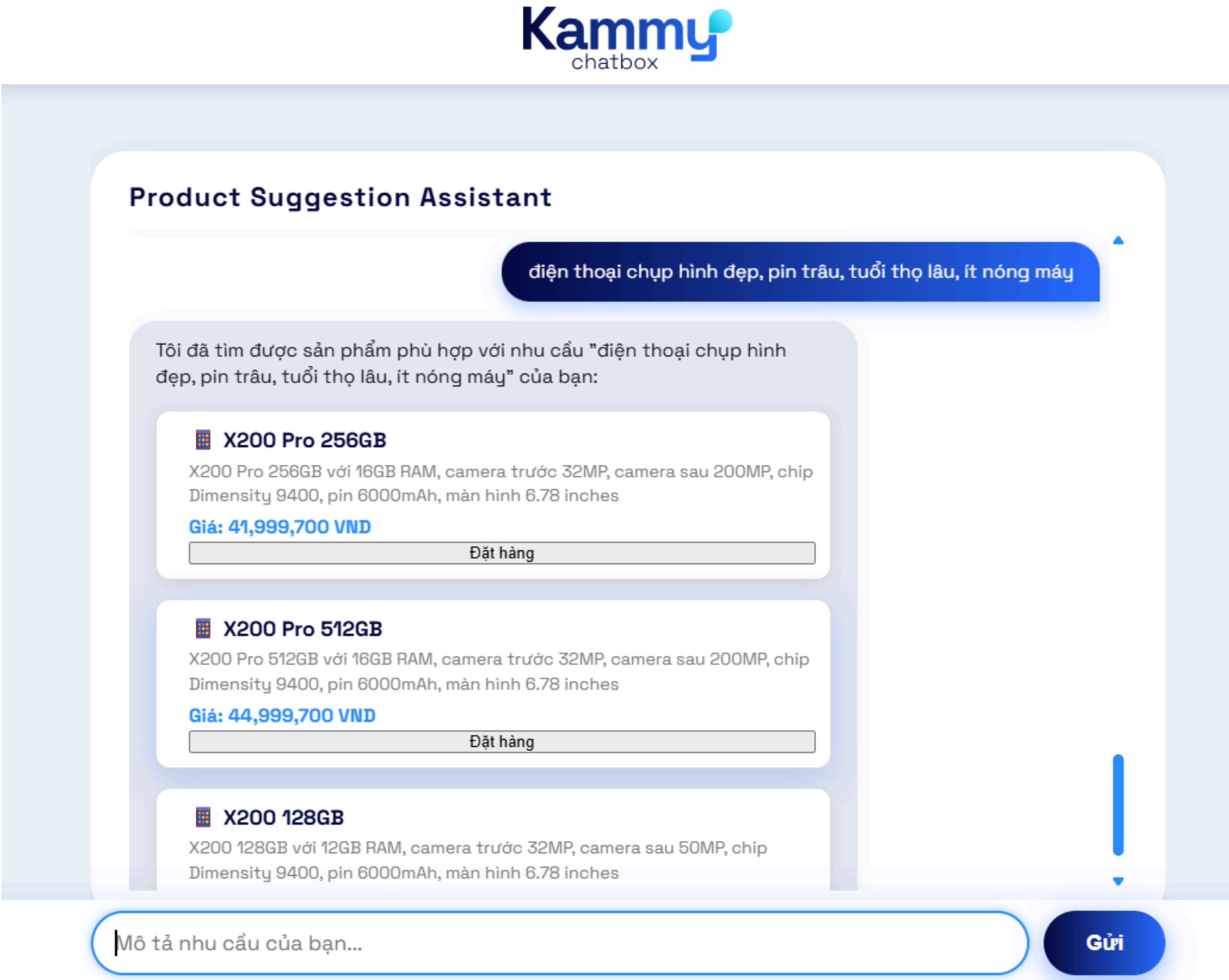

Figure 4.1.4.5: Response of Chabot about performance

ID005. Customers ask about features:

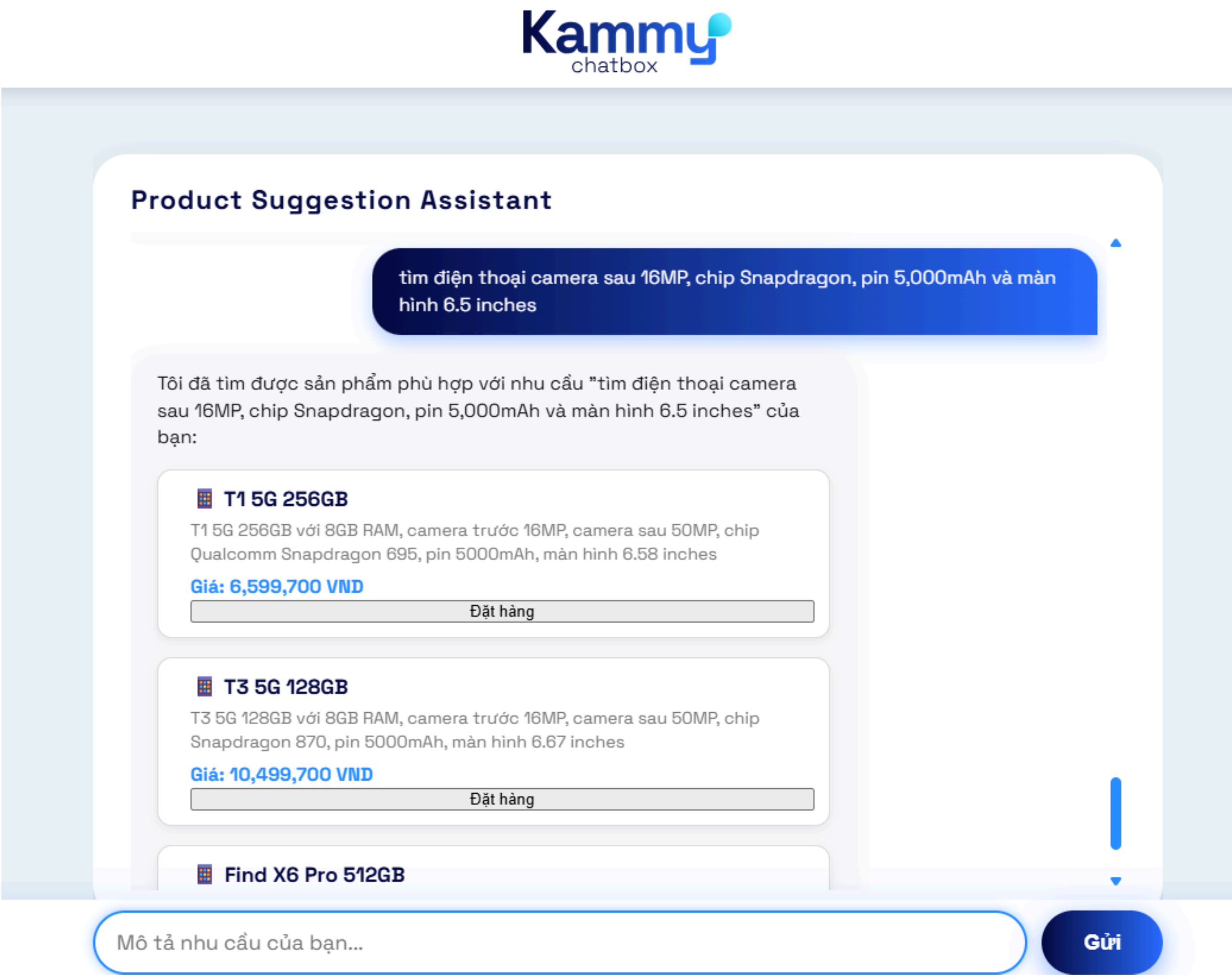

Figure 4.1.4.6: Response of Chabot about features question

b. End-to-Sale Process:

ID006. Orders Functions:

- **ID006.1.** Customers click on the button "order" in list of products chatbot provide, and then the order pop up will be display on the screen, such as:

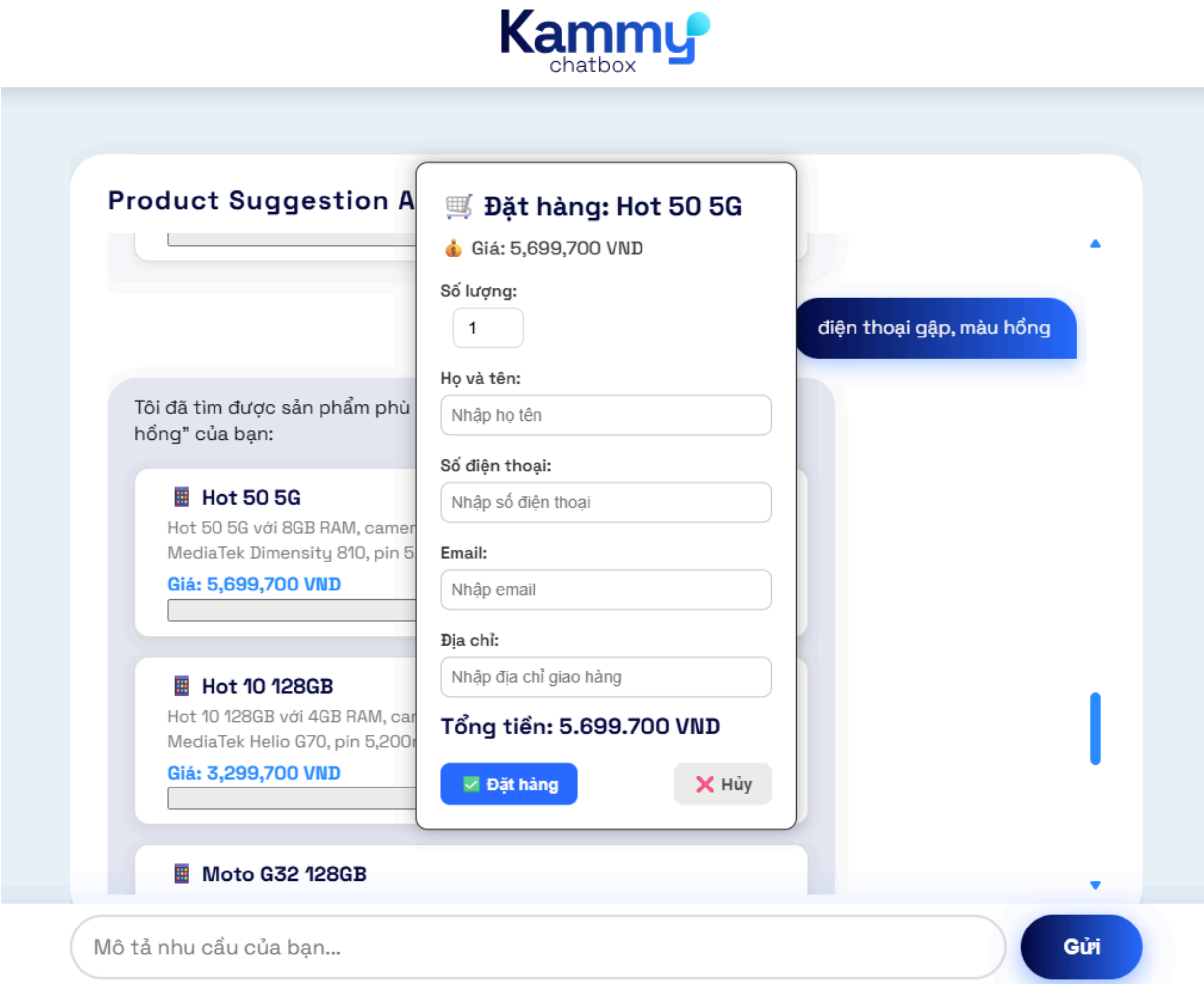

Figure 4.1.4.7: Open Order Popup

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- **ID006.2**: Filling all information to make order via Chatbot:

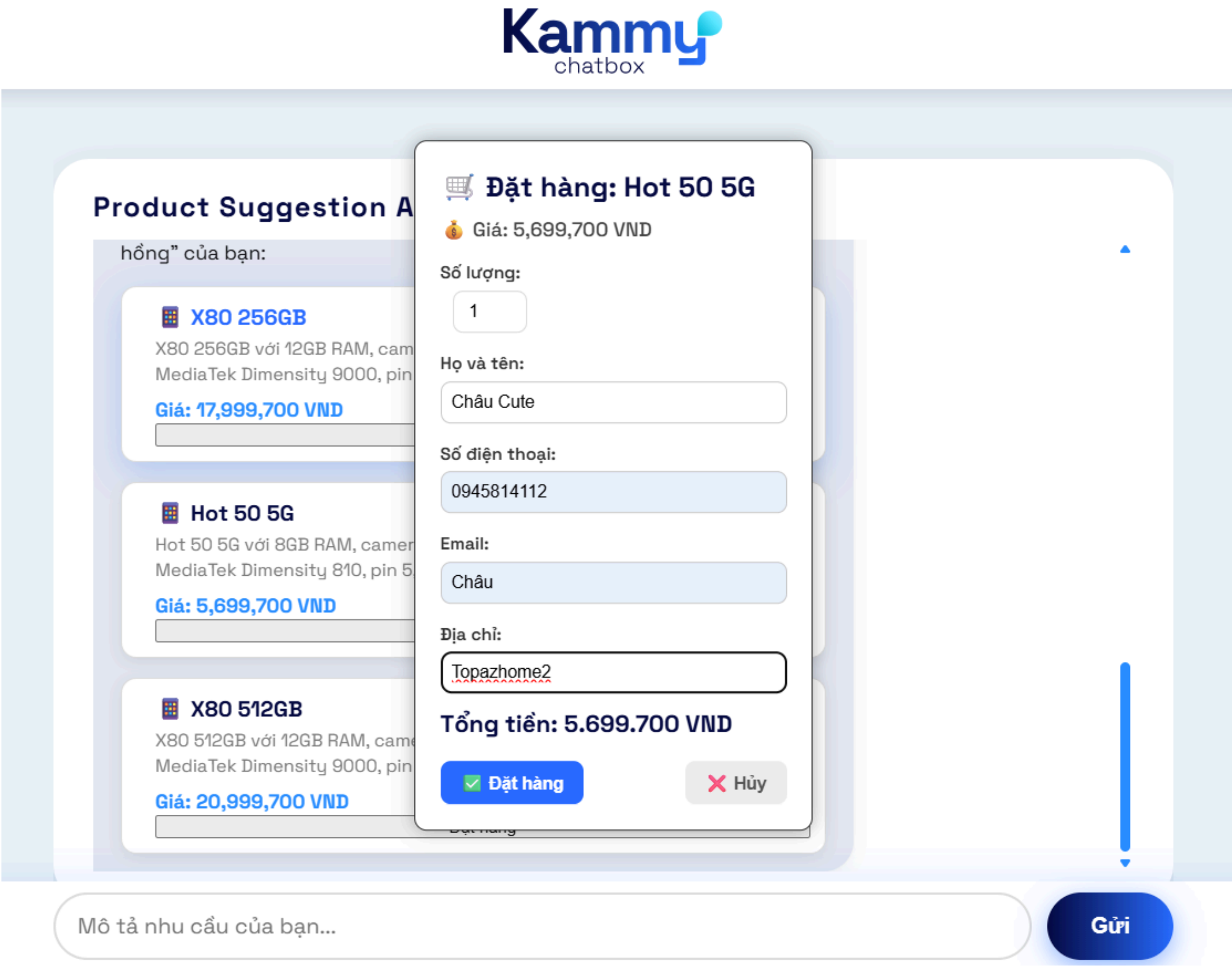

Figure 4.1.4.8: Fill in Information Order Popup

Note that: In this popup, the total of money of order will be calculated by the multiply of price and the quality. If the customer buy more than 2 smartphone, the system will update the voucher 30% or voucher 50%, depending on the number of smartphone and bill, particularly:

+ The product quality is more than 3

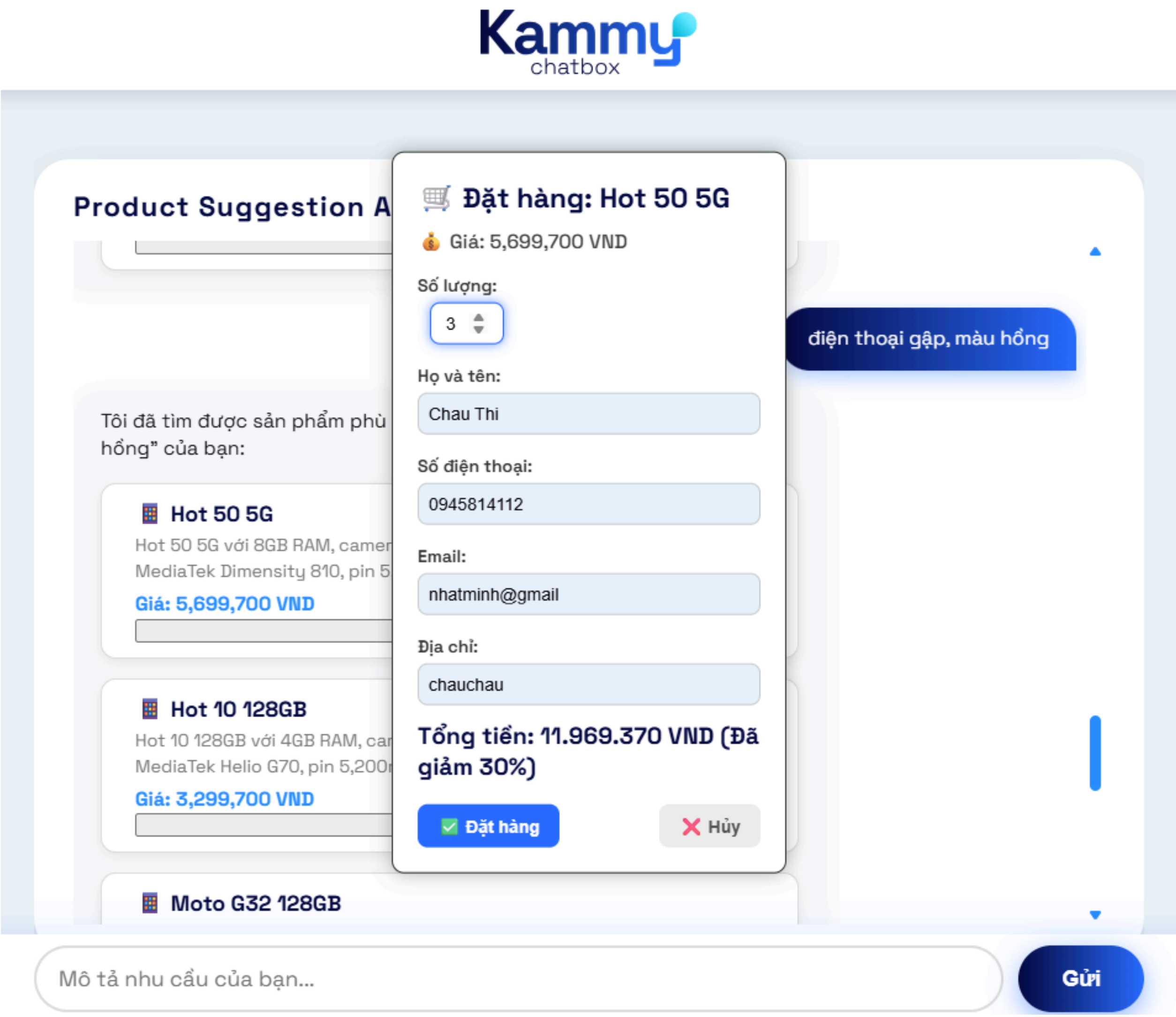

Figure 4.1.4.9: Order Popup with Quantity higher than 3 and lower than 5

+ The product quality is more than 5

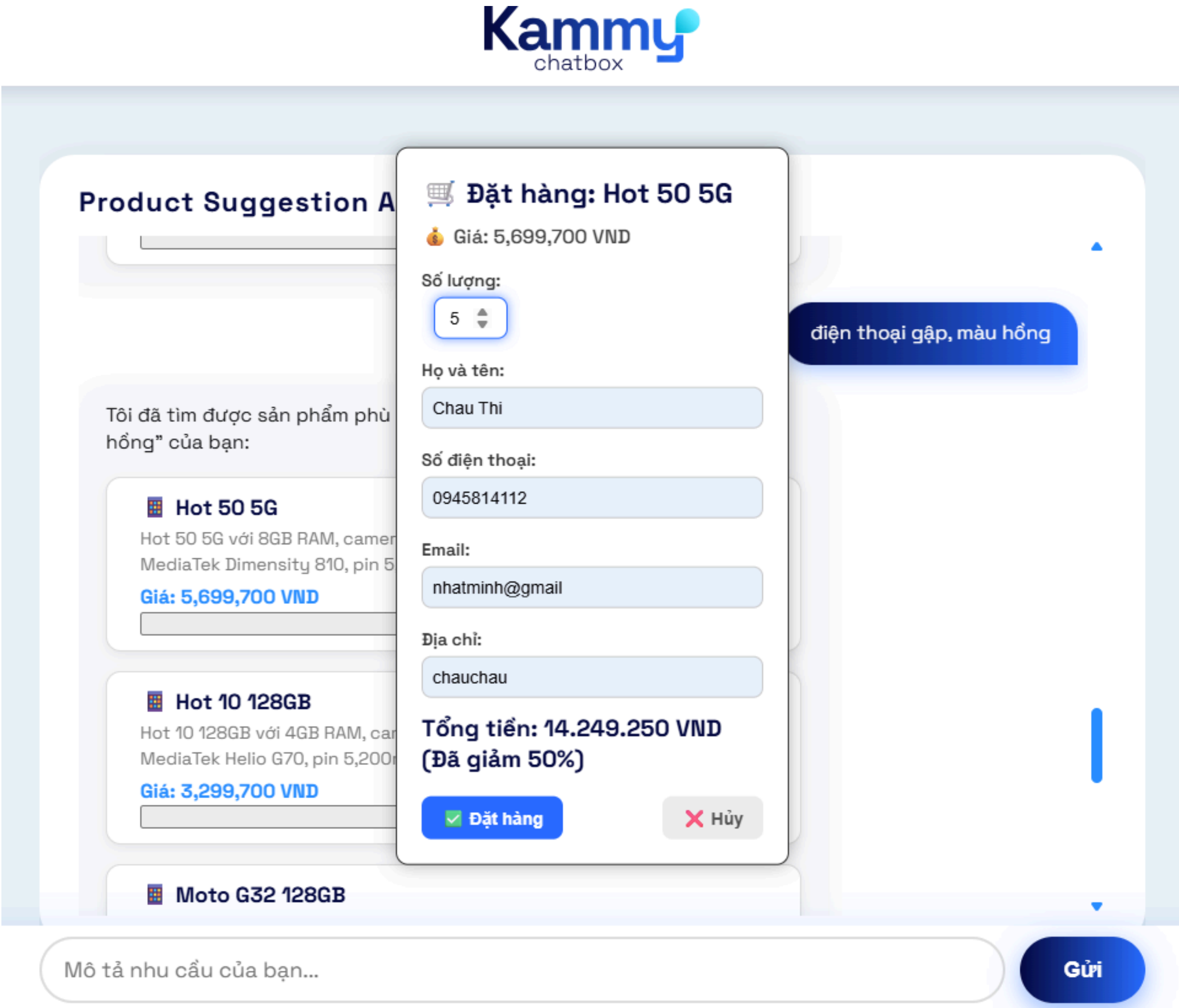

Figure 4.1.4.10: Order Popup with Quantity higher 5

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- **ID006.3**: After filling their information, customer can select "order" or "cancel"
    + **ID006.3.1.** In case of "order", the system will send successfully notification for customer:

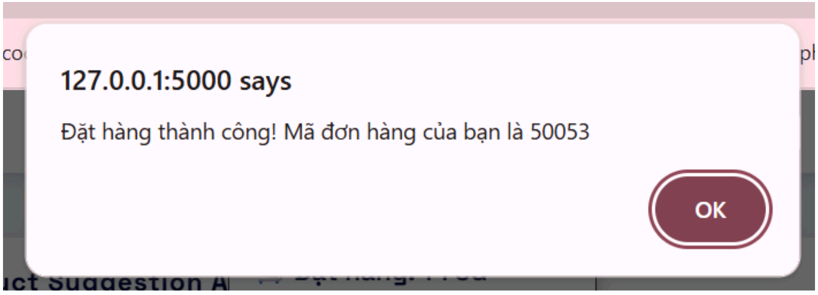

Figure 4.1.4.11: Successfully Order Notification

    + **ID006.3.2.** In case of "Cancel", customers are navigated to the current interface:

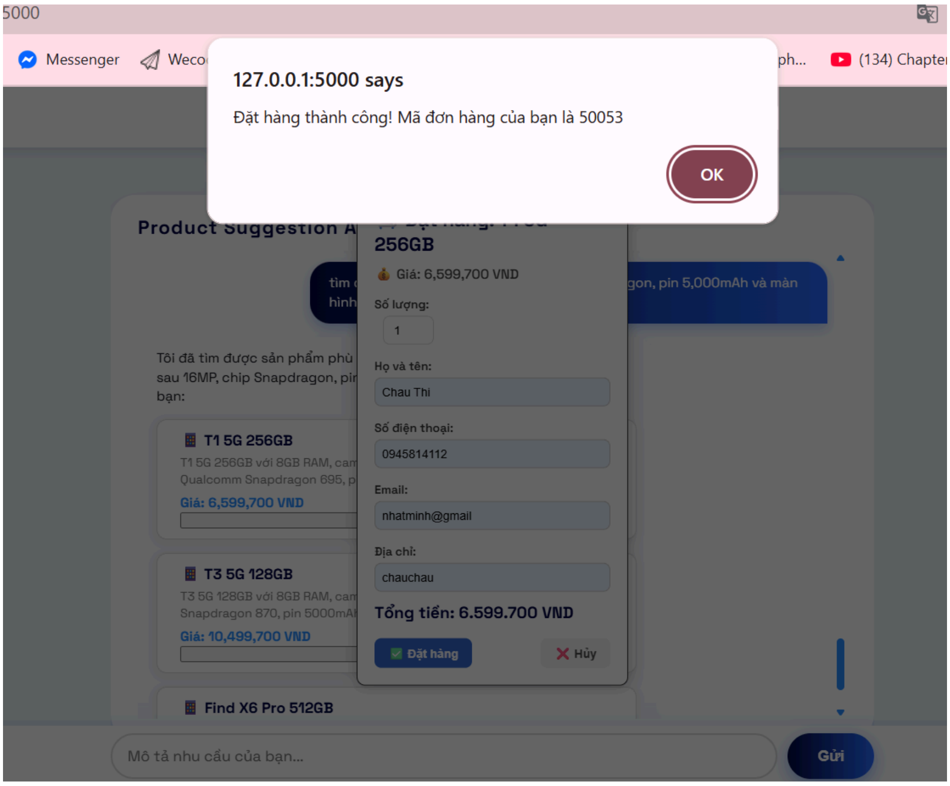

Figure 4.1.4.12: Cancel notification

And when order is finished, the Oracle database has been shown:

	ORDER_ID	CUSTOMER_ID	PRODUCT_ID	MODEL_NAME	PRICE	QUANTITY	TOTAL_PRICE	ORDER_STATUS	STATUS
1	50052	10052	UNKNOWN	Hot 50 Pro+	7499700	1	7499700	pending	pending
2	50053	10053	UNKNOWN	T1 5G 256GB	6599700	1	6599700	pending	pending
3	50041	10041	UNKNOWN	Play 5T	4499700	5	22498500	pending	pending
4	50042	10042	UNKNOWN	Reno8 Pro+ 5G 256GB	14099700	4	56398800	pending	pending
5	50043	10043	UNKNOWN	Reno8 5G 256GB	10499700	1	10499700	pending	pending
6	50044	10044	UNKNOWN	Play 7T	4499700	1	4499700	pending	cancel
7	50045	10045	UNKNOWN	Play 5T	4499700	1	4499700	pending	pending
8	50046	10046	UNKNOWN	Smart 5 64GB	2249700	1	2249700	pending	cancel
9	50047	10047	UNKNOWN	X7 128GB	6899700	1	6899700	pending	cancel
10	50048	10048	UNKNOWN	Razr 50 256GB	23999700	5	119998500	pending	pending

Figure 4.1.4.13: New orders being updated in Database

ID007. Payment Function:

- **ID007.1.** Customers send message “Payment” to the Chatbot Kammy, then it will response as:

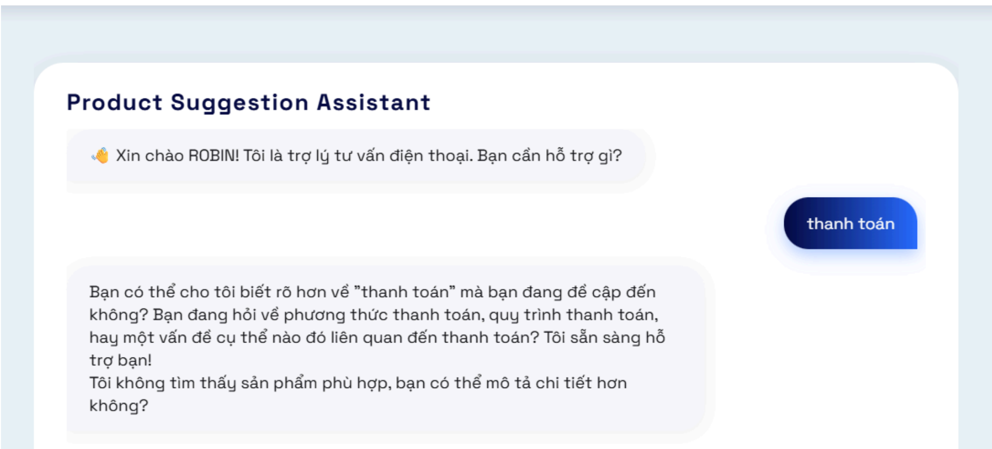

Figure 4.1.4.14: Payment prompts

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- **ID007.2.** System pop up the Payments window, including payment information and customers need to provide Order ID so the Chatbot can calculate automatically the total money that they have to paid and show on the screen like:

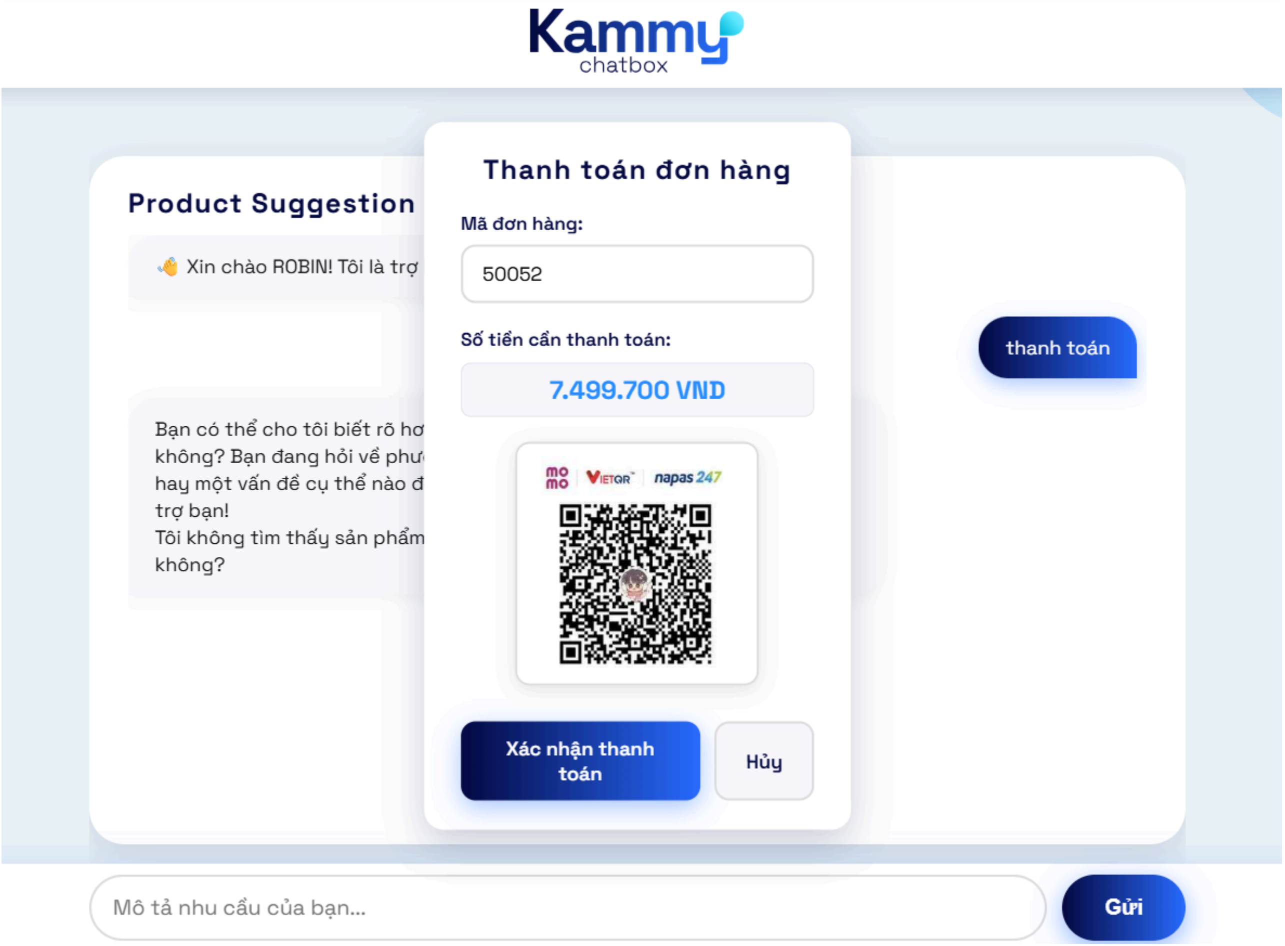

Figure 4.1.4.15: Payment Popup

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- **ID007.3.** If customers scan QR code for payment, the system will wait 5-10 seconds for processing the transaction.

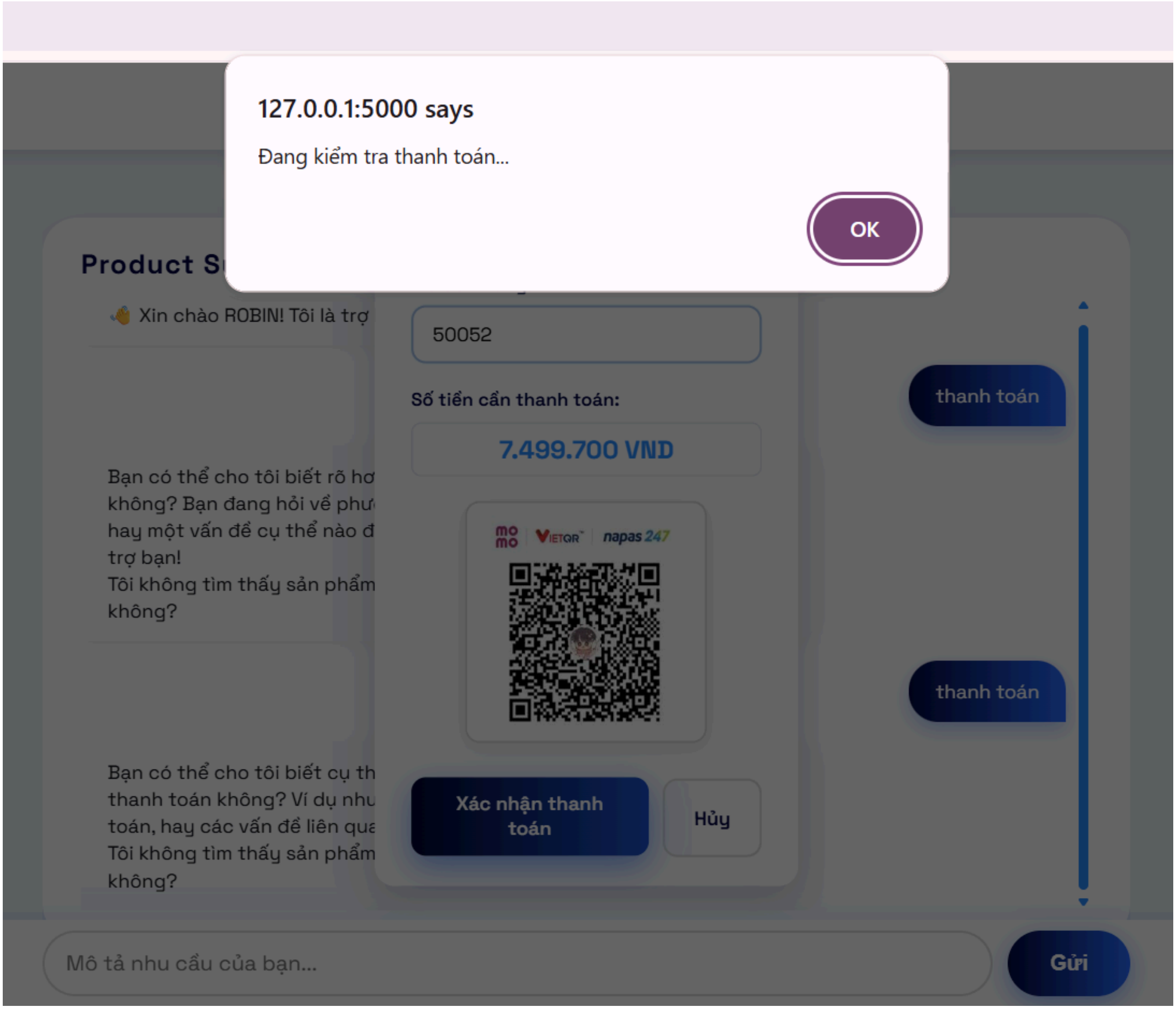

Figure 4.1.4.16: Checking Payment Popup

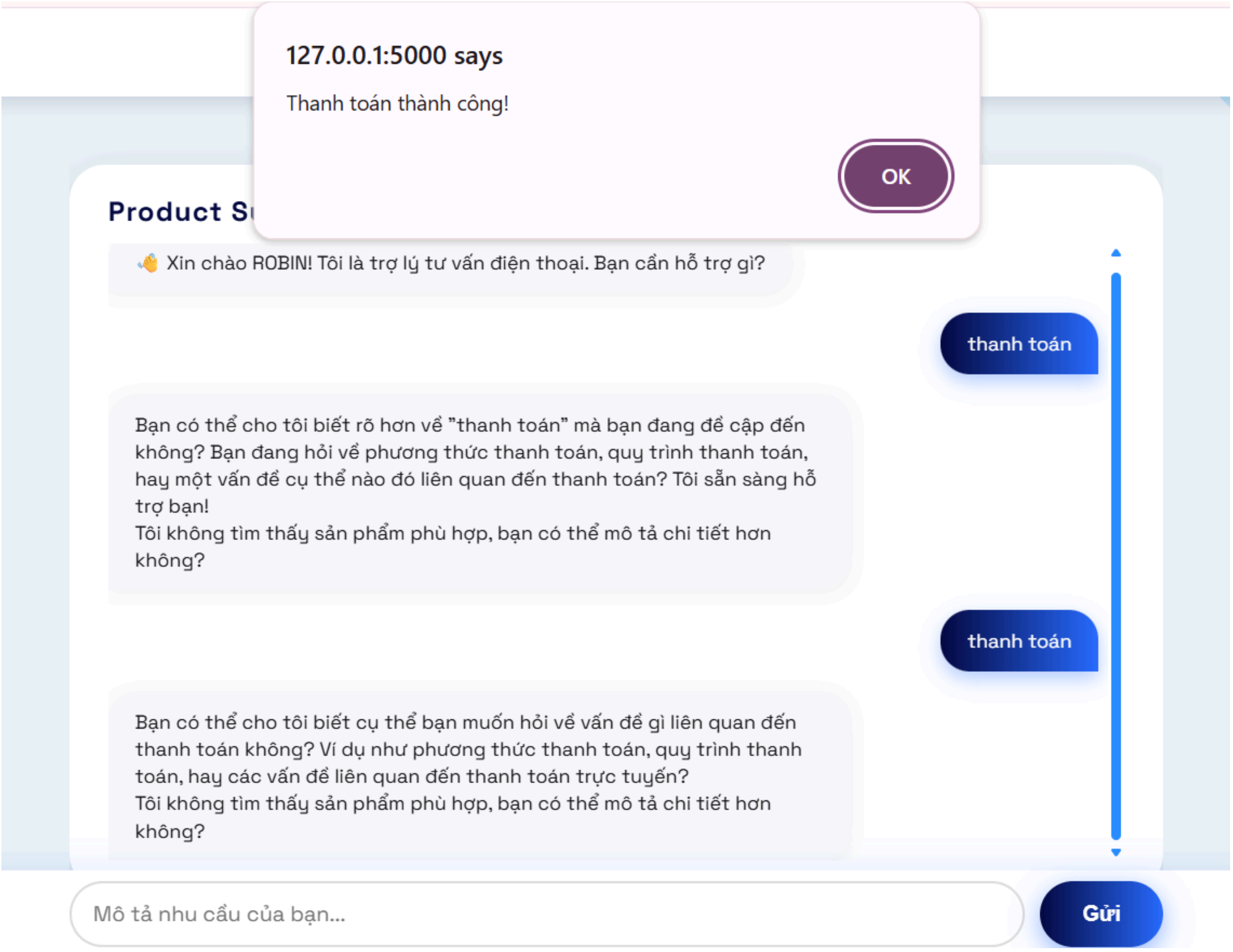

Figure 4.1.4.17: Successfully Payment Notification

The Payment information will be store in Paymentable with Payment Information, including date, amount, order id and status

	PAYMENT_ID	ORDER_ID	CUSTOMER_ID	PAYMENT_DATE	PAYMENT_METHOD	AMOUNT	PAYMENT_STATUS
1	1	50052	10052	05-JUN-25	QR Code	7499700	Paid

Figure 4.1.4.18: Payment Result in Database

ID008. Cancel Function

- **ID008.1.** Customers send message "Cancel" to the Chatbot Kammy, then it will response as:

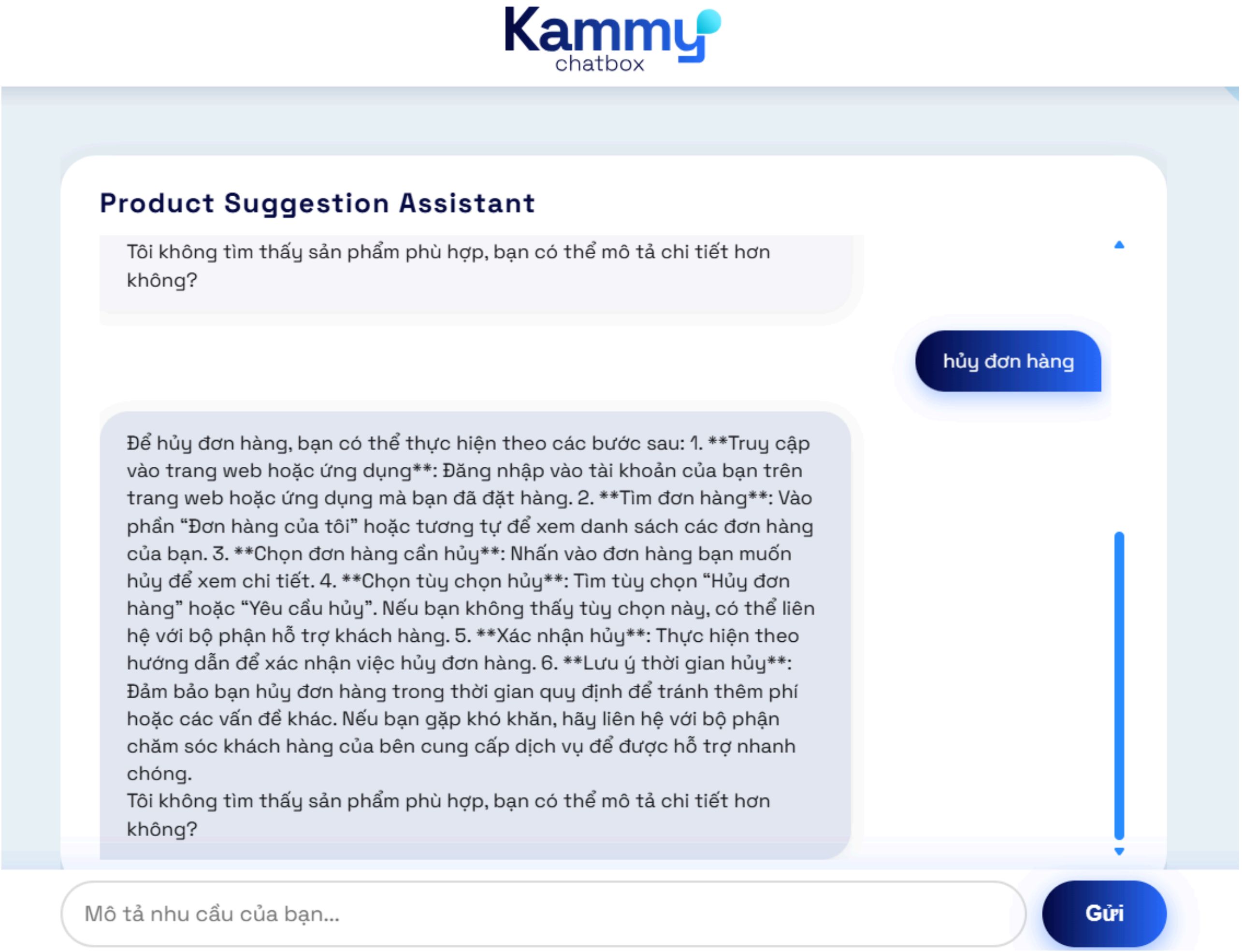

Figure 4.1.4.19: Result of Cancel Prompts

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- **ID008.2.** Customers enter their order ID that they want to cancel and their reason:

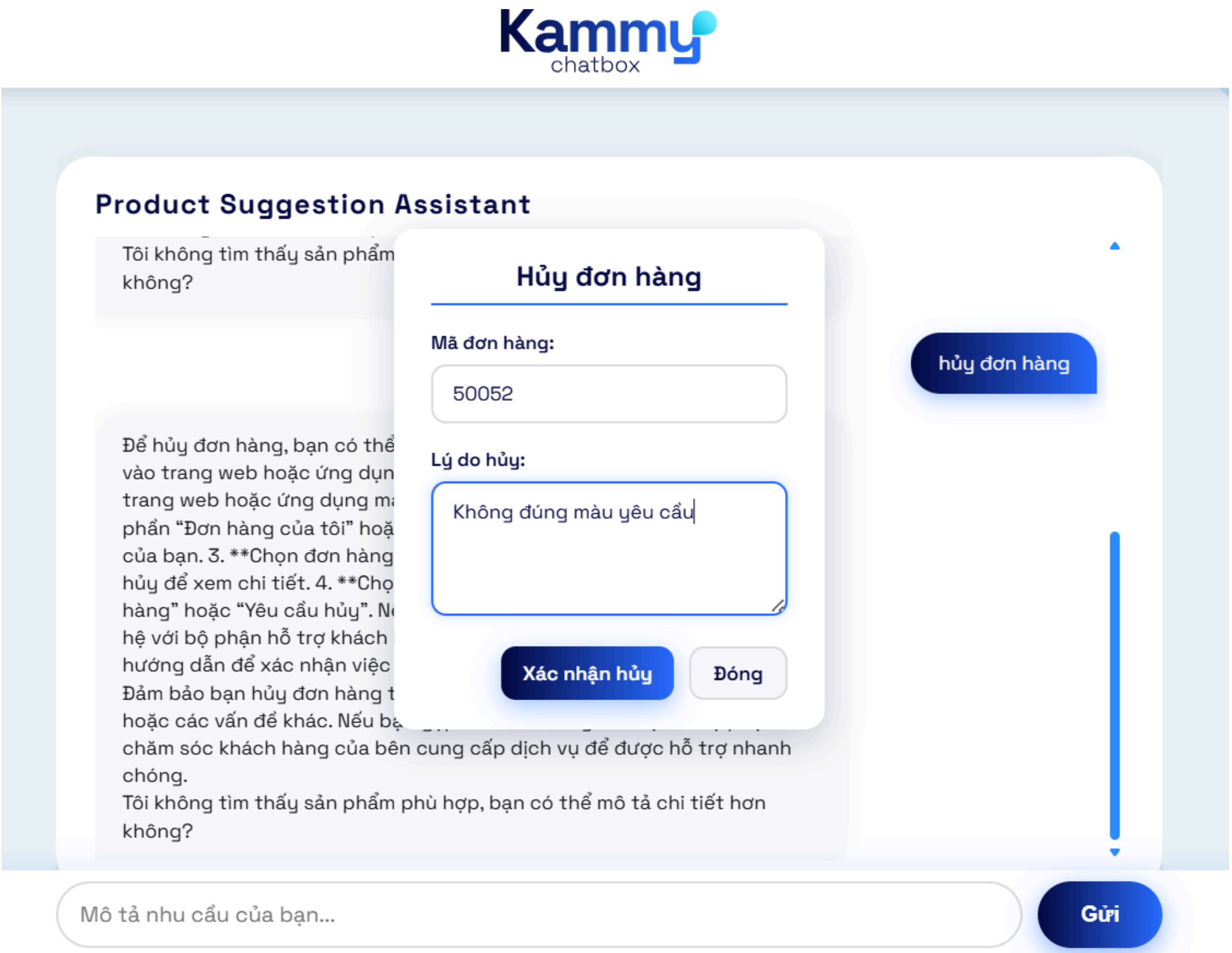

Figure 4.1.4.20: Cancel popup

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- **ID008.3.** If Customers choose “Cancel”, this action will be recorded in the Cancel table with status “pending” and sent to staff for checking and process. Besides, the attribute “Order_Status” of the Order table will be updated as “cancel”:

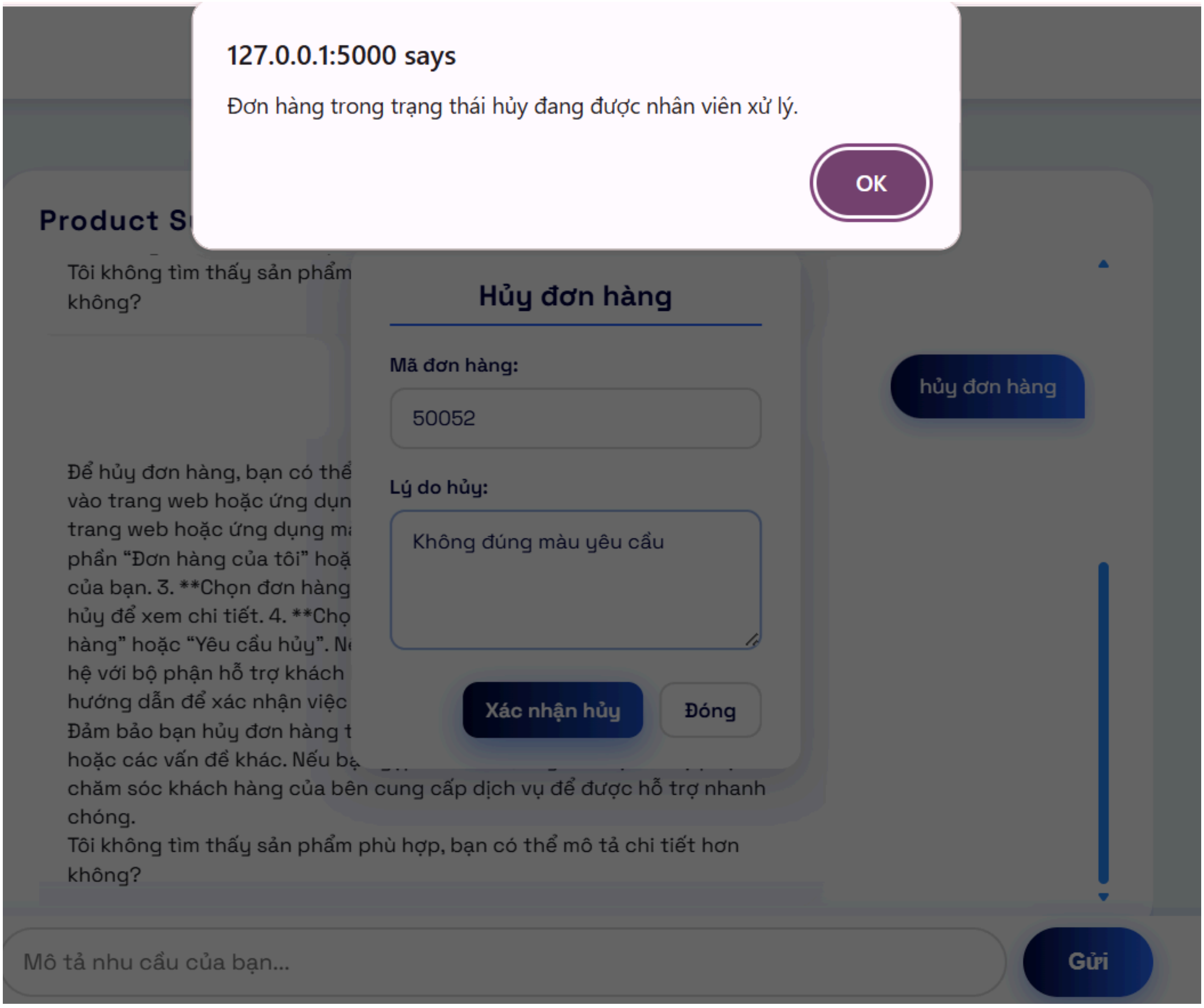

Figure 4.1.4.21: Notification about cancel order status

Then, in Order Table the ORDER_STATUS is updated as cancel

	ORDER_ID	CUSTOMER_ID	PRODUCT_ID	MODEL_NAME	PRICE	QUANTITY	TOTAL_PRICE	ORDER_STATUS	STATUS
1	50052	10052	UNKNOWN	Hot 50 Pro+	7499700	1	7499700	pending	cancel
2	50053	10053	UNKNOWN	T1 5G 256GB	6599700	1	6599700	pending	pending
3	50041	10041	UNKNOWN	Play 5T	4499700	5	22498500	pending	pending
4	50042	10042	UNKNOWN	Reno8 Pro+ 5G 256GB	14099700	4	56398800	pending	pending
5	50043	10043	UNKNOWN	Reno8 5G 256GB	10499700	1	10499700	pending	pending
6	50044	10044	UNKNOWN	Play 7T	4499700	1	4499700	pending	cancel

Figure 4.1.4.22: Result of cancel prompt in Order table

And the information of cancel order will be updated in Cancel with need to be checked by staff before official cancellation.

	CANCEL_ID	ORDER_ID	CANCEL_REASON	CANCEL_DATE
1	8	50052	Không đúng ...	05-JUN-25
2	7	50047	ghéc	05-JUN-25
3	1	50041	Tại ghéc	04-JUN-25
4	2	50042	ghéc	04-JUN-25
5	3	50043	ghéc a	04-JUN-25
6	5	50044	..	04-JUN-25
7	6	50046	chán	04-JUN-25

Figure 4.1.4.23: Result of cancel prompt in Cancel table

4.2. Solution Evaluation:

4.2.1. Competition analysis

a. Our competitions:

Compared with other chatbots in Vietnam, we find that Kammy can provide some features, such as order, cancellation, and payment that give it a significant advantage over its competitors.

Table 4.2.1.1: Competition analysis

Term	AhaChat	Haraven	Fchat	Kammy
Product Recommendation	Yes	Yes	Yes	Yes
Order features	*Yes*	*No*	*No*	*Yes*
Cancel features	*No*	*No*	*No*	*Yes*
Payment features	*No*	*No*	*No*	*Yes*
Automatic Send Notification to Customers	*Yes*	*Yes*	*Yes*	*Yes*
Send Vouchers	*Yes*	*Yes*	*Yes*	*Yes*
Calculating the number of price for each orders	*No*	*No*	*No*	*Yes*
Open AI API features	Yes	Yes	Yes	Yes

Making Report	Yes	Yes	Yes	No
Integrated with Social Media Platforms (such as Messenger, Tiktok, Zalo, etc)	Yes	Yes	Yes	Yes
Automatic Sending Messages via Email, Facebook, Zalo, etc	Yes	Yes	Yes	No
Record customer Information and Analyze their need	Yes	Yes	Yes	Yes

Thanks to this section, our group gets more understanding about our strength and weakness of our solution, compared with these competitors. Firstly, we need to integrate our chatbot with the social media of the company, including Tiktok Channels, Facebook, Instagram, etc in order to allow companies to respond and support their customers effectively and conveniently. Secondly, Kammy does not include automatic Sending Messages via Email, Facebook, Zalo, etc function and automatic report so it can be a weakness when comparing with these solutions.

b. Core Values:

As we use API from Open AI and retrain models with sample dataset to this solution, the investment in building a full AI system and data processing infrastructure can be reduced, saving costs while still optimizing customer support. Besides, by providing a full process O-2-C and product recommendation system for Cellphone S company, they are able to save time for labourers and speed up the transition from customer interest to actual buying decisions.

4.2.2. SWOT analysis

Table 4.2.2.1: SWOT analysis

STRENGTH	WEAKNESS
- Chatbot provides full O-2-C Process so customers do not need to switch many tabs for searching, payments, ordering and so on. - Chatbot can response customer 24/7, adapting with their personal character, and need, fostering their satisfaction during purchasing - Chatbot can reduce the workload for staff, leading to save time, reduce human cost, etc	- “Kammy” lacks some functions that support in CRM, which companies can use to propose their business strategies, such as: record customers’ behaviours, orders information.
OPPORTUNITIES	**THREAT**
- The advent of Technology can be advantage for improve accuracy and human aspect of AI - The investment in AI to enhance enterprise processes has become popular in recent years.	- External factors that may prevent the chatbot from providing support include power outages, hacking attacks, network failures, and hardware malfunctions

CHAPTER 5: CONCLUSIONS AND FUTURE DIRECTIONS

Chapter 5 covers the fundamental elements of ERP systems throughout the implementation process, beginning with an introduction to its theoretical groundwork. It then explores the technologies applied in the development of the system and the historical progression of these technologies. Finally, the hands-on applications and limitations of ERP integrations are illustrated through real-world problems.

5.1 Conclusion:

This project has exhaustively described the perplexing terrain of ERPs, and tracked their application from the theoretical foundations to the actual implementation, and its associated problems. In particular, this study started by providing a solid theoretical foundation for ERP and then considered technological developments that have influenced system evolution. The study makes a substantial contribution with its in-depth analysis of the real-world implications and constraints faced during ERP integrations and specifically from the context of an AI-induced Sales assistant, made for helping with the "paradox of choice".

Several important observations have emerged from the construction and study of the Kammy solution. Firstly, the project successfully demonstrated the viability of leveraging existing AI APIs, such as those from OpenAI, in conjunction with re-trained models to optimize costs while still delivering effective customer support. This approach highlights a pragmatic pathway for businesses to adopt advanced AI capabilities without necessitating massive investments in proprietary AI infrastructure. Secondly, the project underscored the immense potential of an AI-powered sales assistant to streamline the sales process, particularly in facilitating the Order-to-Cash (O2C) cycle and enhancing product recommendations. It therefore provides directly to a reduction of the time spent with labor and of the warranting time of conversion time and leads to shortening the conversion time, so as a result of that, it can work to create beneficial effects to firms, such as Cellphone S mentioned in our contribution, and lastly this study of SWOT analysis helped us a lot in order for us to clarify clearly or systematic and logical whether any strengths of weaknesses are in the developed solutions and whether the further automated integration is needed to be added anywhere else for making the developed solutions more high impact and competitive at the marketplace.

5.2 Future directions:

Leveraging the learnings and foundation set in this project, future work should centre around several key areas to continue driving the progression and wide-spread adoption of AI-enabled sales assistants in the ERP domain. First of all, the most fundamental path is to embed the AI chatbot to intelligently sync with popular social media platforms like TikTok, Facebook, Instagram, etc. This integration is important because it allows companies to meet customers where they are — providing them a return of instant, accessible and highly effective service, right through the communication channel where they are spending most of their time. Secondly, the absence of the feature to automatically sending messages through the most important communication channels: email, Facebook Messenger, Zalo deserve improvements opportunities. Adopting these aspects would not only automate routine communications, but also allow proactive customer communication and follow-ups to enhance customer experience. Additionally, while the proposed solution is a cost effective solution which leverages OpenAI APIs, it will become interesting in the future to look into the development of more customized and proprietary AI systems and specialized data processing infrastructures. This move could have resulted in better personalization, performance, and control of the AI capabilities. Ultimately, in order to really empower companies such as Cellphone S, it is necessary to extend the AI-enable sales assistant's functionality to cover both an end-to-end Order-to-Cash (O2C) process, and and advanced product recommendation system.